Single Variable Calculus

Mihir Rao

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Lecture 1: Derivatives, slope, velocity, rate of change

The derivative has two readings: geometrically it is the slope of the tangent line to a graph, and physically it is an instantaneous rate of change.

1.1 The derivative as the slope of a tangent line

A tangent line is *not* just a line meeting the graph at one point. It is the limit of secant lines. Fix $P = (x_0, f(x_0))$ on the graph of f and let $Q = (x_0 + \Delta x, f(x_0 + \Delta x))$ be a nearby point. The secant line PQ has slope $\Delta f / \Delta x$, where $\Delta f = f(x_0 + \Delta x) - f(x_0)$. The tangent line at P is the limit of the secant lines PQ as $Q \to P$ with P fixed.

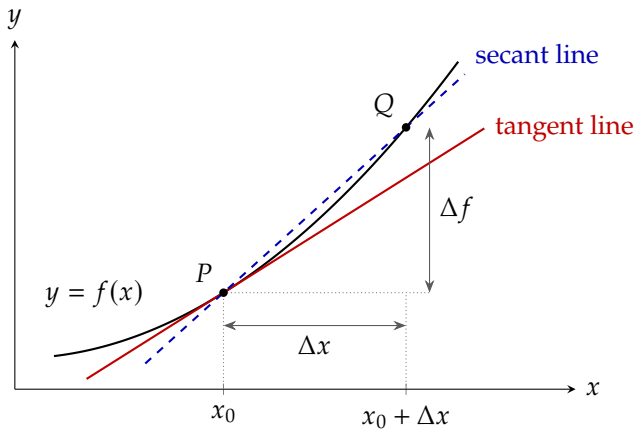


Figure 1: The secant line through P and Q has slope $\Delta f / \Delta x$. As $\Delta x \to 0$, Q slides to P and the secant lines approach the tangent line at P.

Definition 1.1 [Derivative]. The derivative of f at x_0 is the limit of the difference quotient

$$f'(x_0) = \lim_{\Delta x \to 0} \frac{\Delta f}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x},$$

when this limit exists. It is the slope of the tangent line to the graph at $(x_0, f(x_0))$.

Example: *The derivative of* $f(x) = 1/x$. Setting $\Delta x = 0$ directly gives the meaningless $0/0$, so cancel first. For $x_0 \neq 0$,

$$\frac{\Delta f}{\Delta x} = \frac{1}{\Delta x}\left(\frac{1}{x_0 + \Delta x} - \frac{1}{x_0}\right) = \frac{1}{\Delta x} \cdot \frac{x_0 - (x_0 + \Delta x)}{(x_0 + \Delta x)x_0} = \frac{-1}{(x_0 + \Delta x)x_0}.$$

Letting $\Delta x \to 0$ gives $f'(x_0) = -1/x_0^2 < 0$, matching the downward slope of the graph.

1.2 The tangent line and a triangle of constant area

The tangent line at (x_0, y_0), with $y_0 = f(x_0)$, is

$$y - y_0 = f'(x_0)(x - x_0).$$

For $f(x) = 1/x$ with $x_0 > 0$, this is $y - 1/x_0 = -(x - x_0)/x_0^2$.

Example: *The area under a tangent line to* $y = 1/x$. Setting $y = 0$ gives

$$-\frac{1}{x_0} = -\frac{x}{x_0^2} + \frac{1}{x_0} \quad \Longrightarrow \quad \frac{x}{x_0^2} = \frac{2}{x_0} \quad \Longrightarrow \quad x = 2x_0,$$

so the x-intercept is $2x_0$. The equations $y = 1/x$ and $x = 1/y$ are the same, so the graph is symmetric under exchanging x and y, and the y-intercept is $2y_0$ (setting $x = 0$ confirms this). The tangent line and the axes therefore bound a triangle of area

$$\tfrac{1}{2}(2x_0)(2y_0) = 2x_0 y_0 = 2x_0 \cdot \frac{1}{x_0} = 2,$$

independent of the point of tangency.

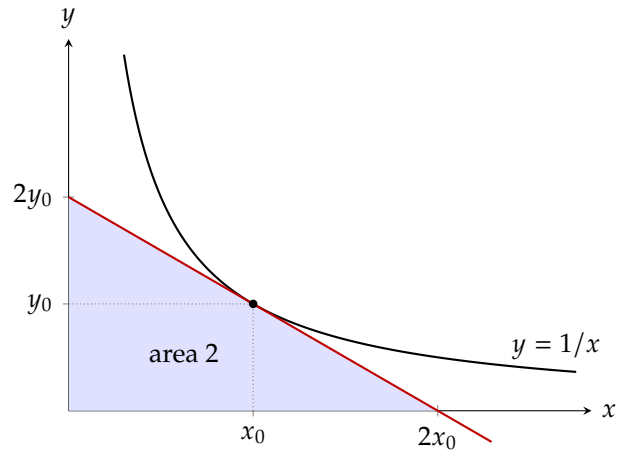


Figure 2: The tangent line to $y = 1/x$ at (x_0, y_0), drawn for $x_0 = 1.2$, meets the axes at $2x_0$ and $2y_0$. The shaded triangle has area 2 for every $x_0 > 0$.

1.3 Notation

With $y = f(x)$, write $\Delta y = \Delta f = f(x_0 + \Delta x) - f(x_0)$, the change in y. The two difference quotients $\Delta y/\Delta x$ and $\Delta f/\Delta x$ are equal, and as $\Delta x \to 0$,

$$\frac{\Delta y}{\Delta x} \to \frac{dy}{dx} \quad \text{(Leibniz's notation)}, \qquad \frac{\Delta f}{\Delta x} \to f'(x_0) \quad \text{(Newton's notation)}.$$

Leibniz's notation hides the point of evaluation, here $x = x_0$, which must be remembered. Other notations for the derivative of f are df/dx, f', and Df.

1.4 Derivatives of powers

Proposition 1.2 [Power rule]. *For $n = 1, 2, 3, \ldots$,*

$$\boxed{\frac{d}{dx}x^n = nx^{n-1}}.$$

Proof. Write x for the base point. Multiplying out the n factors of $(x+\Delta x)^n = (x+\Delta x)\cdots(x+\Delta x)$ gives

$$(x + \Delta x)^n = x^n + n(\Delta x)x^{n-1} + O\big((\Delta x)^2\big),$$

where $O((\Delta x)^2)$ stands for the terms containing $(\Delta x)^2, (\Delta x)^3, \ldots, (\Delta x)^n$. The term $nx^{n-1}\Delta x$ collects the n ways to choose Δx from exactly one factor; this is part of the binomial theorem. Hence

$$\frac{\Delta y}{\Delta x} = \frac{(x + \Delta x)^n - x^n}{\Delta x} = nx^{n-1} + O(\Delta x) \longrightarrow nx^{n-1} \quad \text{as } \Delta x \to 0. \qquad \square$$

The result extends to polynomials term by term, for example $\frac{d}{dx}(x^2 + 3x^{10}) = 2x + 30x^9$.

1.5 The derivative as a rate of change

The difference quotient is an average rate of change, and the derivative is the instantaneous rate of change. Speed is the basic example.

Example: *Dropping a pumpkin.* An object dropped from a 400-foot roof has height $y = 400 - 16t^2$ feet after t seconds. It reaches the ground when $400 - 16t^2 = 0$, at $t = 5$, so its average speed is $|\Delta y|/\Delta t = 400/5 = 80$ ft/s. Its instantaneous velocity at impact is $y'(5) = -32t\big|_{t=5} = -160$ ft/s, about 110 mph; it is negative because the height decreases.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 1, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 2: Limits, continuity; trigonometric limits

2.1 Rates of change

As $\Delta x \to 0$, the average rate of change $\Delta y/\Delta x$ tends to the instantaneous rate of change dy/dx. For example, if q is charge, s distance, and T temperature, then dq/dt is electrical current, ds/dt is speed, and dT/dx is the temperature gradient. Derivatives also measure the sensitivity of one quantity to another: in GPS, radio signals determine a distance h to a satellite up to measurement error, and the ratio $\Delta L/\Delta h$ tells how accurately this determines the position L on the ground.

2.2 Limits and continuity

In an easy limit, the answer is obtained by plugging in the limiting value:

$$\lim_{x\to 3} \frac{x^2+x}{x+1} = \frac{3^2+3}{3+1} = 3.$$

The limit $\lim_{\Delta x\to 0} \Delta f/\Delta x$ defining a derivative is never an easy limit, because $\Delta x = 0$ is not allowed. In general, $\lim_{x\to x_0}$ is computed under the assumption $x \neq x_0$.

Definition 2.1 [Continuity]. A function f is continuous at x_0 if

$$\lim_{x\to x_0} f(x) = f(x_0).$$

Definition 2.2 [One-sided limits]. The right-hand limit $\lim_{x\to x_0^+} f(x)$ is the limit as $x \to x_0$ through values $x > x_0$, and the left-hand limit $\lim_{x\to x_0^-} f(x)$ is the limit through values $x < x_0$.

Example: *A discontinuous function.* Let

$$f(x) = \begin{cases} x+1, & x > 0, \\ -x, & x \leq 0. \end{cases}$$

Then $\lim_{x\to 0^+} f(x) = 1$ but $f(0) = 0$, so f is discontinuous at 0. It is continuous from the left at 0, since $\lim_{x\to 0^-} f(x) = 0 = f(0)$, but not from the right.

Discontinuities come in several types.

1. A removable discontinuity: the one-sided limits are equal, $\lim_{x\to x_0^+} f(x) = \lim_{x\to x_0^-} f(x)$, but this common value is not $f(x_0)$, or $f(x_0)$ is undefined. For example, $\sin x/x$ is undefined at $x = 0$; its limit there is computed below.

2. A JUMP DISCONTINUITY: both one-sided limits exist but are not equal.

3. An INFINITE DISCONTINUITY: for example $1/x$ at 0, where $\lim_{x\to 0^+} 1/x = \infty$ and $\lim_{x\to 0^-} 1/x = -\infty$.

4. Other discontinuities, such as a function that oscillates infinitely often as $x \to x_0$. It does not tend to $\pm\infty$ or to anything else; the limit DOES NOT EXIST.

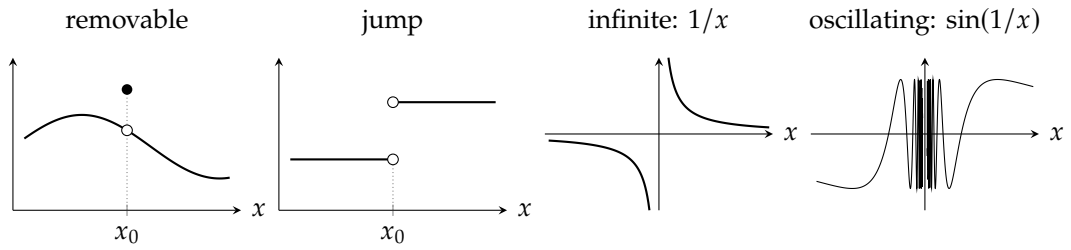


Figure 3: The four types of discontinuity at a point x_0 (at $x_0 = 0$ in the last two panels). The last panel shows $\sin(1/x)$ as an example of a function oscillating infinitely often near 0.

2.3 Picturing the derivative

The graph of f' does not look like the graph of f. For $f(x) = 1/x$, the derivative $f'(x) = -1/x^2$ is negative everywhere. Here f is odd while f' is even; in general the derivative of an odd function is even, and vice versa. Similarly, for a pumpkin thrown up and over a building, with height $y(t) = 400 - 16t^2$ for $-5 \le t \le 5$, the derivative $y'(t) = -32t$ is positive while the pumpkin rises, zero at the top, and negative while it falls.

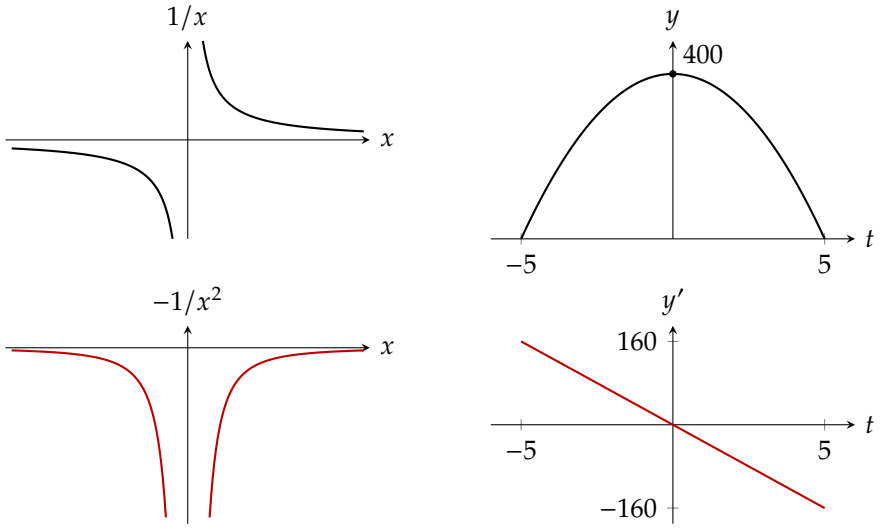


Figure 4: Top: the functions $1/x$ and $y(t) = 400 - 16t^2$. Bottom: their derivatives $-1/x^2$ and $y'(t) = -32t$, aligned below them.

2.4 Two trigonometric limits

With θ measured in radians,

$$\boxed{\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1}, \qquad \boxed{\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}=0}.$$

On the unit circle, an angle θ cuts off an arc of length θ. The vertical segment from the end of the arc to the horizontal axis has length $\sin\theta$, and the gap between its foot and the circle has length $1-\cos\theta$. As $\theta\to 0$, the vertical segment and the arc become indistinguishable, so $\sin\theta/\theta\to 1$, while the short gap $1-\cos\theta$ becomes much smaller than the arc, so $(1-\cos\theta)/\theta\to 0$.

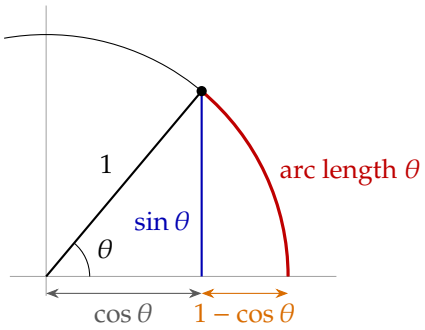


Figure 5: An angle θ in the unit circle. The arc has length θ, the vertical segment has length $\sin\theta$, and the gap at the bottom has length $1-\cos\theta$.

Completion of the argument. Let $0<\theta<\pi/2$. The triangle with vertices $(0,0)$, $(1,0)$, $(\cos\theta,\sin\theta)$ lies inside the circular sector of angle θ, which lies inside the right triangle with legs 1 and $\tan\theta$. Comparing areas,

$$\tfrac{1}{2}\sin\theta<\tfrac{1}{2}\theta<\tfrac{1}{2}\tan\theta, \qquad \text{so} \qquad \cos\theta<\frac{\sin\theta}{\theta}<1.$$

As $\theta\to 0^+$, $\cos\theta\to 1$, so $\sin\theta/\theta\to 1$. Since $\sin\theta/\theta$ is even, the same holds as $\theta\to 0^-$. For the second limit,

$$\frac{1-\cos\theta}{\theta}=\frac{1-\cos^2\theta}{\theta(1+\cos\theta)}=\frac{\sin\theta}{\theta}\cdot\frac{\sin\theta}{1+\cos\theta}\longrightarrow 1\cdot\frac{0}{2}=0. \qquad \square$$

2.5 Differentiable implies continuous

Theorem 2.3 [Differentiable implies continuous]. *If f is differentiable at x_0, then f is continuous at x_0.*

Proof. For $x \neq x_0$, multiply and divide by $x - x_0$:

$$\lim_{x \to x_0} \big(f(x) - f(x_0)\big) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}\,(x - x_0) = f'(x_0) \cdot 0 = 0.$$

Multiplying by $(x - x_0)/(x - x_0)$ is legitimate because the limit $x \to x_0$ only involves $x \neq x_0$. □

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 2, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 3: Derivatives of products, quotients, sine, cosine

There are two kinds of derivative formulas: *specific* ones, such as $\frac{d}{dx}x^n$ or $\frac{d}{dx}\frac{1}{x}$, and *general* ones, which build derivatives of combinations from those of the pieces. Throughout, $(u+v)(x) = u(x)+v(x)$ and $(uv)(x) = u(x)v(x)$.

3.1 Sums and constant multiples

Proposition 3.1 [Linearity]. *If u and v are differentiable and c is a constant, then*

$$(u+v)' = u' + v', \qquad (cu)' = cu'.$$

Proof. By the definition of the derivative,

$$(u+v)'(x) = \lim_{\Delta x\to 0}\left\{\frac{u(x+\Delta x)-u(x)}{\Delta x} + \frac{v(x+\Delta x)-v(x)}{\Delta x}\right\} = u'(x)+v'(x).$$

The same procedure, factoring out c, gives $(cu)' = cu'$. □

3.2 Derivatives of sine and cosine

The trigonometric limits of Lecture 2 are exactly the derivatives of sin and cos at 0:

$$\frac{d}{dx}\sin x\Big|_{x=0} = \lim_{\Delta x\to 0}\frac{\sin\Delta x}{\Delta x} = 1, \qquad \frac{d}{dx}\cos x\Big|_{x=0} = \lim_{\Delta x\to 0}\frac{\cos\Delta x - 1}{\Delta x} = 0.$$

The addition formulas transfer these values to every x.

Proposition 3.2 [Derivatives of sine and cosine].

$$\boxed{\frac{d}{dx}\sin x = \cos x}, \qquad \boxed{\frac{d}{dx}\cos x = -\sin x}.$$

Proof. Using $\sin(a+b) = \sin a\cos b + \cos a\sin b$,

$$\frac{\sin(x+\Delta x)-\sin x}{\Delta x} = \sin x\,\frac{\cos\Delta x - 1}{\Delta x} + \cos x\,\frac{\sin\Delta x}{\Delta x} \longrightarrow \sin x\cdot 0 + \cos x\cdot 1 = \cos x.$$

Similarly, $\cos(a+b) = \cos a \cos b - \sin a \sin b$ gives

$$\frac{\cos(x+\Delta x)-\cos x}{\Delta x} = \cos x \,\frac{\cos \Delta x - 1}{\Delta x} - \sin x \,\frac{\sin \Delta x}{\Delta x} \longrightarrow -\sin x. \qquad \square$$

3.3 The product rule

Theorem 3.3 [Product rule]. *If u and v are differentiable, then*

$$\boxed{(uv)' = u'v + uv'}.$$

Proof. Adding $u(x+\Delta x)v(x) - u(x+\Delta x)v(x) = 0$ to the numerator of the difference quotient and rearranging,

$$\frac{(uv)(x+\Delta x)-(uv)(x)}{\Delta x} = \frac{u(x+\Delta x)-u(x)}{\Delta x}\, v(x) + u(x+\Delta x)\,\frac{v(x+\Delta x)-v(x)}{\Delta x}.$$

The limit of a sum is the sum of the limits. As $\Delta x \to 0$, the first term tends to $u'(x)v(x)$. In the second, $u(x+\Delta x) \to u(x)$ because u is differentiable and hence continuous (Lecture 2), so it tends to $u(x)v'(x)$. $\square$

The rule has a geometric picture. Write $u = u(x)$, $v = v(x)$, $\Delta u = u(x+\Delta x) - u(x)$, and $\Delta v = v(x+\Delta x) - v(x)$. The change in the area of a $u \times v$ rectangle is

$$(u+\Delta u)(v+\Delta v) - uv = u\,\Delta v + v\,\Delta u + \Delta u\,\Delta v.$$

The first two terms are the side strips, and the corner term $\Delta u\,\Delta v$ is negligibly small. Dividing by Δx and letting $\Delta x \to 0$ gives $(uv)' = uv' + vu'$; the corner contributes $\Delta u \cdot (\Delta v/\Delta x) \to 0 \cdot v' = 0$.

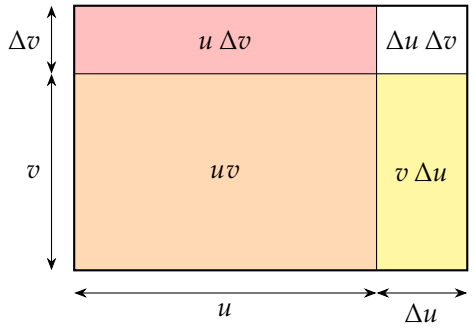


Figure 6: The product rule as a change in area. Increasing v with u fixed adds $u\,\Delta v$; increasing u with v fixed adds $v\,\Delta u$; the corner $\Delta u\,\Delta v$ is of smaller order.

3.4 The quotient rule

Theorem 3.4 [Quotient rule]. *If u and v are differentiable and $v \neq 0$, then*

$$\boxed{\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}}.$$

Proof. With the notation above, put the difference over a common denominator and cancel $uv - uv$:

$$\frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} = \frac{(u + \Delta u)v - u(v + \Delta v)}{(v + \Delta v)v} = \frac{(\Delta u)v - u(\Delta v)}{(v + \Delta v)v}.$$

Dividing by Δx,

$$\frac{1}{\Delta x}\left(\frac{u + \Delta u}{v + \Delta v} - \frac{u}{v}\right) = \frac{\frac{\Delta u}{\Delta x}v - u\frac{\Delta v}{\Delta x}}{(v + \Delta v)v} \longrightarrow \frac{u'v - uv'}{v^2} \quad \text{as } \Delta x \to 0,$$

since $\Delta v \to 0$ by continuity of v. □

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 3, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 4: Chain rule; higher derivatives

4.1 The chain rule

Sums, differences, and products have differentiation rules; the remaining basic operation is composition. The composition of f and g is $(f \circ g)(x) = f(g(x))$: first apply g, then f.

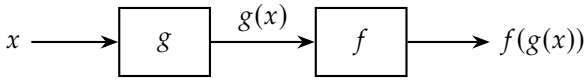


Figure 7: The composition $f \circ g$ feeds the output of g into f.

Composition is not commutative. With $f(x) = \sin x$ and $g(x) = x^2$,

$$(f \circ g)(x) = \sin(x^2), \qquad (g \circ f)(x) = \sin^2 x.$$

To differentiate $y = f(g(t))$, set $x = g(t)$ and follow a change through both variables. Starting from t_0, $x_0 = g(t_0)$, $y_0 = f(x_0)$, let

$$t = t_0 + \Delta t, \qquad x = g(t) = x_0 + \Delta x, \qquad y = f(x) = y_0 + \Delta y.$$

Then

$$\frac{\Delta y}{\Delta t} = \frac{\Delta y}{\Delta x} \cdot \frac{\Delta x}{\Delta t}.$$

As $\Delta t \to 0$, also $\Delta x \to 0$ because g is continuous, so each factor tends to a derivative.

Theorem 4.1 [Chain rule]. *If g is differentiable at t and f is differentiable at $g(t)$, then*

$$\boxed{\frac{dy}{dt} = \frac{dy}{dx}\,\frac{dx}{dt}}, \qquad \textit{that is,} \qquad \frac{d}{dt} f(g(t)) = f'(g(t))\, g'(t).$$

The factorization divides by Δx, so as written the argument needs $\Delta x \neq 0$ for all small $\Delta t \neq 0$, which holds, for example, when $g'(t) \neq 0$. The rule itself holds without this restriction.

Example: *Three applications.*

1. For $y = \sin(t^2)$, take $x = t^2$: $dx/dt = 2t$ and $dy/dx = \cos x$, so

$$\frac{d}{dt} \sin(t^2) = (\cos x)(2t) = 2t \cos(t^2).$$

2. For $y = \cos(1/x)$, take $u = 1/x$: $dy/du = -\sin u$ and $du/dx = -1/x^2$, so
$$\frac{d}{dx}\cos\left(\frac{1}{x}\right) = (-\sin u)\left(-\frac{1}{x^2}\right) = \frac{\sin(1/x)}{x^2}.$$

3. For x^{-n} with n a positive integer, there are two routes. Writing $x^{-n} = (1/x)^n$ and using $\frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}$,
$$\frac{d}{dx}\left(\frac{1}{x}\right)^n = n\left(\frac{1}{x}\right)^{n-1}\left(-\frac{1}{x^2}\right) = -nx^{-(n-1)}x^{-2} = -nx^{-n-1}.$$
Writing $x^{-n} = 1/u$ with $u = x^n$ instead,
$$\frac{d}{dx}\frac{1}{x^n} = -\frac{1}{u^2}\frac{du}{dx} = -\frac{nx^{n-1}}{x^{2n}} = -nx^{-n-1}.$$
Either way, the power rule $\frac{d}{dx}x^m = mx^{m-1}$ extends to negative integers m.

4.2 Higher derivatives

Definition 4.2 [Higher derivatives]**.** Higher derivatives are derivatives of derivatives: if $g = f'$, then $g' = (f')' = f''$ is the second derivative of f, and so on. The notations are

$f'(x)$	Df	$\dfrac{df}{dx}$
$f''(x)$	D^2f	$\dfrac{d^2f}{dx^2}$
$f'''(x)$	D^3f	$\dfrac{d^3f}{dx^3}$
$f^{(n)}(x)$	D^nf	$\dfrac{d^nf}{dx^n}$

Proposition 4.3. *For every positive integer n, $D^n x^n = n!$, where $n! = n(n-1)\cdots 2\cdot 1$.*

The first cases suggest the pattern: $Dx = 1$, $D^2x^2 = D(2x) = 2$, $D^3x^3 = D^2(3x^2) = D(6x) = 6$, and $D^4x^4 = D^3(4x^3) = D^2(12x^2) = D(24x) = 24$.

Proof. By induction on n. The case $n = 1$ is $Dx = 1$. If $D^n x^n = n!$, then, since D^n commutes with multiplication by constants,
$$D^{n+1}x^{n+1} = D^n\left(Dx^{n+1}\right) = D^n\left((n+1)x^n\right) = (n+1)D^n x^n = (n+1)\,n! = (n+1)!. \qquad \square$$

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 4, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 5: Implicit differentiation, inverses

5.1 Implicit differentiation

In implicit differentiation, we differentiate both sides of an equation relating x and y, treating y as a function of x and using the chain rule, then solve for dy/dx.

Proposition 5.1 [Power rule for rational exponents]. *For a rational number a and $x > 0$,*

$$\boxed{\frac{d}{dx}x^a = ax^{a-1}}.$$

The formula was proved for $a = 0, 1, 2, \ldots$ by direct computation and extended to $a = -1, -2, \ldots$ in Lecture 4.

Proof. Write $a = m/n$ with integers m and $n > 0$, and let $y = x^{m/n}$. Then $y^n = x^m$. Differentiating both sides with respect to x, using the chain rule on the left,

$$ny^{n-1}\frac{dy}{dx} = mx^{m-1}, \qquad \text{so} \qquad \frac{dy}{dx} = \frac{m}{n}\frac{x^{m-1}}{y^{n-1}}.$$

Substituting $y = x^{m/n}$,

$$\frac{dy}{dx} = \frac{m}{n}\frac{x^{m-1}}{x^{m(n-1)/n}} = \frac{m}{n}x^{(n(m-1)-m(n-1))/n} = \frac{m}{n}x^{(m-n)/n} = \frac{m}{n}x^{m/n-1}. \qquad \square$$

The argument assumes that $y = x^{m/n}$ is differentiable, and computes its derivative.

Example: *The unit circle.* The circle $x^2 + y^2 = 1$ has upper half $y = \sqrt{1-x^2} = (1-x^2)^{1/2}$. Directly,

$$\frac{dy}{dx} = \tfrac{1}{2}(1-x^2)^{-1/2}(-2x) = \frac{-x}{\sqrt{1-x^2}} = \frac{-x}{y}.$$

Implicitly, differentiating $x^2 + y^2 = 1$ gives

$$2x + 2y\frac{dy}{dx} = 0, \qquad \text{so} \qquad \frac{dy}{dx} = -\frac{x}{y},$$

the same answer, without solving for y.

Example: *An equation that is hard to solve.* For $y^3 + xy^2 + 1 = 0$, it is not easy to write y as a function

of x. Implicit differentiation still gives

$$3y^2\frac{dy}{dx} + y^2 + 2xy\frac{dy}{dx} = 0, \qquad \text{so} \qquad \frac{dy}{dx} = \frac{-y^2}{3y^2 + 2xy},$$

in terms of x and y.

5.2 Derivatives of inverse functions

Definition 5.2 [Inverse function]. If $y = f(x)$ and $g(y) = x$, then g is the inverse function of f, written $g = f^{-1}$, so that $x = g(y) = f^{-1}(y)$.

Proposition 5.3 [Derivative of an inverse]. *If $y = f(x)$, then*

$$\boxed{\frac{d}{dy}f^{-1}(y) = \frac{1}{dy/dx}}.$$

Proof. Differentiate $f^{-1}(y) = x$ implicitly with respect to x. By the chain rule,

$$\frac{d}{dy}\left(f^{-1}(y)\right)\frac{dy}{dx} = \frac{d}{dx}(x) = 1,$$

and dividing by dy/dx gives the formula. □

Example: *The arctangent.* Let $y = \arctan x$, so $\tan y = x$. Differentiating implicitly, with $\frac{d}{dy}\tan y = 1/\cos^2 y$,

$$\frac{1}{\cos^2 y}\frac{dy}{dx} = 1, \qquad \text{so} \qquad \frac{dy}{dx} = \cos^2 y = \cos^2(\arctan x).$$

To simplify, draw a right triangle with angle y, adjacent side 1, and opposite side x, so that $\tan y = x$. By the Pythagorean theorem its hypotenuse is $\sqrt{1+x^2}$, so $\cos y = 1/\sqrt{1+x^2}$ and

$$\boxed{\frac{d}{dx}\arctan x = \frac{1}{1+x^2}}.$$

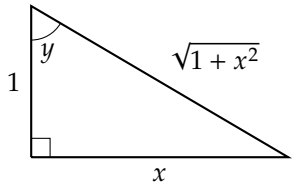


Figure 8: The triangle for $y = \arctan x$: $\tan y = x$ and $\cos y = 1/\sqrt{1+x^2}$.

5.3 Graphing an inverse function

To graph $g = f^{-1}$ on the same axes as f, write it as a function of x. If $g(x) = y$, then $x = f(y)$: we have traded the roles of x and y. Thus a point (a, b) lies on the graph of f exactly when (b, a) lies on the graph of f^{-1}, and the graph of f^{-1} is the reflection of the graph of f across the line $y = x$. The defining identities are

$$f^{-1}(f(x)) = x, \qquad f(f^{-1}(x)) = x,$$

that is, $f^{-1} \circ f$ and $f \circ f^{-1}$ are both the identity.

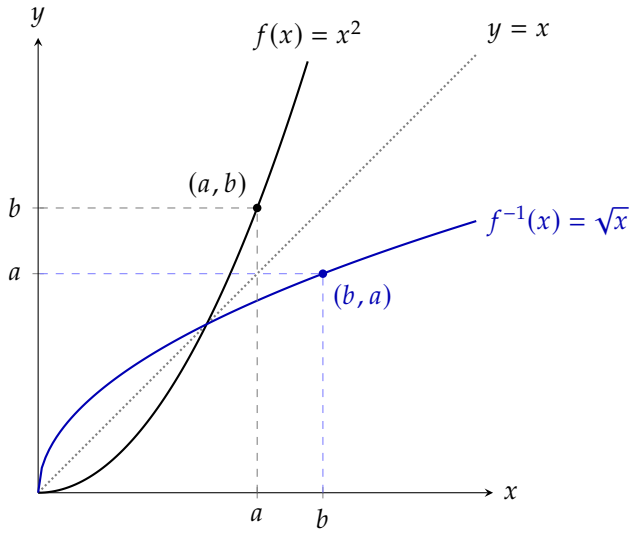


Figure 9: Reflection across $y = x$, illustrated with $f(x) = x^2$ for $x \geq 0$ and $f^{-1}(x) = \sqrt{x}$. The point (a, b) with $b = f(a)$ corresponds to (b, a) on the graph of f^{-1}.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 5, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 6: Exponential and log; logarithmic differentiation

6.1 Exponentials and the number e

Fix a base $a > 1$. Starting from $a^0 = 1$, $a^1 = a$, $a^2 = a \cdot a, \ldots$, exponentials obey

$$a^{x_1+x_2} = a^{x_1}a^{x_2}, \qquad (a^{x_1})^{x_2} = a^{x_1 x_2}, \qquad a^{p/q} = \sqrt[q]{a^p} \quad (p, q \text{ integers}),$$

and a^r for real r is filled in by continuity. To differentiate a^x, factor a^x out of the difference quotient:

$$\frac{d}{dx}a^x = \lim_{\Delta x \to 0} \frac{a^{x+\Delta x} - a^x}{\Delta x} = a^x \lim_{\Delta x \to 0} \frac{a^{\Delta x} - 1}{\Delta x}.$$

Definition 6.1 [The constant $M(a)$]**.** Let

$$M(a) = \lim_{\Delta x \to 0} \frac{a^{\Delta x} - 1}{\Delta x}, \qquad \text{so that} \qquad \frac{d}{dx}a^x = M(a)\, a^x.$$

Analytically, $M(a) = \frac{d}{dx}a^x\big|_{x=0}$, since the defining limit is the difference quotient of a^x at 0. Geometrically, $M(a)$ is the slope of the graph $y = a^x$ at $x = 0$.

The trick is to choose the base for which this slope is 1. As a increases, the graph of a^x gets steeper. For $a = 2$, the secant line from $(0, 1)$ to $(1, 2)$ has slope 1, and the convex graph lies below this secant between them, so the tangent at $x = 0$ is less steep: $M(2) < 1$. For $a = 4$, the secant line from $(-\frac{1}{2}, \frac{1}{2})$ to $(0, 1)$ has slope 1, and now the tangent at $x = 0$ is steeper: $M(4) > 1$. Somewhere between 2 and 4 there is a base whose slope at $x = 0$ is 1.

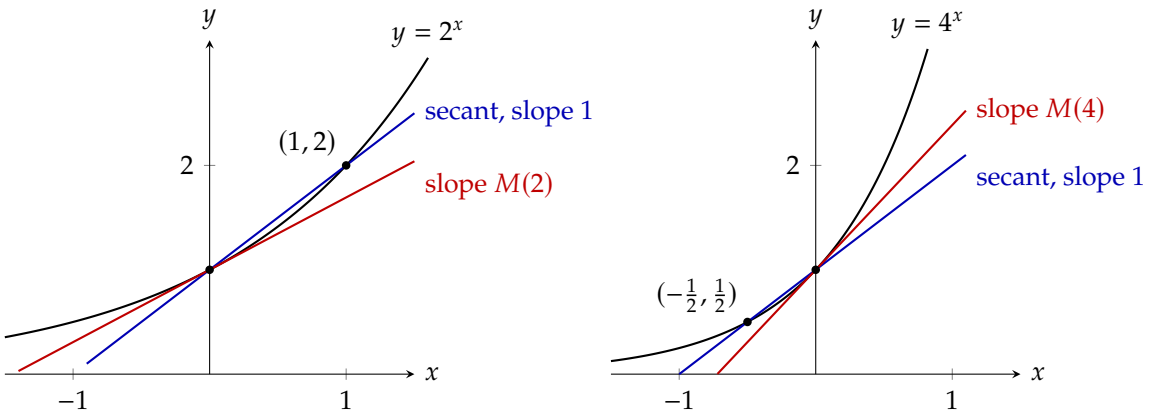


Figure 10: Left: the secant of 2^x from $(0, 1)$ to $(1, 2)$ has slope 1, more than the tangent slope $M(2)$ at $x = 0$. Right: the secant of 4^x from $(-\frac{1}{2}, \frac{1}{2})$ to $(0, 1)$ has slope 1, less than $M(4)$.

Definition 6.2 [The number e]**.** The number e is the unique base with $M(e) = 1$; equivalently,

$\lim_{h\to 0}(e^h - 1)/h = 1$, or $\frac{d}{dx}e^x = 1$ at $x = 0$.

Consequently $\dfrac{d}{dx}e^x = M(e)e^x = \boxed{e^x}$.

6.2 The natural logarithm

Definition 6.3 [Natural logarithm]. The natural logarithm ln is the inverse function of e^x: if $y = e^x$, then $\ln y = x$; equivalently, if $w = \ln x$, then $e^w = x$.

Since $e^x > 0$ for every x, even negative x, $\ln x$ is defined for $x > 0$. Also $\ln 1 = 0$, $\ln x < 0$ for $0 < x < 1$, $\ln x > 0$ for $x > 1$, and $\ln(x_1 x_2) = \ln x_1 + \ln x_2$.

Proposition 6.4 [Derivative of the logarithm].

$$\boxed{\frac{d}{dx}\ln x = \frac{1}{x}}.$$

Proof. Let $w = \ln x$, so $e^w = x$. Differentiating implicitly with respect to x,

$$\frac{d}{dw}\left(e^w\right)\frac{dw}{dx} = e^w\frac{dw}{dx} = 1, \qquad \text{so} \qquad \frac{dw}{dx} = \frac{1}{e^w} = \frac{1}{x}. \qquad \square$$

6.3 Derivatives of general exponentials

Proposition 6.5 [Derivative of a^x].

$$\boxed{\frac{d}{dx}a^x = (\ln a)\,a^x}, \qquad \textit{so} \qquad M(a) = \ln a.$$

Method 1: rewrite in base e. Write $a = e^{\ln a}$, so $a^x = (e^{\ln a})^x = e^{(\ln a)x}$. Here $\ln a$ is a constant, so by the chain rule, just as $\frac{d}{dx}e^{3x} = 3e^{3x}$,

$$\frac{d}{dx}e^{(\ln a)x} = (\ln a)\,e^{(\ln a)x} = (\ln a)\,a^x.$$

Comparing with $\frac{d}{dx}a^x = M(a)a^x$ gives $M(a) = \ln a$. $\square$

Even starting from another base, the natural logarithm appears; for example, $\frac{d}{dx}10^x = (\ln 10)10^x$.

The second method is logarithmic differentiation: find $\frac{d}{dx}f$ by first finding $\frac{d}{dx}\ln f$. With $u = f(x)$,

the chain rule gives $\frac{d}{dx}\ln u = \frac{1}{u}\frac{du}{dx}$, that is,

$$\boxed{(\ln f)' = \frac{f'}{f}}, \qquad \text{or} \qquad f' = f\,(\ln f)'.$$

Method 2: logarithmic differentiation. For $f(x) = a^x$, $\ln f(x) = x \ln a$, so $(\ln f)' = \ln a$, a constant. Hence $f' = f\,(\ln f)' = (\ln a)\,a^x$. □

With a variable ("moving") exponent, use base e or logarithmic differentiation.

Example: *The derivative of* x^x. For $f = x^x$, $\ln f = x \ln x$, so

$$(\ln f)' = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1, \qquad \frac{d}{dx}x^x = f\,(\ln f)' = x^x(\ln x + 1).$$

Writing $f = e^{x \ln x}$ and using the chain rule gives the same answer with a little more writing.

6.4 A limit formula for e

Proposition 6.6.

$$\boxed{\lim_{k\to\infty}\left(1 + \frac{1}{k}\right)^k = e}.$$

Proof. Because the exponent k changes, take logarithms first:

$$\ln\left[\left(1 + \frac{1}{k}\right)^k\right] = k \ln\left(1 + \frac{1}{k}\right).$$

The two parts compete: $k \to \infty$ while $\ln(1 + 1/k) \to 0$. With $h = 1/k$ and $\ln 1 = 0$, this is a difference quotient:

$$k \ln\left(1 + \frac{1}{k}\right) = \frac{\ln(1+h) - \ln 1}{h} \longrightarrow \frac{d}{dx}\ln x\Big|_{x=1} = 1 \quad \text{as } h = \tfrac{1}{k} \to 0.$$

So $a_k = \ln\left[(1+1/k)^k\right] \to 1$, and by continuity of the exponential, $(1+1/k)^k = e^{a_k} \to e^1 = e$. □

Logarithmic derivatives measure relative change. A drop of 50 points in a market average means different things if the average was 300 or 10,000; what matters is the relative rate of change $f'(t)/f(t) = \frac{d}{dt}\ln f(t)$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 6, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 7: Hyperbolic functions; Exam 1 review

The exponential function produces two more functions, the hyperbolic sine and cosine, which behave like sin and cos with a hyperbola in place of the circle. Exponentials and logarithms also complete the power rule for arbitrary real exponents.

7.1 Hyperbolic sine and cosine

Definition 7.1 [Hyperbolic sine and cosine]. The hyperbolic sine and hyperbolic cosine are

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}.$$

Since $(e^{-x})' = -e^{-x}$, differentiating either one gives the other:

$$\frac{d}{dx}\sinh x = \frac{e^x - (-e^{-x})}{2} = \cosh x, \qquad \frac{d}{dx}\cosh x = \frac{e^x + (-e^{-x})}{2} = \sinh x.$$

Unlike $\frac{d}{dx}\cos x = -\sin x$, there is no minus sign.

Proposition 7.2 [Hyperbolic identity]. *For every real x, $\cosh^2 x - \sinh^2 x = 1$.*

Proof. Expand both squares, using $e^x e^{-x} = 1$:

$$\cosh^2 x - \sinh^2 x = \frac{1}{4}\big(e^{2x} + 2 + e^{-2x}\big) - \frac{1}{4}\big(e^{2x} - 2 + e^{-2x}\big) = \frac{1}{4}(2+2) = 1.$$

□

The name comes from this identity. With $u = \cosh x$ and $v = \sinh x$, the point (u, v) satisfies $u^2 - v^2 = 1$, the equation of a hyperbola; since $\cosh x > 0$, it lies on the right branch. The ordinary trigonometric functions are *circular*: $u = \cos x$, $v = \sin x$ satisfy $u^2 + v^2 = 1$, the unit circle.

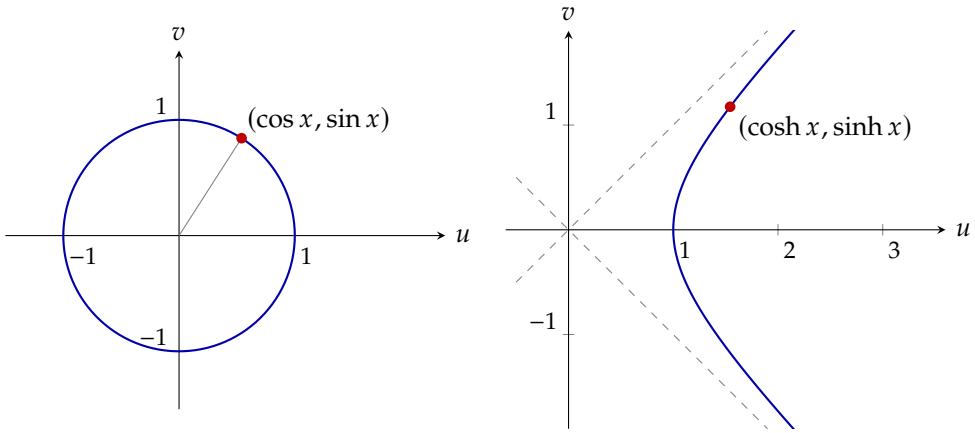


Figure 11: Left: $(\cos x, \sin x)$ lies on the circle $u^2 + v^2 = 1$. Right: $(\cosh x, \sinh x)$ lies on the right branch of the hyperbola $u^2 - v^2 = 1$. Both points are drawn for $x = 1$.

7.2 The power rule for real exponents

Implicit differentiation proves $\frac{d}{dx}x^r = rx^{r-1}$ only for rational r. For an arbitrary real r and $x > 0$, there are two proofs.

Proposition 7.3 [Power rule]. *For every real r and $x > 0$, $\dfrac{d}{dx}x^r = rx^{r-1}$.*

Proof. Base e. Write $x = e^{\ln x}$, so $x^r = e^{r \ln x}$. By the chain rule,

$$\frac{d}{dx}x^r = e^{r \ln x}\,\frac{d}{dx}(r \ln x) = x^r \cdot \frac{r}{x} = rx^{r-1}.$$

Logarithmic differentiation. For $f = x^r$, $\ln f = r \ln x$, so $(\ln f)' = r/x$. Since $(\ln f)' = f'/f$,

$$f' = f\,(\ln f)' = x^r \cdot \frac{r}{x} = rx^{r-1}.$$

□

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 7, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 9: Linear and quadratic approximations

The first application of differentiation is approximation: near a point, a function is close to its tangent line, and closer still to a well-chosen parabola.

9.1 Linear approximation

Definition 9.1 [Linear approximation]. The linear approximation to f at the base point x_0 is its tangent line:

$$\boxed{f(x) \approx f(x_0) + f'(x_0)(x - x_0)} \qquad (x \approx x_0).$$

Since $f'(x_0)$ is the limit of the difference quotients $\big(f(x) - f(x_0)\big)/(x - x_0)$, the error of this approximation is small compared with $|x - x_0|$ as $x \to x_0$. It is good near x_0 and grows less accurate farther away.

Example: *Logarithm at* $x_0 = 1$. For $f(x) = \ln x$, $f(1) = 0$ and $f'(1) = 1/x\big|_{x=1} = 1$, so $\ln x \approx x - 1$ near $x = 1$. Changing variables to $x = 1 + u$ moves the base point to $u_0 = 0$: $\ln(1 + u) \approx u$.

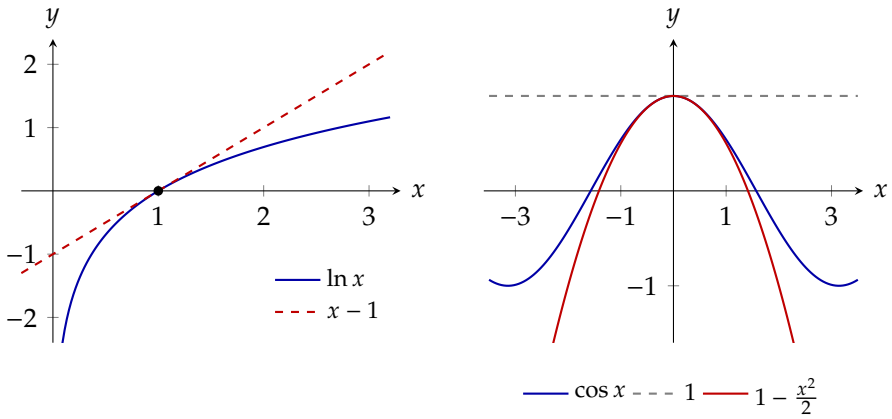


Figure 12: Left: $\ln x$ and its tangent line $x - 1$ at the base point $x_0 = 1$. Right: $\cos x$ with its linear approximation 1 and its quadratic approximation $1 - x^2/2$ at $x_0 = 0$.

Most approximations are taken at the base point $x_0 = 0$, for $|x| \ll 1$. The basic list is:

1. $\sin x \approx x$;
2. $\cos x \approx 1$;
3. $e^x \approx 1 + x$;
4. $\ln(1 + x) \approx x$;
5. $(1 + x)^r \approx 1 + rx$.

Each follows from $f(x) \approx f(0) + f'(0)x$. For example, $\sin 0 = 0$ and $\sin' 0 = \cos 0 = 1$ give (1), and item (4) is the example above. For (5), $f(0) = 1$ and $f'(0) = r(1+x)^{r-1}\big|_{x=0} = r$.

9.2 Combining basic approximations

Algebraic combinations of the basic list are often quicker than differentiating.

Example: *Linear approximation of $e^{-2x}/\sqrt{1+x}$ near 0.* With $u = -2x$ in (3) and $r = -\frac{1}{2}$ in (5),

$$\frac{e^{-2x}}{\sqrt{1+x}} = e^{-2x}(1+x)^{-1/2} \approx (1-2x)\Big(1 - \frac{1}{2}x\Big) = 1 - \frac{5}{2}x + x^2 \approx 1 - \frac{5}{2}x.$$

The x^2 term is dropped because other terms of order x^2 were already discarded in the factors; for $|x| \ll 1$ they are all negligible compared with x. As a by-product, $f(0) = 1$ and $f'(0) = -5/2$, read off without computing $f'(x)$.

Example: *A limit.* For $f(x) = (1+2x)^{10}$, the limit

$$\lim_{x\to 0} \frac{(1+2x)^{10} - 1}{x} = \lim_{x\to 0} \frac{f(x) - f(0)}{x} = f'(0) = 20,$$

since $f'(x) = 20(1+2x)^9$. Linear approximation gives the same answer: $(1+2x)^{10} \approx 1 + 20x$ by (5) with $u = 2x$, so the quotient is approximately $20x/x = 20$.

Example: *Time dilation.* A satellite passes overhead with speed v. If T is the time measured on board and T' the time measured on the ground, special relativity gives $T' = T(1 - v^2/c^2)^{-1/2}$. By (5) with $u = -v^2/c^2$ and $r = -\frac{1}{2}$,

$$T' \approx T\Big(1 + \frac{1}{2}\,\frac{v^2}{c^2}\Big).$$

For $v = 4$ km/s and $c = 3 \times 10^5$ km/s, $v^2/c^2 \approx 10^{-10}$: the two clocks barely differ. Corrections of exactly this kind are nevertheless used to calibrate the transmitters on GPS satellites.

9.3 Quadratic approximation

Definition 9.2 [Quadratic approximation]. The quadratic approximation to f at x_0 is

$$\boxed{f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2} \qquad (x \approx x_0).$$

Geometrically, this is the best-fit parabola at x_0. For $f(x) = \cos x$ at $x_0 = 0$, $f(0) = 1$, $f'(0) = -\sin 0 = 0$, and $f''(0) = -\cos 0 = -1$, so $\cos x \approx 1 - \frac{1}{2}x^2$ (Figure 12).

The factor $\frac{1}{2}$ makes the approximation exact for quadratics. If $f(x) = a + bx + cx^2$, then $f'(x) = b + 2cx$

and $f''(x) = 2c$, so

$$a = f(0), \qquad b = f'(0), \qquad c = \frac{f''(0)}{2}.$$

At $x_0 = 0$, the basic quadratic approximations are:

1. $\sin x \approx x$;
2. $\cos x \approx 1 - \dfrac{x^2}{2}$;
3. $e^x \approx 1 + x + \dfrac{x^2}{2}$;
4. $\ln(1+x) \approx x - \dfrac{x^2}{2}$;
5. $(1+x)^r \approx 1 + rx + \dfrac{r(r-1)}{2}x^2$.

Each comes from computing $f(0)$, $f'(0)$, and $f''(0)$. For (4), $f'(x) = 1/(1+x)$ and $f''(x) = -1/(1+x)^2$, so $f(0) = 0$, $f'(0) = 1$, and $f''(0) = -1$.

Example: *Time dilation, second order.* Item (5) with $x = -v^2/c^2$ and $r = -\frac{1}{2}$, where $r(r-1)/2 = \frac{3}{8}$, gives

$$\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\left(\frac{v^2}{c^2}\right)^2.$$

For the satellite, the quadratic term is of order 10^{-20}, far below what any clock can measure, so the linear approximation suffices.

Example: *Quadratic approximation of $e^{-2x}/\sqrt{1+x}$.* Multiply the quadratic approximations of the two factors and drop terms of order x^3 and higher:

$$e^{-2x}(1+x)^{-1/2} \approx \left(1 - 2x + 2x^2\right)\left(1 - \frac{1}{2}x + \frac{3}{8}x^2\right) \approx 1 - \frac{5}{2}x + \left(2 + 1 + \frac{3}{8}\right)x^2 = 1 - \frac{5}{2}x + \frac{27}{8}x^2.$$

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 9, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 10: Curve sketching

The goal is a qualitatively correct graph of f, not necessarily to scale, from the behavior of f' and f''.

10.1 Critical points and the sketching rubric

Definition 10.1 [Critical point]**.** A critical point of f is a point x_0 with $f'(x_0) = 0$; the value $y_0 = f(x_0)$ is the corresponding critical value.

In the typical picture, the sign of f' changes at x_0. If $f' < 0$ for $x < x_0$ and $f' > 0$ for $x > x_0$, then f decreases and then increases, and y_0 is a minimum value; with the signs reversed, y_0 is a maximum value.

To sketch $y = f(x)$:

1. Plot the discontinuities of f, especially the infinite ones.
2. Find the critical points, where $f'(x) = 0$.
3. Plot the critical points and critical values (when easy), and decide the sign of f' between them.
4. Find and plot the zeros of f (when easy).
5. Determine the behavior at the endpoints, or as $x \to \pm\infty$.

The sign of f' in step 3 also serves as a consistency check on the finished sketch.

10.2 Examples

Example: $y = 3x - x^3$. There are no discontinuities. Since $y' = 3 - 3x^2 = 3(1 - x^2)$, the critical points are $x = \pm 1$, with critical values $y(1) = 2$ and $y(-1) = -2$. The zeros of $y = x(3 - x^2)$ are 0 and $\pm\sqrt{3}$. As $x \to \infty$ the $-x^3$ term dominates, so $y \to -\infty$; likewise $y \to \infty$ as $x \to -\infty$. Consistently, $y' > 0$ for $|x| < 1$ and $y' < 0$ for $|x| > 1$: the function increases between -1 and 1 and decreases elsewhere (Figure 13).

Example: $y = 1/x$*: check discontinuities first.* Here $y' = -1/x^2 < 0$ wherever it is defined, which might suggest a single decreasing curve. In fact y decreases on each of $(-\infty, 0)$ and $(0, \infty)$ but jumps from $-\infty$ to $+\infty$ at $x = 0$. This is why step 1 comes first.

Example: $y = x^3 - 3x^2 + 3x$*: a critical point without a turn.* Here $y' = 3x^2 - 6x + 3 = 3(x - 1)^2$, so $x = 1$ is the only critical point, with value $y(1) = 1$. Since $y' > 0$ on both sides of $x = 1$, the function increases everywhere; the graph only levels off, with a horizontal tangent at $(1, 1)$.

Example: $y = (\ln x)/x$ *for* $x > 0$. Near 0, take $x = 2^{-n}$: then $y = (-n \ln 2)\, 2^n \to -\infty$ as $n \to \infty$, so $y \to -\infty$ as $x \to 0^+$. By the quotient rule,

$$y' = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2},$$

which vanishes only when $\ln x = 1$, that is, at $x = e$, with critical value $y(e) = 1/e$. The only zero is $x = 1$. As $x \to \infty$, take $x = 2^n$: then $y = n \ln 2/2^n \to 0$. Consistently, $y' > 0$ for $0 < x < e$ and $y' < 0$ for $x > e$, so the graph rises to $(e, 1/e)$ and then decays toward 0.

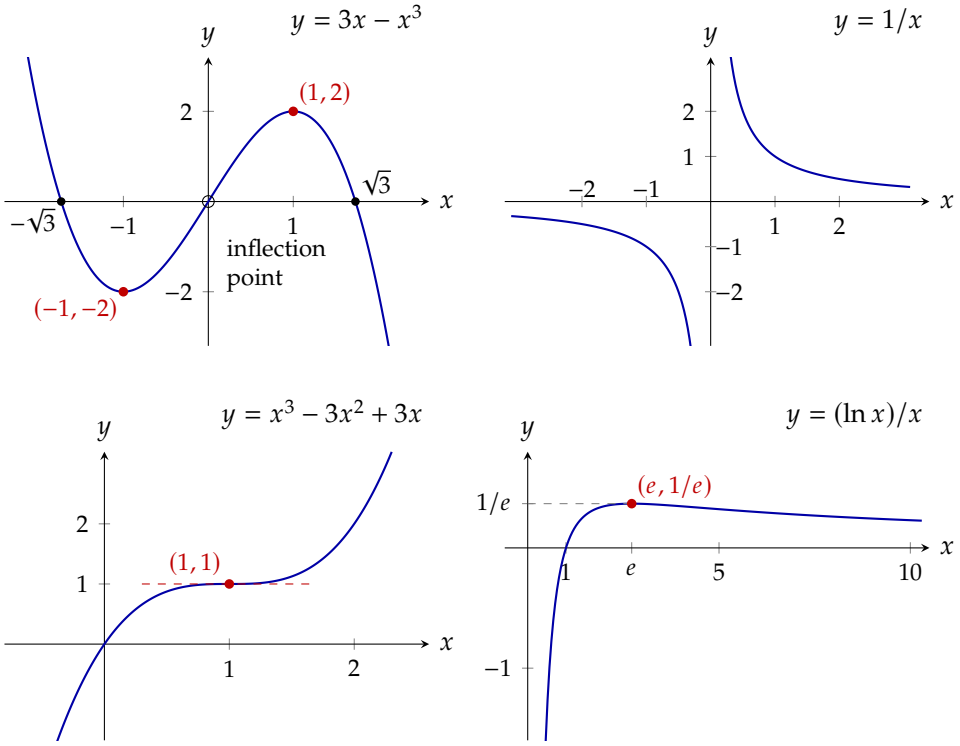


Figure 13: The four examples: $y = 3x - x^3$ with its critical points, zeros, and inflection point; $y = 1/x$; $y = x^3 - 3x^2 + 3x$ with its horizontal tangent at $(1, 1)$; and $y = (\ln x)/x$ with its maximum at $(e, 1/e)$.

10.3 Second derivative information

If $f'' > 0$ on an interval, then f' is increasing there and the graph is CONVEX (concave up): the slope increases, for instance from negative through zero to positive. If $f'' < 0$, then f' is decreasing and the graph is CONCAVE DOWN. The second derivative is used in two ways.

1. *Second derivative test.* Let $f'(x_0) = 0$. If $f''(x_0) < 0$, then x_0 is a local maximum; if $f''(x_0) > 0$, then x_0 is a local minimum.
2. *Concavity.* The sign of f'' determines how the graph bends between the plotted points.

Definition 10.2 [INFLECTION POINT]. An INFLECTION POINT is a point where $f'' = 0$. Usually the graph

changes there from concave up to concave down, or vice versa.

For $y = 3x - x^3$, $y'' = -6x$, so the inflection point is $x = 0$: the graph is concave up for $x < 0$, concave down for $x > 0$, and passes through the origin between the minimum $(-1, -2)$ and the maximum $(1, 2)$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 10, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 11: Max-min problems

Maximum and minimum values are found by comparing a short list of candidates, which avoids sketching the whole graph.

11.1 Maximum value versus maximum point

For $y = (\ln x)/x$ (Lecture 10), the critical point is $x = e$, and the *maximum value* is the critical value $y = 1/e$, not e. Both numbers matter: the point $(e, 1/e)$ is where the graph turns around.

11.2 Where extrema occur

The maximum and minimum of f on an interval occur among the *critical points* and the *endpoints* (including the behavior as $x \to \pm\infty$ on an unbounded interval), and one must also watch out for discontinuities. So it suffices to compare the values of f at these candidates; a full sketch is unnecessary. The maximum or minimum may well be at an endpoint rather than at a critical point.

A word problem is set up in steps:

1. draw a picture and name the variables;
2. express the constraint and the quantity to optimize as formulas;
3. use the constraint to write the quantity in terms of one variable;
4. compare the values at the critical points and the endpoints.

11.3 The open-topped can

Example: *Least surface area for a fixed volume.* An open-topped can of radius r and height h has volume and surface area

$$V = \pi r^2 h \text{ (constant)}, \qquad S = \pi r^2 + 2\pi r h.$$

Since $h = V/(\pi r^2)$,

$$S = \pi r^2 + \frac{2V}{r}, \qquad 0 < r < \infty, \qquad \frac{dS}{dr} = 2\pi r - \frac{2V}{r^2}.$$

Setting $dS/dr = 0$ gives $\pi r^3 = V$, so $r = (V/\pi)^{1/3}$. At the ends, $S \to \infty$ as $r \to 0$ (from $2V/r$) and as $r \to \infty$ (from πr^2). Hence the minimum is at the critical point, where

$$h = \frac{V}{\pi r^2} = \frac{V}{\pi}\left(\frac{V}{\pi}\right)^{-2/3} = \left(\frac{V}{\pi}\right)^{1/3}, \qquad S = \pi\left(\frac{V}{\pi}\right)^{2/3} + 2V\left(\frac{\pi}{V}\right)^{1/3} = 3\pi^{1/3}V^{2/3}.$$

The answer is often best stated as a proportion: the optimal can has $h/r = 1$.

11.4 The wire cut into two squares

Example: *Endpoints matter.* A wire of length 1 is cut at x, and each piece is bent into a square, with sides $x/4$ and $(1-x)/4$. The total area is

$$A(x) = \left(\frac{x}{4}\right)^2 + \left(\frac{1-x}{4}\right)^2, \qquad A'(x) = \frac{2x}{16} - \frac{2(1-x)}{16} = \frac{2x-1}{8}, \qquad 0 \le x \le 1.$$

The only critical point is $x = \frac{1}{2}$, with $A(\frac{1}{2}) = \frac{1}{32}$. At the endpoints, $A(0) = A(1) = \frac{1}{16}$. So the critical point gives the *minimum* area; the maximum, $\frac{1}{16}$, occurs at the endpoints, using all of the wire for one square. Looking only at critical points would have produced the worst answer rather than the best.

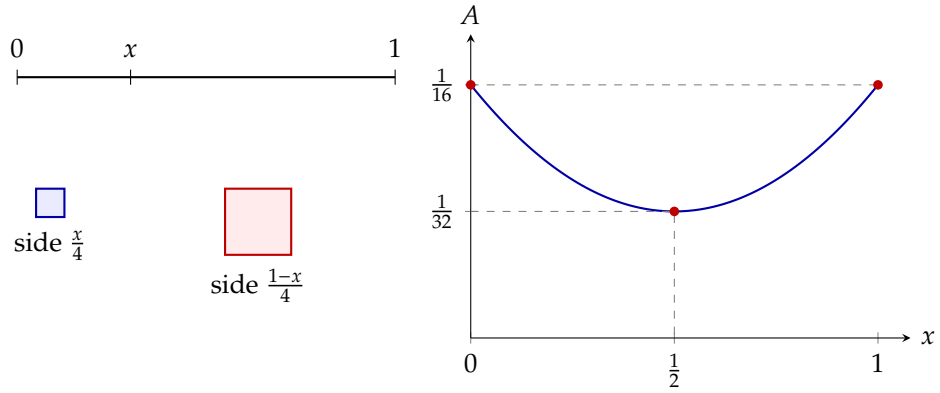


Figure 14: Left: the wire cut at x and the two squares, drawn to scale for $x = 0.3$. Right: the total area $A(x)$. The critical point $x = \frac{1}{2}$ is the minimum; the maximum is at the endpoints $x = 0$ and $x = 1$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 11, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 12: Related rates

When several quantities are linked by an equation and change with time, differentiating the equation with respect to t links their rates of change.

12.1 The radar problem

Example: *Are you speeding?* A police radar is 30 feet from the road. It measures your car approaching at 80 ft/s when the car is 50 feet from the radar. The speed limit is 65 mph, about 95 ft/s.

Let x be the distance along the road from the car to the point nearest the radar, and D the distance from the car to the radar (Figure 15). Both change with time, and by Pythagoras

$$30^2 + x^2 = D^2.$$

At the instant in question, $D = 50$, so $x = 40$ (a 3-4-5 triangle), and $D' = dD/dt = -80$, negative because D is decreasing. Differentiating with respect to t first,

$$2xx' = 2DD' \quad \Longrightarrow \quad x' = \frac{D}{x}\,D' = \frac{50}{40}(-80) = -100 \text{ ft/s}.$$

The car's speed is 100 ft/s, above the limit of 95 ft/s.

One could instead differentiate $D = (30^2 + x^2)^{1/2}$ or $x = (D^2 - 30^2)^{1/2}$ and get the same result with more work; the relation is easiest to differentiate in its simplest algebraic form. The general strategy is:

1. draw a picture, name the variables, and write the equations relating them;
2. differentiate with respect to t;
3. only then substitute the numerical values at the given instant.

Substituting before differentiating would freeze quantities, such as D, that are in fact changing.

12.2 The conical tank

Example: *How fast does the water rise?* An inverted conical tank has top radius 4 ft and height 10 ft, and is filled at 2 ft^3/min. Let r and h be the radius and height of the water surface. The cross-section (Figure 15) gives similar triangles,

$$\frac{r}{h} = \frac{4}{10}, \qquad r = \frac{2}{5}h,$$

so the volume of water is

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{2}{5}h\right)^2 h = \frac{4}{75}\pi h^3, \qquad \frac{dV}{dt} = \frac{4}{25}\pi h^2 h'.$$

With $dV/dt = 2$ and $h = 5$, $2 = \frac{4}{25}\pi(25)h' = 4\pi h'$, so the level rises at $h' = 1/(2\pi)$ ft/min.

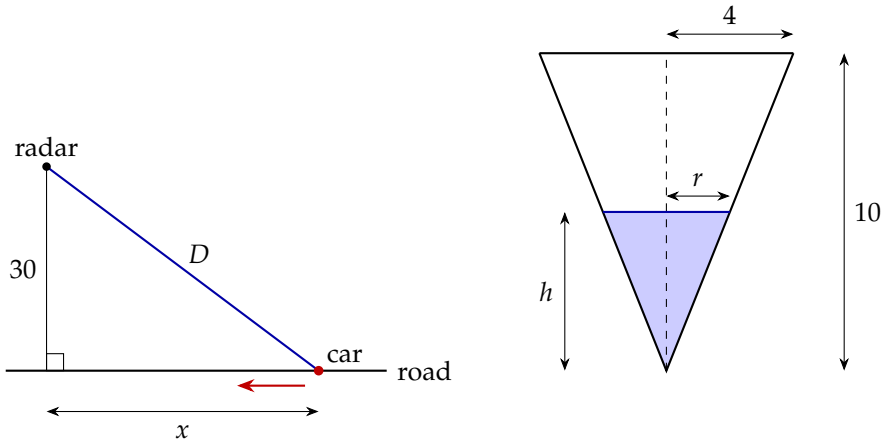


Figure 15: Left: the radar problem, with the radar 30 ft from the road and the car at road distance x and straight-line distance D (drawn at $x = 40$, $D = 50$). Right: cross-section of the conical tank (top radius 4, height 10) with water of height h and surface radius r.

12.3 Sensitivity of measurements

The same differentiation estimates how an error in one measured quantity propagates to another. For a satellite at height c above the ground, let h be its distance to a ground point at horizontal distance L, so that $L^2 + c^2 = h^2$ with c fixed. Then $2LL' = 2hh'$, and

$$\frac{\Delta L}{\Delta h} \approx \frac{L'}{h'} = \frac{h}{L}.$$

An error Δh in the measured distance therefore causes an error of about $(h/L)\Delta h$ in L.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 12, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 13: Newton's method and other applications

Newton's method solves $f(x) = 0$ numerically by following tangent lines. The second application is a constrained minimization, the ring on a string, solved by implicit differentiation.

13.1 Newton's method

Start from a guess x_0, and let x_1 be the point where the tangent line at (x_0, y_0), $y_0 = f(x_0)$, crosses the x-axis. The tangent line is $y - y_0 = m(x - x_0)$ with slope $m = f'(x_0)$; setting $y = 0$ gives $x_1 - x_0 = -y_0/m$. Repeating this step is Newton's method.

Definition 13.1 [Newton's method]. Given x_0, define the iterates

$$\boxed{x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}}, \qquad k = 0, 1, 2, \ldots.$$

Example: *Computing* $\sqrt{3}$. For $f(x) = x^2 - 3$, $f'(x) = 2x$, and

$$x_{k+1} = x_k - \frac{x_k^2 - 3}{2x_k} = \frac{1}{2}x_k + \frac{3}{2x_k}.$$

From $x_0 = 1$:

k	x_k	decimal	$\lvert x_k - \sqrt{3}\rvert$
0	1	1	7×10^{-1}
1	2	2	3×10^{-1}
2	7/4	1.75	2×10^{-2}
3	97/56	1.7321428…	9×10^{-5}
4	18817/10864	1.7320508100…	2×10^{-9}

The number of correct digits roughly doubles at each step. If the iterates converge to a limit $\bar{x}$, letting $k \to \infty$ in the recursion gives

$$\bar{x} = \frac{1}{2}\bar{x} + \frac{3}{2\bar{x}} \quad \Longrightarrow \quad \frac{1}{2}\bar{x} = \frac{3}{2\bar{x}} \quad \Longrightarrow \quad \bar{x}^2 = 3,$$

so the limit is indeed a root.

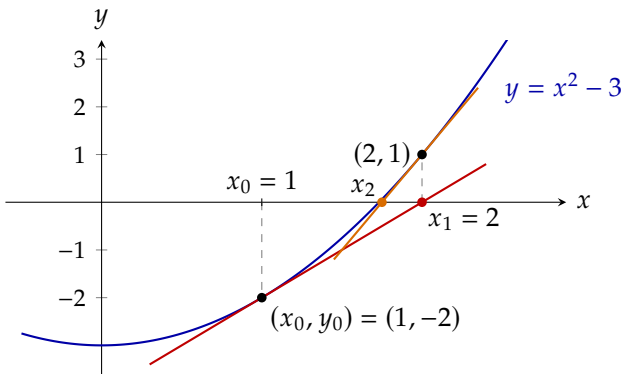


Figure 16: Newton's method for $f(x) = x^2 - 3$ from $x_0 = 1$: the tangent at $(1, -2)$ meets the axis at $x_1 = 2$, and the tangent at $(2, 1)$ meets it at $x_2 = 7/4$, already close to $\sqrt{3}$.

Newton's method has two failure modes.

- *It can find an unexpected root.* For $x^2 - 3$ with $x_0 = -1$, the tangent leads to the left, and the iterates converge to $-\sqrt{3}$ instead of $\sqrt{3}$.
- *It can fail completely.* The iterates can cycle, with $x_2 = x_0$, $x_3 = x_1$, and so on, never converging: the tangent at x_0 leads to x_1, and the tangent at x_1 leads back to x_0.

13.2 The ring on a string

A string of length L is fixed at $(0, 0)$ and (a, b), and a ring slides freely on it. The ring settles at the lowest point, so the problem is to minimize y over the positions (x, y) allowed by the constraint

$$\sqrt{x^2 + y^2} + \sqrt{(x - a)^2 + (y - b)^2} = L.$$

This curve is an ellipse with foci $(0, 0)$ and (a, b), but that fact is not needed. The constraint determines $y = y(x)$ implicitly, and since the ends of the curve are higher than its middle, the lowest point is a critical point, $y'(x) = 0$. Differentiating the constraint implicitly,

$$\frac{x + yy'}{\sqrt{x^2 + y^2}} + \frac{x - a + (y - b)y'}{\sqrt{(x - a)^2 + (y - b)^2}} = 0,$$

and at $y' = 0$ this becomes

$$\frac{x}{\sqrt{x^2 + y^2}} = \frac{a - x}{\sqrt{(x - a)^2 + (y - b)^2}}.$$

Let α and β be the angles the left and right parts of the string make with the vertical (Figure 17). The left side of this equation is $\sin\alpha$ and the right side is $\sin\beta$, so the critical point is characterized by the balance condition

$$\alpha = \beta.$$

Physically, resolving the forces shows the tensions in the two halves of the string are equal, an equilibrium. Geometrically, it is the reflection property of the ellipse: a light ray from the focus $(0,0)$ reflects off the ellipse, with angle of incidence equal to angle of reflection, and passes through the other focus (a,b).

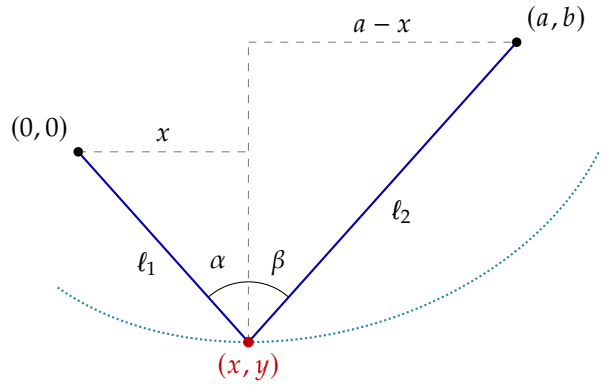


Figure 17: The ring on a string, drawn to scale for $a = 4$, $b = 1$, $L = 6$. The dotted curve is part of the constraint ellipse; at its lowest point (x, y), the two parts of the string make equal angles $\alpha = \beta$ with the vertical.

Proposition 13.2 [Position of the ring]. *The lowest point is*

$$x = \frac{a}{2}\left(1 - \frac{b}{\sqrt{L^2 - a^2}}\right), \qquad y = \frac{1}{2}\left(b - \sqrt{L^2 - a^2}\right).$$

Proof. Write $\ell_1 = \sqrt{x^2 + y^2}$ and $\ell_2 = \sqrt{(x-a)^2 + (y-b)^2}$, so $\ell_1 + \ell_2 = L$. The horizontal legs of the two right triangles are $x = \ell_1 \sin\alpha$ and $a - x = \ell_2 \sin\alpha$ (using $\beta = \alpha$); adding gives $a = L\sin\alpha$, so $\sin\alpha = a/L$. The vertical legs are $-y = \ell_1\cos\alpha$ (note $y < 0$) and $b - y = \ell_2\cos\alpha$; adding gives $b - 2y = L\cos\alpha = L\sqrt{1 - a^2/L^2} = \sqrt{L^2 - a^2}$, which is the formula for y. For x, the similar right triangles give

$$\tan\alpha = \frac{x}{-y} = \frac{a-x}{b-y} \quad\Longrightarrow\quad x(b-y) = -y(a-x) \quad\Longrightarrow\quad (b-2y)x = -ay,$$

so

$$x = \frac{-ay}{b-2y} = \frac{a\left(\sqrt{L^2-a^2} - b\right)}{2\sqrt{L^2-a^2}} = \frac{a}{2}\left(1 - \frac{b}{\sqrt{L^2-a^2}}\right).$$

□

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 13, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 14: Mean value theorem; inequalities

The mean value theorem converts information about the derivative, an infinitesimal quantity, into information about the values of a function over a whole interval. It is the theoretical underpinning of calculus.

14.1 The mean value theorem

Theorem 14.1 [Mean value theorem]. *If f is continuous on $a \le x \le b$ and differentiable on $a < x < b$, then*

$$\boxed{\frac{f(b)-f(a)}{b-a} = f'(c)} \qquad \text{for some } c,\ a < c < b.$$

The left side is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$; the right side is the slope of a tangent line. The theorem says some tangent line between a and b is parallel to the secant.

Geometric argument. Take lines parallel to the secant line, start with one lying below the graph over $[a, b]$, and shift it up until it first touches the graph. If the first contact is at an interior point c, the line touches the graph there without crossing it, so it is the tangent line at c and $f'(c)$ equals the secant slope. If the first contact is only at the endpoints, then the graph lies on or above the secant line; start instead with a parallel line above the graph and move it down, which gives an interior point of contact unless the graph is the secant line itself, in which case every c works. □

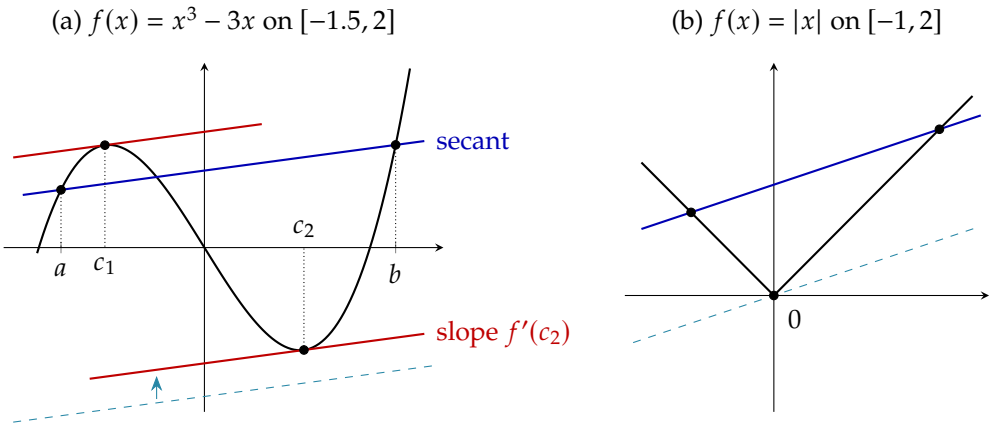


Figure 18: (a) For $f(x) = x^3 - 3x$ on $[-1.5, 2]$ the secant slope is $1/4$, and both $c_{1,2} = \mp\sqrt{13/12}$ have $f'(c) = 3c^2 - 3 = 1/4$. The dashed parallel line rises until it touches the graph at c_2. (b) For $f(x) = |x|$ every parallel line moving up first touches the graph at the corner $x = 0$, where f' is undefined. Here the secant slope is $1/3$, while $f' = \pm 1$ wherever it exists.

Differentiability is essential. For $f(x) = |x|$ on $[-1, 2]$ the argument breaks down: the moving line always first touches the graph at $x = 0$, whatever its slope, and $f'(0)$ does not exist. The conclusion fails too, since $f'(x) = \pm 1$ never equals the secant slope $1/3$.

Example: *A road trip.* Drive the 1000 miles from Boston to Chicago in exactly 3 hours, and let $f(t)$ be the distance from Boston at time t. With $a = 0$, $b = 3$, $f(0) = 0$, and $f(3) = 1000$, the theorem gives a time c with speed $f'(c) = 1000/3$ mph exactly.

14.2 Other forms, and comparison with linear approximation

Multiplying through by $b - a$, and then renaming b as x, gives

$$f(b) = f(a) + f'(c)(b - a), \qquad \boxed{f(x) = f(a) + f'(c)(x - a)} \quad \text{for some } c \text{ between } a \text{ and } x.$$

The theorem does not say what c is; it depends on f, a, and x. Compare the linear approximation

$$f(x) \approx f(a) + f'(a)(x - a) \qquad (x \text{ near } a).$$

The tangent line there has a definite slope $f'(a)$ but is only approximate. The mean value formula is exact, and it hides its lack of specificity in the slope $f'(c)$, which could be the slope of f at any point between a and x.

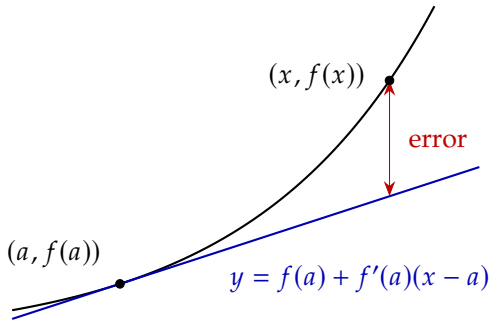


Figure 19: Linear approximation (drawn for $f(x) = e^x$, $a = 0$): the tangent line at a misses $f(x)$ by the error shown. The mean value theorem replaces the slope $f'(a)$ by $f'(c)$ for some c between a and x, and then the formula is exact.

14.3 Increasing, decreasing, and constant functions

A function is INCREASING if $a < b$ implies $f(a) < f(b)$, and DECREASING if $a < b$ implies $f(a) > f(b)$.

Corollary 14.2 [SIGN OF THE DERIVATIVE]. *Let f be differentiable on an interval.*

(1) If $f'(x) > 0$ throughout, then f is increasing.

(2) If $f'(x) < 0$ throughout, then f is decreasing.

(3) If $f'(x) = 0$ throughout, then f is constant.

Proof. Let $a < b$ in the interval. By the mean value theorem, $f(b) = f(a) + f'(c)(b-a)$ for some c with $a < c < b$. In (1), $f'(c) > 0$ and $b - a > 0$, so $f(b) > f(a)$. Case (2) is the same with the inequality reversed. In (3), $f(b) = f(a) + 0 \cdot (b-a) = f(a)$. □

These conclusions look obvious, but they are not. The derivative is defined by a limit over infinitesimal changes, and it takes a theorem to connect it with the non-infinitesimal behavior of the function.

14.4 Inequalities

The property $f' > 0 \Rightarrow f$ increasing yields many inequalities. Take $e^x > 0$ for granted.

Proposition 14.3 [Lower bounds for e^x]. *For $x > 0$,*

$$e^x > 1, \qquad e^x > 1 + x, \qquad e^x > 1 + x + \frac{x^2}{2}, \qquad e^x > 1 + x + \frac{x^2}{2} + \frac{x^3}{3!}.$$

Proof. Let $f_1(x) = e^x - 1$. Then $f_1(0) = 0$ and $f_1'(x) = e^x > 0$, so f_1 is increasing and $f_1(x) > f_1(0) = 0$ for $x > 0$.
Next let $f_2(x) = e^x - (1 + x)$. Then $f_2(0) = 0$ and $f_2'(x) = e^x - 1 = f_1(x) > 0$ for $x > 0$. By the mean value theorem on $[0, x]$, $f_2(x) = f_2(0) + f_2'(c)x > 0$ for some $0 < c < x$. Hence $e^x > 1 + x$.
Each step uses the previous one: $f_3(x) = e^x - (1 + x + x^2/2)$ has $f_3(0) = 0$ and $f_3' = f_2 > 0$ for $x > 0$, and likewise for the next polynomial. □

Continuing indefinitely suggests the infinite sum $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \cdots$, which is its Taylor series.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 14, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 15: Differentials, antiderivatives

15.1 Differentials

For $y = f(x)$, write $\boxed{dy = f'(x)\,dx}$. Both dy and $f'(x)\,dx$ are called DIFFERENTIALS, and $dy/dx = f'(x)$ can be read as a quotient of differentials. One use is linear approximation: $\Delta y/\Delta x \approx dy/dx$.

Example: *Approximating $65^{1/3}$, three ways.* Each method rests on the same two facts, $64^{1/3} = 4$ and the derivative of $y = x^{1/3}$ at $x = 64$,

$$\frac{dy}{dx} = \frac{1}{3}x^{-2/3}\Big|_{x=64} = \frac{1}{3}\cdot\frac{1}{16} = \frac{1}{48}.$$

- *Linear approximation.* With $f(x) = x^{1/3}$ and base point $a = 64$, $f(x) \approx f(a) + f'(a)(x-a)$ gives $65^{1/3} \approx 4 + \frac{1}{48}(65-64) = 4 + \frac{1}{48} \approx 4.02$. Similarly $(64.1)^{1/3} \approx 4 + \frac{1}{480}$.
- *Factoring out the base.* $65^{1/3} = \left[64(1+\frac{1}{64})\right]^{1/3} = 4\big(1+\frac{1}{64}\big)^{1/3}$, and $(1+x)^r \approx 1 + rx$ with $r = \frac{1}{3}$, $x = \frac{1}{64}$ gives $4\big(1+\frac{1}{3\cdot 64}\big) = 4 + \frac{1}{48}$.
- *Differentials.* At $x = 64$, $y = 4$ and $dy = \frac{1}{48}\,dx$. To reach $x + dx = 65$ take $dx = 1$, so $dy = \frac{1}{48}$ and $65^{1/3} \approx 4 + \frac{1}{48}$.

15.2 Antiderivatives

Definition 15.1 [ANTIDERIVATIVE]. $F(x) = \int f(x)\,dx$ means that F is an ANTIDERIVATIVE of f, that is, $F'(x) = f(x)$, or equivalently $dF = f(x)\,dx$.

Each formula below, with c an arbitrary constant, is checked by differentiating the right side:

$$\int \sin x\,dx = -\cos x + c, \quad \int x^n\,dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1), \quad \int \frac{dx}{x} = \ln|x| + c,$$
$$\int \sec^2 x\,dx = \tan x + c, \quad \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c, \quad \int \frac{dx}{1+x^2} = \tan^{-1} x + c.$$

Here $\sin^{-1}$ is the inverse function arcsin, not $1/\sin x$. The formula for $1/x$ covers the exceptional case $n = -1$, and the absolute value makes it correct for $x < 0$ as well: there $\ln|x| = \ln(-x)$, and the chain rule with $u = -x$ gives

$$\frac{d}{dx}\ln(-x) = \frac{1}{u}\cdot\frac{du}{dx} = \frac{1}{-x}(-1) = \frac{1}{x}.$$

Proposition 15.2 [Uniqueness up to a constant]. *If $F'(x) = f(x)$ and $G'(x) = f(x)$ on an interval, then $G(x) = F(x) + c$ for some constant c.*

Proof. $(G - F)' = f - f = 0$, and a function with zero derivative on an interval is constant, a corollary of the mean value theorem (Lecture 14). So $G - F = c$. □

15.3 Substitution and advanced guessing

Substitute $u = u(x)$, $du = u'(x)\,dx$ to reach a known form, or guess the answer and fix it by differentiating.

Example: *Substitution.* For $\int x^3(x^4+2)^5\,dx$, put $u = x^4 + 2$, so $du = 4x^3\,dx$ and $x^3\,dx = \frac{1}{4}du$:

$$\int x^3(x^4+2)^5\,dx = \frac{1}{4}\int u^5\,du = \frac{u^6}{24} + c = \frac{1}{24}(x^4+2)^6 + c.$$

Example: *Advanced guessing.* Write $\int \frac{x}{\sqrt{1+x^2}}\,dx = \int x(1+x^2)^{-1/2}\,dx$ and guess $(1+x^2)^{1/2}$. Its derivative is $\frac{1}{2}(1+x^2)^{-1/2}(2x) = x(1+x^2)^{-1/2}$, so the integral is $\sqrt{1+x^2} + c$. Similarly, guessing e^{6x} and e^{-x^2} and checking $(e^{6x})' = 6e^{6x}$, $(e^{-x^2})' = -2xe^{-x^2}$ gives

$$\int e^{6x}\,dx = \frac{1}{6}e^{6x} + c, \qquad \int xe^{-x^2}\,dx = -\frac{1}{2}e^{-x^2} + c.$$

Example: *Two answers that differ by a constant.* Both

$$\int \sin x \cos x\,dx = \frac{1}{2}\sin^2 x + c \qquad \text{and} \qquad \int \sin x \cos x\,dx = -\frac{1}{2}\cos^2 x + c$$

are correct, since $\frac{d}{dx}\sin^2 x = 2\sin x\cos x$ and $\frac{d}{dx}\cos^2 x = -2\cos x \sin x$. There is no contradiction:

$$\frac{1}{2}\sin^2 x - \left(-\frac{1}{2}\cos^2 x\right) = \frac{1}{2}(\sin^2 x + \cos^2 x) = \frac{1}{2},$$

so the two families are the same, with the constant shifted by $\frac{1}{2}$.

Example: *A logarithm inside a logarithm.* For $x > 0$, put $u = \ln x$, so $du = dx/x$:

$$\int \frac{dx}{x\ln x} = \int \frac{du}{u} = \ln|u| + c = \ln|\ln x| + c,$$

which is $\ln(\ln x) + c$ for $x > 1$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 15, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 16: Differential equations, separation of variables

16.1 Ordinary differential equations

An ordinary differential equation relates an unknown function $y(x)$ to its derivatives. The simplest kind, $dy/dx = f(x)$, is solved by integration, $y = \int f(x)\,dx$; we regard such equations as solved.

Example: *The equation* $\left(\frac{d}{dx} + x\right)y = 0$, that is, $dy/dx = -xy$. The key step is to separate variables, putting all y-dependence on one side and all x-dependence on the other:

$$\frac{dy}{y} = -x\,dx.$$

Take antiderivatives of both sides (one constant suffices) and exponentiate:

$$\ln|y| = -\frac{x^2}{2} + c, \qquad |y| = e^c e^{-x^2/2}, \qquad y = ae^{-x^2/2} \quad (a = \pm e^c).$$

Although $e^c \neq 0$, the value $a = 0$ is also allowed: $y \equiv 0$ solves the equation (it is the case excluded when dividing by y). The initial condition fixes a: $y(0) = a$ gives $y = ae^{-x^2/2}$, and in particular $y(0) = 1$ gives $y = e^{-x^2/2}$.

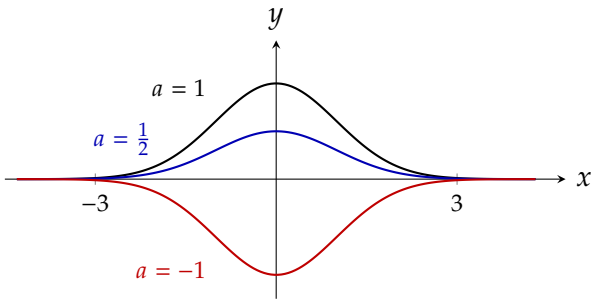


Figure 20: Solutions $y = ae^{-x^2/2}$ of $dy/dx = -xy$ with $y(0) = a$, for $a = 1, \frac{1}{2}, -1$.

16.2 The general method

Proposition 16.1 [Separation of variables]. *Consider* $\dfrac{dy}{dx} = f(x)g(y)$. *Write* $h(y) = 1/g(y)$, *so that* $h(y)\,dy = f(x)\,dx$, *and let* H *and* F *be antiderivatives,* $H' = h$, $F' = f$. *Then solutions satisfy the implicit equation*

$$\boxed{H(y) = F(x) + c} \qquad \text{so} \qquad y = H^{-1}\big(F(x) + c\big),$$

where H^{-1} *is the inverse function.*

Proof. Where $g(y) \neq 0$, the equation reads $h(y)\,y' = f(x)$. By the chain rule the left side is $\frac{d}{dx}H(y(x))$, so $H(y(x))$ and $F(x)$ have the same derivative and differ by a constant (Lecture 15). □

In the previous example, $f(x) = -x$, $F(x) = -x^2/2$, $g(y) = y$, $h(y) = 1/y$, and $H(y) = \ln|y|$.

16.3 Geometric examples

Example: *Tangent slope twice the ray slope.* Find curves whose tangent line at (x, y) has twice the slope of the ray from $(0, 0)$ to (x, y):

$$\frac{dy}{dx} = 2\,\frac{y}{x}.$$

Separating, $\dfrac{dy}{y} = \dfrac{2\,dx}{x}$, so $\ln|y| = 2\ln|x| + c$ and $|y| = e^c x^2$ (using $e^{2\ln|x|} = x^2$). Hence

$$y = ax^2,$$

with $a > 0$, $a < 0$, and $a = 0$ all acceptable: for example $y = x^2$, $y = 2x^2$, $y = -x^2$, $y = -2x^2$, $y = 100x^2$, and $y = 0$.

Example: *Orthogonal curves.* Find the curves perpendicular to all of these parabolas. At (x, y) the parabola through that point has slope $2y/x$, so a perpendicular curve has slope

$$\frac{dy}{dx} = \frac{-1}{2y/x} = -\frac{x}{2y}.$$

Separating, $y\,dy = -\frac{x}{2}\,dx$, so $\frac{y^2}{2} = -\frac{x^2}{4} + c$, that is,

$$\frac{x^2}{4} + \frac{y^2}{2} = c.$$

This is a family of ellipses whose x-semi-axis is $\sqrt{2}$ times their y-semi-axis ($2\sqrt{c}$ versus $\sqrt{2c}$). Separation of variables usually gives y only implicitly, but here one can solve: $y = \pm\sqrt{2\left(c - x^2/4\right)}$.

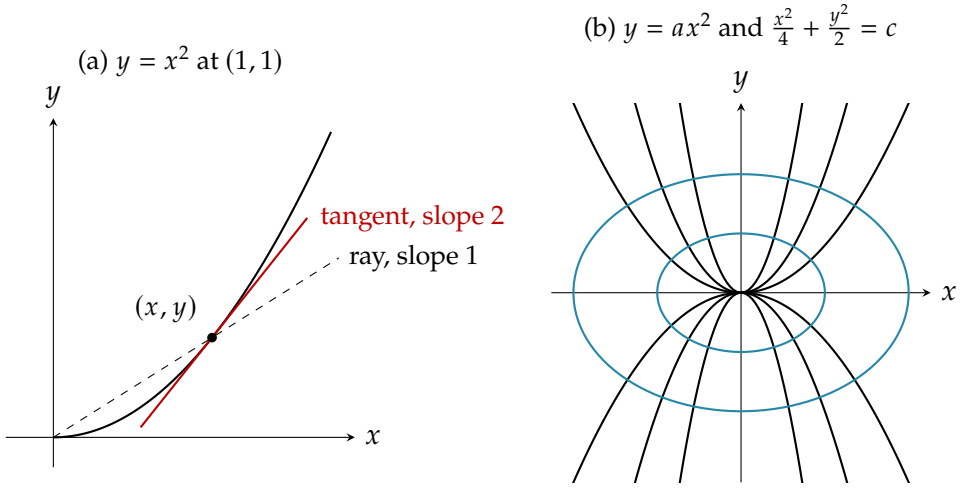


Figure 21: (a) On $y = x^2$ at $(1, 1)$ the ray from the origin has slope 1 and the tangent line has slope 2. (b) The parabolas $y = ax^2$ ($a = \pm 0.4, \pm 1, \pm 3$) and the orthogonal ellipses $x^2/4 + y^2/2 = c$ ($c = \frac{1}{2}, 2$).

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 16, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 18: Definite integrals

Integrals compute cumulative totals, averages, and areas. To find the area under a curve: divide the region into rectangles, add up their areas, and take the limit as the rectangles become thin.

18.1 The area under a parabola

Example: $f(x) = x^2$ *on* $[0, b]$. Divide $[0, b]$ into n intervals of length b/n, the common base of the rectangles. Use the right endpoints $x = b/n, 2b/n, \ldots, nb/n$, so the ith rectangle has height $(ib/n)^2$. The total area of the rectangles is

$$\frac{b}{n}\left(\frac{b}{n}\right)^2 + \frac{b}{n}\left(\frac{2b}{n}\right)^2 + \cdots + \frac{b}{n}\left(\frac{nb}{n}\right)^2 = \frac{b^3}{n^3}\left(1^2 + 2^2 + \cdots + n^2\right).$$

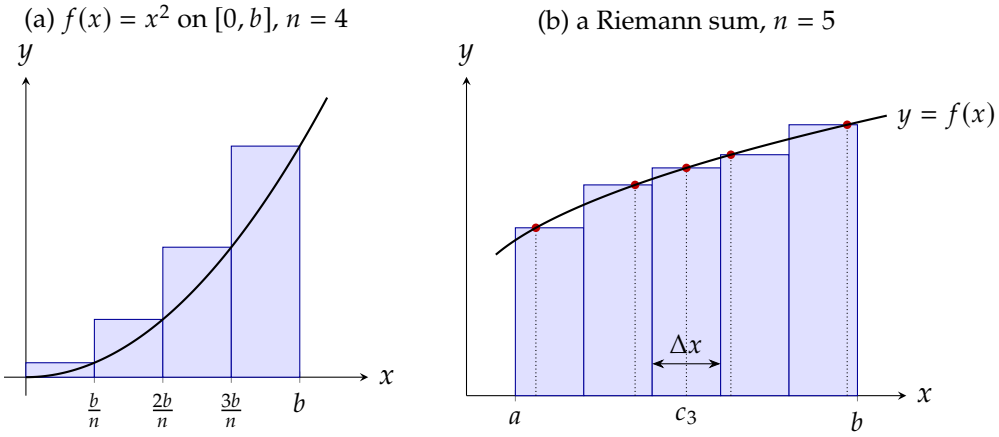


Figure 22: (a) Right-endpoint rectangles under $y = x^2$ on $[0, b]$. (b) A general Riemann sum (schematic, $f(x) = 1 + \sqrt{x}$): on each subinterval of length Δx the height is $f(c_i)$ at an arbitrary point c_i.

Lemma 18.1 [Sum of squares]. *For every* $n \geq 1$,

$$\frac{1}{3}n^3 < 1^2 + 2^2 + \cdots + n^2 < \frac{1}{3}(n+1)^3.$$

Proof. Build a staircase pyramid of unit-height layers: the bottom layer is an $n \times n$ square slab of volume n^2, the next is $(n-1)\times(n-1)$, and so on up to 1×1, all centered on the same vertical axis. Its volume is $n^2 + (n-1)^2 + \cdots + 1$.
At height z in the kth layer, $k - 1 \leq z \leq k$, the cross-section is a square of side $n - k + 1$. The pyramid with an $n \times n$ base and height n has cross-section of side $n - z \leq n - k + 1$ there, so it

lies inside the staircase. The pyramid with an $(n+1)\times(n+1)$ base and height $n+1$ has side $n+1-z \geq n-k+1$, so it contains the staircase. A pyramid has volume $\frac{1}{3}$(base)(height), so the volumes are $\frac{1}{3}n^2 \cdot n$ and $\frac{1}{3}(n+1)^2(n+1)$, and the inequalities are strict because the solids differ by regions of positive volume. □

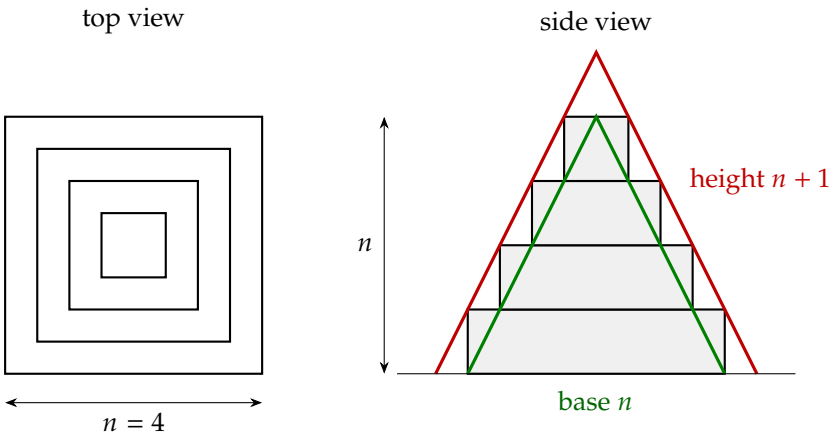


Figure 23: Staircase pyramid for $n=4$. The inner pyramid (base n, height n) passes through the bottom corners of the layers; the outer pyramid (base $n+1$, height $n+1$) passes through their top corners.

Dividing by n^3,

$$\frac{1}{3} < \frac{1^2+2^2+\cdots+n^2}{n^3} < \frac{1}{3}\cdot\frac{(n+1)^3}{n^3} \longrightarrow \frac{1}{3},$$

so

$$\lim_{n\to\infty}\frac{b^3}{n^3}\left(1^2+2^2+\cdots+n^2\right) = \frac{b^3}{3},$$

and the area under x^2 from 0 to b is $\boxed{b^3/3}$.

Similar but easier reasoning for $f(x)=x$ gives rectangle areas $\frac{b^2}{n^2}(1+2+\cdots+n) \to \frac{1}{2}b^2$, the area of the triangle with base and height b.

Notice the pattern: $\frac{d}{db}\left(\frac{b^3}{3}\right) = b^2$ and $\frac{d}{db}\left(\frac{b^2}{2}\right) = b$. The area $A(b)$ under f from 0 to b should satisfy $A'(b) = f(b)$.

18.2 Riemann sums and the definite integral

In general, divide $[a,b]$ into n equal pieces of length $\Delta x = (b-a)/n$, pick *any* point c_i in the ith piece, and use $f(c_i)$ as the height of the ith rectangle. The total area $\sum_{i=1}^{n} f(c_i)\,\Delta x = f(c_1)\Delta x + \cdots + f(c_n)\Delta x$ is a Riemann sum.

Definition 18.2 [Definite integral]**.** The definite integral of f from a to b is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(c_i)\,\Delta x.$$

For $f \geq 0$ it is the area under the curve $y = f(x)$ above $[a, b]$.

18.3 An integral that is not an area

Example: *Borrowing.* A principal P borrowed at simple interest rate r per year (for example $r = 0.06$ for 6%) grows to $P(1 + rt) = P + Prt$ after t years. Let $f(t)$ be a *borrowing rate* in dollars per year, allowed to vary over time; borrowing \$1000 per month is $f(t) = 12{,}000$ per year. With $\Delta t = 1/12$ year (one month) and $t_i = i/12$, $i = 1, \ldots, 12$, the amount borrowed in month i is about $f(t_i)\Delta t$, so the total borrowed in the year is about $\sum_{i=1}^{12} f(t_i)\Delta t$. Letting $\Delta t \to 0$,

$$\text{total borrowed in one year} = \int_0^1 f(t)\,dt \quad \text{dollars.}$$

The amount borrowed at time t_i accrues interest for the remaining $1 - t_i$ years, so at the end of the year it is owed as $f(t_i)\big(1 + r(1 - t_i)\big)\Delta t$. In the limit, the total owed at the end of the year is

$$\int_0^1 f(t)\big(1 + r(1 - t)\big)\,dt.$$

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 18, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 19: First fundamental theorem of calculus

19.1 The first fundamental theorem

Theorem 19.1 [Fundamental theorem of calculus, FTC 1]. *If f is continuous and $F'(x) = f(x)$, then*

$$\boxed{\int_a^b f(x)\,dx = F(b) - F(a)}$$

So a definite integral, a limit of Riemann sums, is computed by any antiderivative. We write

$$F(x)\Big|_a^b = F(x)\Big|_{x=a}^{x=b} = F(b) - F(a).$$

Example: *Three integrals.* With $F(x) = x^3/3$, $F'(x) = x^2$,

$$\int_a^b x^2\,dx = \frac{x^3}{3}\Big|_a^b = \frac{b^3}{3} - \frac{a^3}{3},$$

which recovers the area $b^3/3$ for $a = 0$ found from Riemann sums in Lecture 18. The area under one hump of $\sin x$ is

$$\int_0^\pi \sin x\,dx = -\cos x\Big|_0^\pi = -\cos\pi - (-\cos 0) = 1 + 1 = 2,$$

and $\int_0^1 x^5\,dx = \frac{x^6}{6}\Big|_0^1 = \frac{1}{6}$.

19.2 Interpretation: speedometer and odometer

Let $x(t)$ be a position and $v(t) = x'(t) = dx/dt$ the velocity. FTC 1 says

$$\int_a^b v(t)\,dt = x(b) - x(a).$$

The right side is how far $x(t)$ went from $t = a$ to $t = b$, the difference between two odometer readings. The left side is built from speedometer readings: the Riemann sum $\sum_{i=1}^n v(t_i)\Delta t$ adds the approximate distances traveled over the short times Δt, and the approximation is accurate when $v(t)$ stays close to $v(t_i)$ on the ith interval.

If v changes sign, $x(b) - x(a)$ is the *net* distance, not the odometer reading. On a round trip $x(b) - x(a) = 0$, because positive and negative velocities cancel, whereas an odometer measures the total distance traveled. In the same way an integral counts area above the x-axis as positive and

area below it as negative:

$$\int_0^{2\pi} \sin x \, dx = -\cos x \Big|_0^{2\pi} = -\cos 2\pi - (-\cos 0) = 0.$$

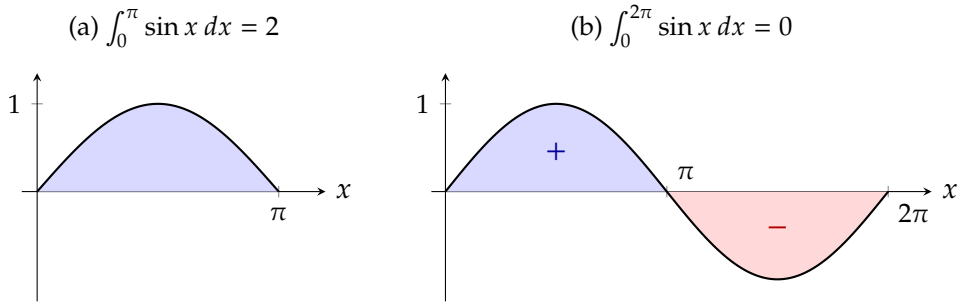


Figure 24: (a) One hump of $\sin x$ has area 2. (b) Over a full period, the area above the axis cancels the area below it.

19.3 Properties of the definite integral

Proposition 19.2 [Additivity].

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx.$$

For $a < b < c$ this says the area over $[a, c]$ is the sum of the areas over $[a, b]$ and $[b, c]$. We also *define*

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

With this convention, FTC 1 holds whether $a < b$ or $b < a$, and additivity holds for a, b, c in any order. (Both also follow at once from FTC 1, since $[F(b) - F(a)] + [F(c) - F(b)] = F(c) - F(a)$.)

Proposition 19.3 [Estimation]. *If $f(x) \le g(x)$ on $[a, b]$ and $a < b$, then* $\displaystyle\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$

The hypothesis $a < b$ matters: reversing the limits reverses the inequality. The proposition follows from comparing Riemann sums term by term, since $\Delta x > 0$.

Example: *Bounds on e.* Since $1 \le e^x$ for $x \ge 0$,

$$1 = \int_0^1 1\,dx \le \int_0^1 e^x\,dx = e^x\Big|_0^1 = e - 1,$$

so $e \geq 2$. Using the sharper inequality $1 + x \leq e^x$ from Lecture 14,

$$\frac{3}{2} = \left(x + \frac{x^2}{2}\right)\Big|_0^1 = \int_0^1 (1 + x)\, dx \leq \int_0^1 e^x\, dx = e - 1,$$

so $e \geq \frac{5}{2}$.

19.4 Change of variable

If $f(x) = g(u(x))$, write $du = u'(x)\, dx$. For indefinite integrals,

$$\int g(u)\, du = \int g(u(x))\, u'(x)\, dx = \int f(x)\, u'(x)\, dx.$$

Proposition 19.4 [Substitution in definite integrals]. *For u with continuous derivative and g continuous,*

$$\int_{x_1}^{x_2} g(u(x))\, u'(x)\, dx = \int_{u_1}^{u_2} g(u)\, du, \qquad u_1 = u(x_1),\ u_2 = u(x_2).$$

Proof. Let $G' = g$. By the chain rule $\frac{d}{dx} G(u(x)) = g(u(x))u'(x)$, so by FTC 1 both sides equal $G(u_2) - G(u_1)$. □

The limits change with the variable, so there is no need to return to x.

Example: *A definite substitution.* For $\int_1^2 (x^3+2)^4 x^2\, dx$, let $u = x^3+2$, so $du = 3x^2\, dx$ and $x^2\, dx = du/3$. The limits $x_1 = 1$, $x_2 = 2$ become $u_1 = 3$, $u_2 = 10$, and

$$\int_1^2 (x^3 + 2)^4 x^2\, dx = \int_3^{10} u^4\, \frac{du}{3} = \frac{u^5}{15}\Big|_3^{10} = \frac{10^5 - 3^5}{15}.$$

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 19, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 20: Second fundamental theorem

20.1 New functions defined by integrals

FTC 1 (Lecture 19) says that if f is continuous and $F' = f$, then

$$\int_a^b f(x)\,dx = F(b) - F(a) = \int f(x)\,dx \Big|_{x=a}^{x=b}.$$

Does every continuous function have an antiderivative? Yes. For instance $\int e^{-x^2}\,dx$ exists, but it is not a function met before; it is a new function, defined by an integral:

$$F(x) = \int_0^x e^{-t^2}\,dt, \qquad \text{with} \qquad F'(x) = e^{-x^2}.$$

Antiderivatives of e^{-x^2}, $x^{1/2}e^{-x^2}$, $\frac{\sin x}{x}$, $\sin(x^2)$, $\cos(x^2)$, and many others are new functions of this kind.

20.2 The second fundamental theorem

Theorem 20.1 [Fundamental theorem of calculus, FTC 2]. *If f is continuous and $F(x) = \int_a^x f(t)\,dt$, then*

$$\boxed{F'(x) = f(x)}$$

Proof. For $f \geq 0$, $F(x)$ is the area under the curve between a and x. The increment $\Delta F = F(x + \Delta x) - F(x)$ is the area of a thin strip of base Δx and height about $f(x)$, so $\Delta F \approx (\Delta x) f(x)$ and $\Delta F/\Delta x \approx f(x)$.
More precisely, for any sign of f, additivity gives

$$\frac{\Delta F}{\Delta x} = \frac{1}{\Delta x}\Big[\int_a^{x+\Delta x} f(t)\,dt - \int_a^x f(t)\,dt\Big] = \frac{1}{\Delta x}\int_x^{x+\Delta x} f(t)\,dt,$$

the average value of f on the interval between x and $x + \Delta x$. By continuity, $f(t)$ lies within any $\varepsilon > 0$ of $f(x)$ on that interval once $|\Delta x|$ is small, and then so does the average (by estimation of integrals, Lecture 19). As $\Delta x \to 0$ the average tends to $f(x)$, and by the definition of the derivative, $F'(x) = \lim_{\Delta x \to 0} \Delta F/\Delta x = f(x)$. □

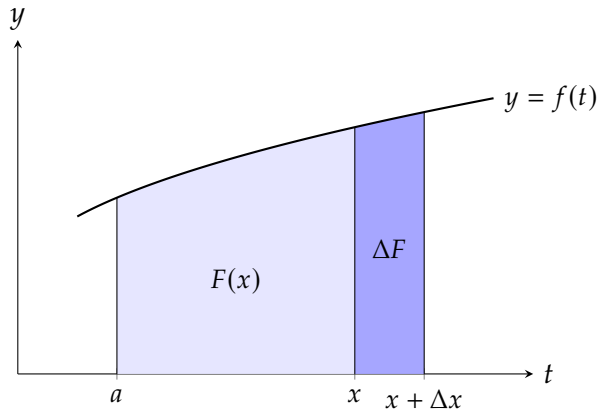


Figure 25: $F(x)$ is the area from a to x (schematic). The strip ΔF has base Δx and height about $f(x)$.

20.3 FTC 1 from FTC 2

Proof of FTC 1. Let f be continuous and $F' = f$. Define $G(x) = \int_a^x f(t)\,dt$. By FTC 2, $G' = f$, so $(F - G)' = f - f = 0$ and $F - G$ is a constant c, a consequence of the mean value theorem. Since $G(a) = 0$,

$$\int_a^b f(t)\,dt = G(b) - G(a) = [F(b) - c] - [F(a) - c] = F(b) - F(a). \qquad \square$$

In this proof G is a specific definite integral, while F can be any antiderivative. For example, with $f(x) = \cos x$ and $a = 0$, $G(x) = \int_0^x \cos t\,dt = \sin t\big|_0^x = \sin x$ and $G(0) = 0$. Another antiderivative is $F(x) = \sin x + 21$, and it gives the same answer,

$$\int_a^b \cos x\,dx = (\sin b + 21) - (\sin a + 21) = \sin b - \sin a.$$

Every $F = G + c$ works in FTC 1.

20.4 Examples of new functions

Definition 20.2 [Error function]. The error function, used in probability and statistics, is

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt, \qquad \lim_{x\to\infty} \operatorname{erf}(x) = 1.$$

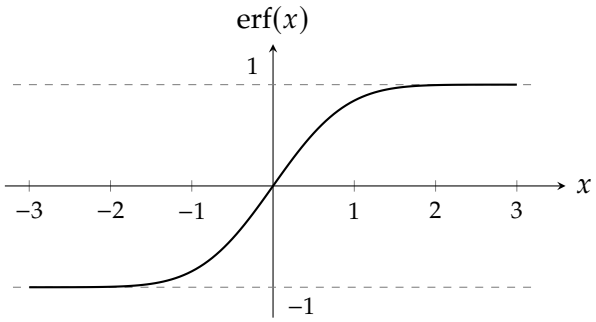


Figure 26: The error function, which increases from -1 to 1.

The LOGARITHMIC INTEGRAL

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\ln t}$$

approximates the number of primes less than x. For example, it estimates how many 200-digit primes there are, which matters for encryption schemes whose security depends on a secret prime being hard to guess. Since

$$\ln 10^{200} = 200 \ln 10 \approx 200(2.3) = 460, \qquad \ln 10^{201} = 201 \ln 10 \approx 462,$$

to one significant figure $\ln t \approx 500$ for $10^{200} \le t \le 10^{201}$. Using $\int_a^b c\,dx = c(b-a)$, the area of a rectangle,

$$\int_{10^{200}}^{10^{201}} \frac{dt}{\ln t} \approx \int_{10^{200}}^{10^{201}} \frac{dt}{500} = \frac{10^{201} - 10^{200}}{500} = \frac{9 \cdot 10^{200}}{500} \approx 10^{198}.$$

There are very many 200-digit primes.

Two further families are the FRESNEL FUNCTIONS of optics and the BESSEL FUNCTION J_0, which arises in problems with circular symmetry:

$$C(x) = \int_0^x \cos(t^2)\,dt, \qquad S(x) = \int_0^x \sin(t^2)\,dt, \qquad J_0(x) = \frac{1}{\pi}\int_0^\pi \cos(x \sin\theta)\,d\theta.$$

By FTC 2, derivatives of the Fresnel functions are immediate, for example $C'(x) = \cos(x^2)$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 20, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 21: Applications to logarithms and geometry

The second fundamental theorem (FTC2) turns a definite integral with a variable upper limit into a new function whose derivative is known. The simplest new function obtained this way is the logarithm. The first theorem (FTC1) then computes areas and volumes by summing thin slices.

21.1 The logarithm as an integral

The power rule $\int x^n\,dx = x^{n+1}/(n+1) + c$ fails only for $n = -1$, so an antiderivative of $1/x$ must be something other than a power.

Definition 21.1 [Integral logarithm]. For $0 < x < \infty$, let

$$L(x) = \int_1^x \frac{dt}{t}.$$

The definition stops at $x = 0$ because $1/t$ is infinite at $t = 0$. By FTC2 and the choice of lower limit,

$$L'(x) = \frac{1}{x}, \qquad L(1) = \int_1^1 \frac{dt}{t} = 0, \qquad L''(x) = -\frac{1}{x^2}.$$

So L is increasing and concave down, with $L(x) < 0$ for $0 < x < 1$ and $L(x) > 0$ for $x > 1$. The key property, which shows that L is a logarithm, is that it converts multiplication into addition.

Proposition 21.2 [Product rule for L]. *For $a, b > 0$, $L(ab) = L(a) + L(b)$.*

Proof. Split the interval of integration at a:

$$L(ab) = \int_1^{ab} \frac{dt}{t} = \int_1^{a} \frac{dt}{t} + \int_a^{ab} \frac{dt}{t} = L(a) + \int_a^{ab} \frac{dt}{t}.$$

In the last integral substitute $t = au$, so $dt = a\,du$, and t runs from a to ab exactly when u runs from 1 to b:

$$\int_a^{ab} \frac{dt}{t} = \int_1^{b} \frac{a\,du}{au} = \int_1^{b} \frac{du}{u} = L(b). \qquad \square$$

Proposition 21.3 [End behavior]. *$L(x) \to \infty$ as $x \to \infty$, and $L(x) \to -\infty$ as $x \to 0^+$.*

Proof. Applying Proposition 21.2 repeatedly,

$$L(2^n) = L(2 \cdot 2^{n-1}) = L(2) + L(2^{n-1}) = \cdots = nL(2),$$

and $L(2) > 0$, so $L(2^n) \to \infty$. Since L is increasing, it exceeds $nL(2)$ for all $x > 2^n$, which gives the first limit. For the second, $0 = L(1) = L\big(x \cdot \frac{1}{x}\big) = L(x) + L(1/x)$, so $L(x) = -L(1/x)$. As $x \to 0^+$, $1/x \to \infty$, hence $L(x) = -L(1/x) \to -\infty$. □

Thus L increases from $-\infty$ to ∞ on $(0, \infty)$, crossing the axis at $x = 1$ (Figure 27). This gives an alternative route to the exponential and logarithm: *define* $\ln x = L(x)$, define e as the number with $L(e) = 1$, define e^x as the inverse function of L, and define $a^x = e^{xL(a)}$. In the familiar notation the proof above reads $\ln 2^n = n \ln 2$.

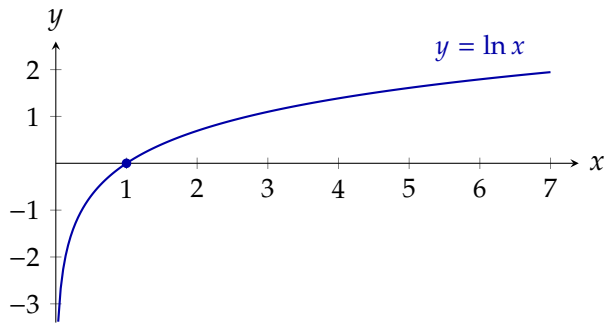


Figure 27: The graph of $y = L(x) = \ln x$: increasing and concave down, through $(1, 0)$, tending to $-\infty$ as $x \to 0^+$ and to $+\infty$ as $x \to \infty$.

21.2 Areas between two curves

If $f \geq g$ on $[a, b]$, where a and b are the crossing points of the two graphs, a vertical strip of width dx has area $(f(x) - g(x))\,dx$, and the area between the curves is

$$A = \int_a^b \big(f(x) - g(x)\big)\,dx = \int (y_{\text{top}} - y_{\text{bottom}})\,dx.$$

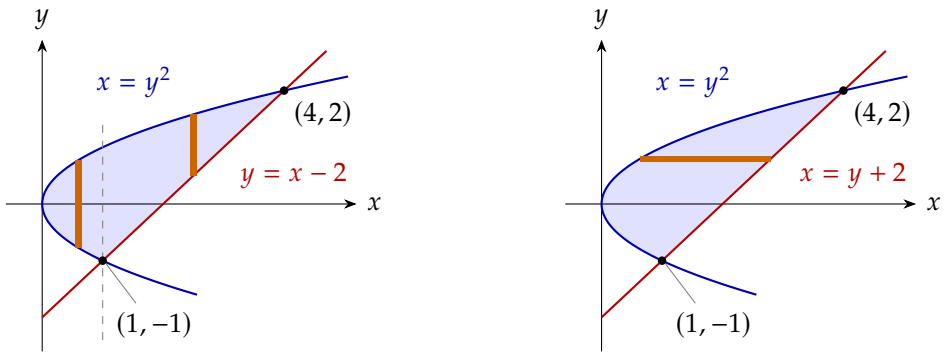
One may equally slice horizontally and integrate $(x_{\text{right}} - x_{\text{left}})\,dy$; the better choice is the one that describes the region with a single formula.

Example: *The region between* $x = y^2$ *and* $y = x - 2$. The curves meet where $y + 2 = x = y^2$, i.e. $y^2 - y - 2 = (y-2)(y+1) = 0$. So $y = -1, 2$, and the crossing points are $(1, -1)$ and $(4, 2)$ (Figure 28). *Vertical slices* need two pieces. For $0 < x < 1$ the strip runs between the two branches $y = \pm\sqrt{x}$ of the parabola; for $1 < x < 4$ it runs from the line up to $y = \sqrt{x}$:

$$A = \int_0^1 \big(\sqrt{x} - (-\sqrt{x})\big)\,dx + \int_1^4 \big(\sqrt{x} - (x-2)\big)\,dx = \frac{4}{3} + \frac{19}{6} = \frac{9}{2}.$$

Horizontal slices need one piece: for $-1 < y < 2$ the strip runs from the parabola $x = y^2$ on the left to the line $x = y + 2$ on the right, so

$$A = \int_{-1}^{2} \big((y+2) - y^2\big)\, dy = \left(\frac{y^2}{2} + 2y - \frac{y^3}{3}\right)\Big|_{-1}^{2} = \left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right) = \frac{9}{2}.$$



vertical strips horizontal strips

Figure 28: The region between $x = y^2$ and $y = x - 2$. Left: a vertical strip; for $x < 1$ its lower end lies on the lower branch of the parabola, so vertical slicing needs two integrals. Right: every horizontal strip runs from the parabola to the line.

21.3 Volumes of solids of revolution: disks

Rotate the region under $y = f(x) \geq 0$ about the x-axis. Slicing perpendicular to the axis cuts the solid into thin DISKS of radius y and thickness dx, so

$$dV = \pi y^2\, dx, \qquad V = \int \pi y^2\, dx \quad \text{(rotation about the } x\text{-axis)}.$$

Example: *The volume of a ball of radius a.* Rotating the upper semicircle $y = \sqrt{a^2 - x^2}$, $-a \leq x \leq a$, about the x-axis gives a ball of radius a, so

$$V = \int_{-a}^{a} \pi(a^2 - x^2)\, dx = \left(\pi a^2 x - \frac{\pi x^3}{3}\right)\Big|_{-a}^{a} = \frac{2}{3}\pi a^3 - \left(-\frac{2}{3}\pi a^3\right) = \frac{4}{3}\pi a^3.$$

The integrand is even, so it is simpler to write $V = 2\int_0^a \pi(a^2 - x^2)\, dx = 2\big(\pi a^3 - \frac{\pi}{3}a^3\big)$, which gives the same $\frac{4}{3}\pi a^3$ with the easier lower limit 0.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 21, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 22: Volumes by disks and shells

A solid of revolution can be cut into disks perpendicular to the axis or into thin cylindrical shells around it. Both methods give the same volume; the choice is a matter of convenience. The same ring-shaped slices also add up quantities other than volume.

22.1 Disks and shells for the cauldron

Example: *A witch's cauldron.* Rotate the parabola $y = x^2$ about the y-axis and fill the resulting bowl up to height $y = a$. Find its volume V.

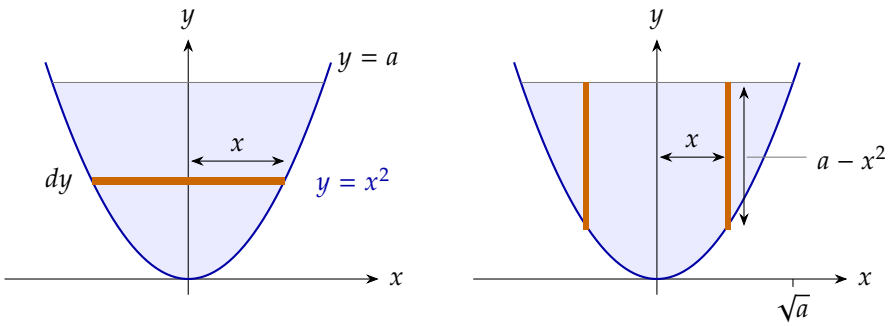
Method 1: disks. The horizontal slice at height y is a disk of radius x, area πx^2, and thickness dy (Figure 29, left). So $dV = \pi x^2\,dy$, and substituting $x^2 = y$,

$$V = \int_0^a \pi x^2\,dy = \int_0^a \pi y\,dy = \pi\frac{y^2}{2}\Big|_0^a = \frac{\pi a^2}{2}.$$

Method 2: shells. This really should be called the CYLINDER METHOD. The water between radii x and $x + dx$ forms a thin cylindrical shell of circumference $2\pi x$, thickness dx, and height $a - y = a - x^2$ (Figure 29, right). The radius runs from 0 to $\sqrt{a}$, so

$$V = \int_0^{\sqrt{a}} (a - x^2)(2\pi x)\,dx = 2\pi\left(a\frac{x^2}{2} - \frac{x^4}{4}\right)\Big|_0^{\sqrt{a}} = 2\pi\left(\frac{a^2}{2} - \frac{a^2}{4}\right) = \frac{\pi a^2}{2},$$

in agreement with the disk method.



disks: $dV = \pi x^2\,dy$ shells: $dV = (a - x^2)\,2\pi x\,dx$

Figure 29: Cross-sections of the cauldron $y = x^2$, $0 \le y \le a$, rotated about the y-axis. Left: a horizontal slice of thickness dy sweeps out a disk of radius $x = \sqrt{y}$. Right: a vertical strip of width dx at radius x sweeps out a cylindrical shell of height $a - x^2$.

Units. With $a = 1$ meter, $V = \frac{\pi}{2}\,\mathrm{m}^3 = \frac{\pi}{2}(100\,\mathrm{cm})^3 = \frac{\pi}{2}\cdot 10^6\,\mathrm{cm}^3 \approx 1600$ liters. With $a = 100$ cm, $V = \frac{\pi}{2}(100)^2\,\mathrm{cm}^3 = \frac{\pi}{2}\cdot 10^4\,\mathrm{cm}^3 \approx 16$ liters, although 100 cm = 1 m. The two bowls are different shapes: in centimeters the rim $100 = x^2$ has radius $x = 10$ cm (a skinny bowl, 20 cm across and 100 cm deep), while in meters the rim $1 = x^2$ has radius 1 m. The equation $y = x^2$ is not scale-invariant, so the shape it describes depends on the units.

22.2 Other quantities summed over slices

Example: *The boiling cauldron.* Take $a = 1$ meter, so $x^2 = y$ for $0 \le y \le 1$, and fill the cauldron with water at 0°C. A fire heats the water to the temperature profile $T = 100 - 30y$ (degrees Celsius), so $T = 100°$ at the bottom and 70° at the top. How much heat does this take?

The temperature is constant on each horizontal disk, so use disks: the disk at height y has volume $\pi x^2\,dy = \pi y\,dy$ and must be raised by $T(y)$ degrees. The total heat, in units of degree·m^3, is

$$H = \int_0^1 T\,(\pi x^2)\,dy = \pi\int_0^1 (100 - 30y)\,y\,dy = \pi\big(50y^2 - 10y^3\big)\Big|_0^1 = 40\pi.$$

Heating 1 cm^3 of water by one degree takes 1 calorie, so

$$H = (40\pi)\,\frac{1\,\mathrm{cal}}{\mathrm{cm}^3\cdot\mathrm{deg}}\left(\frac{100\,\mathrm{cm}}{1\,\mathrm{m}}\right)^3 = 40\pi\cdot 10^6\,\mathrm{cal} \approx 125\times 10^3\,\mathrm{kcal},$$

roughly 500 candy bars at about 250 kcal each.

Example: *Pipe flow.* Poiseuille found that fluid in a pipe of radius R (an artery, a capillary) flows fastest at the center and sticks at the wall, with speed $v = c(R^2 - r^2)$ at distance r from the center. Find the total flow through the pipe (volume per unit time).

Cut the cross-section into thin rings (annuli). The ring between r and $r + dr$ has area $2\pi r\,dr$ (circumference times thickness), and the flow through it is (area)(speed) = $v\cdot 2\pi r\,dr$; the speed plays the role of the height of a shell. Hence

$$\text{flow} = \int_0^R c(R^2 - r^2)\,2\pi r\,dr = 2\pi c\left(\frac{R^2 r^2}{2} - \frac{r^4}{4}\right)\Big|_0^R = \frac{\pi}{2}cR^4.$$

The flow is proportional to R^4: doubling the radius multiplies the flow by 16, a big advantage for thick pipes.

Dart board. If the density of hits at distance r from the center of a dart board is $N = ce^{-r^2}$, then summing over rings, the number of hits with $r_1 < r < r_2$ is $c\int_{r_1}^{r_2} e^{-r^2}\,(2\pi r\,dr)$. Lecture 23 evaluates this and turns it into a probability.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 22, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 23: Work, average value, probability

Integrals compute continuous averages, weighted averages, and probabilities, all as limits of sums. The same slicing ideas evaluate the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx$.

23.1 Average value

Divide $[a, b]$ into n equal pieces, $a = x_0 < x_1 < \cdots < x_n = b$ with $\Delta x = (b-a)/n$, and let $y_i = f(x_i)$. The average of the sample values is approximately the average of f. Since $(y_1 + \cdots + y_n)\Delta x$ is a Riemann sum,

$$\lim_{n\to\infty} (y_1 + \cdots + y_n)\frac{b-a}{n} = \int_a^b f(x)\,dx,$$

and dividing by $b - a$ gives the continuous average.

Definition 23.1 [Average value]. The average value of f on $[a, b]$ is

$$\lim_{n\to\infty} \frac{y_1 + \cdots + y_n}{n} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

The average of a constant is that constant, for example $\frac{1}{b-a}\int_a^b 53\,dx = 53$.

Example: *Average height of a semicircle.* For $y = \sqrt{1-x^2}$ on $-1 \le x \le 1$, the integral is the area $\pi/2$ of the half disk, so the average height is $\frac{1}{2}\int_{-1}^{1}\sqrt{1-x^2}\,dx = \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4}$.

Example: *Average height with respect to arclength.* Weighting the points of the semicircle equally by arclength means averaging in the angle θ, where $y = \sin\theta$, $0 \le \theta \le \pi$:

$$\frac{1}{\pi}\int_0^{\pi} \sin\theta\,d\theta = \frac{1}{\pi}(-\cos\theta)\Big|_0^{\pi} = \frac{2}{\pi}.$$

This differs from $\pi/4$: equal steps in θ near the ends of the semicircle correspond to short steps in x, so the two averages weight the points differently.

23.2 Weighted averages

More generally, giving the value $f(y)$ the importance or weight $w(y) \ge 0$ leads to the weighted average

$$\frac{\int f(y)\,w(y)\,dy}{\int w(y)\,dy}.$$

Example: *Average temperature in the cauldron.* In the cauldron of Lecture 22 ($y = x^2$ rotated about the y-axis, height 1 m), the water temperature is $T(y) = 100 - 30y$, so $T(0) = 100°$ and $T(1) = 70°$. The average over the volume weights the disk at height y by its volume factor $w(y) = \pi x^2 = \pi y$. The denominator is the volume $V = \int_0^1 \pi y \, dy = \pi/2$, and the numerator is

$$\int_0^1 T\,\pi y \, dy = \pi \int_0^1 (100 - 30y)\, y \, dy = \pi\big(50y^2 - 10y^3\big)\Big|_0^1 = 40\pi.$$

So the average temperature is $\frac{40\pi}{\pi/2} = 80°\text{C}$. The average with respect to height is instead

$$\int_0^1 (100 - 30y)\, dy = (100y - 15y^2)\Big|_0^1 = 85°\text{C},$$

which, since T is linear, is the mean $\frac{70+100}{2}$ of the extreme values. The volume-weighted average is lower because more of the water lies in the wide, cooler upper part of the cauldron.

23.3 Probability

When outcomes are spread over a continuum,

$$\text{probability} = \frac{\text{part}}{\text{whole}},$$

where the part and the whole are integrals of the same weight.

Example: *Dart board, revisited.* As in Lecture 22, the density of hits at radius r is ce^{-r^2} (a normal distribution), so the ring between r and $r + dr$ carries weight $ce^{-r^2}\, 2\pi r \, dr$. A little brother stands near the board, approximated by the sector $2r_1 < r < 3r_1$ between 3 and 5 o'clock, where r_1 is the radius of the board (Figure 30). This sector is $\frac{5-3}{12} = \frac{1}{6}$ of each ring it meets. What is the probability that a dart hits him?

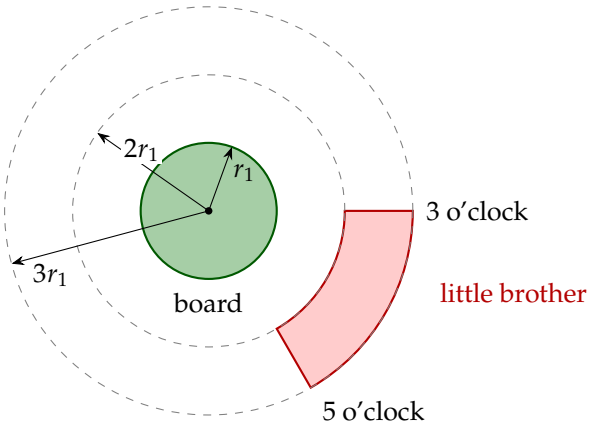


Figure 30: The dart board of radius r_1 and the shaded sector $2r_1 < r < 3r_1$ between 3 and 5 o'clock, $\frac{1}{6}$ of the ring $2r_1 < r < 3r_1$.

Since $\frac{d}{dr}e^{-r^2} = -2re^{-r^2}$ (note e^{-r^2} means $e^{(-r^2)}$, not $(e^{-r})^2$),

$$\int_a^b re^{-r^2}\,dr = -\frac{1}{2}e^{-r^2}\Big|_a^b = \frac{1}{2}\big(e^{-a^2} - e^{-b^2}\big), \qquad \int_0^\infty re^{-r^2}\,dr = \lim_{R\to\infty}\Big(\frac{1}{2} - \frac{1}{2}e^{-R^2}\Big) = \frac{1}{2}.$$

The constants c and 2π cancel between part and whole, so

$$P = \frac{\frac{1}{6}\int_{2r_1}^{3r_1} ce^{-r^2}\,2\pi r\,dr}{\int_0^\infty ce^{-r^2}\,2\pi r\,dr} = \frac{1}{3}\int_{2r_1}^{3r_1} e^{-r^2} r\,dr = \frac{e^{-4r_1^2} - e^{-9r_1^2}}{6}.$$

To calibrate, suppose the thrower hits the board ($0 \le r \le r_1$) half the time:

$$\frac{1}{2} = \int_0^{r_1} 2e^{-r^2} r\,dr = 1 - e^{-r_1^2} \quad\Longrightarrow\quad e^{-r_1^2} = \frac{1}{2}.$$

Then $e^{-4r_1^2} = (1/2)^4 = 1/16$ and $e^{-9r_1^2} = (1/2)^9 \approx 0$, so

$$P \approx \frac{1}{16}\cdot\frac{1}{6} = \frac{1}{96} \approx \frac{1}{100}:$$

about a 1% chance per dart thrown.

23.4 The Gaussian integral

Theorem 23.2 [Gaussian integral].

$$Q = \int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Proof. Compute the volume V under the surface $z = e^{-r^2}$, where $r = \sqrt{x^2+y^2}$, in two ways. *Shells.* The cylinder of radius r under the surface has height e^{-r^2}, circumference $2\pi r$, and thickness dr, so $dV = e^{-r^2}2\pi r\,dr$. As in the dart problem,

$$\int_0^R e^{-r^2}2\pi r\,dr = -\pi e^{-r^2}\Big|_0^R = \pi - \pi e^{-R^2}, \qquad V = \lim_{R\to\infty}\big(\pi - \pi e^{-R^2}\big) = \pi.$$

Slices. Slice by planes with y fixed. If $A(y)$ is the area of the cross-section and the thickness is dy, then $V = \int_{-\infty}^{\infty} A(y)\,dy$. On the slice, y is constant and $e^{-r^2} = e^{-x^2-y^2} = e^{-y^2}e^{-x^2}$, so

$$A(y) = \int_{-\infty}^{\infty} e^{-y^2}e^{-x^2}\,dx = e^{-y^2}Q, \qquad V = \int_{-\infty}^{\infty} e^{-y^2}Q\,dy = Q\int_{-\infty}^{\infty} e^{-y^2}\,dy = Q^2,$$

because the name of the integration variable does not matter. Therefore $Q^2 = V = \pi$, and $Q > 0$ gives $Q = \sqrt{\pi}$. □

Equivalently $\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2}\, dx = 1$. Replacing x by $x/(\sqrt{2}\,\sigma)$ gives the rescaled form

$$\frac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2}\, dx = 1,$$

which is central to probability and statistics: the density $\frac{1}{\sqrt{2\pi}\,\sigma} e^{-x^2/2\sigma^2}$ on $-\infty < x < \infty$ is the NORMAL DISTRIBUTION, and $\sigma > 0$ is its standard deviation.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 23, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 24: Numerical integration

Numerical integration evaluates definite integrals whose integrands are messy, or have no elementary antiderivative, such as

$$\int_0^x \cos(t^2)\,dt \qquad \text{or} \qquad \int_0^3 e^{-x^2}\,dx.$$

It produces numbers rather than formulas. Three methods: Riemann sums, the trapezoidal rule, and Simpson's rule.

Throughout, divide $[a, b]$ into n equal intervals,

$$a = x_0 < x_1 < \cdots < x_n = b, \qquad x_i - x_{i-1} = \Delta x = \frac{b-a}{n}, \qquad y_i = f(x_i).$$

24.1 Riemann sums and the trapezoidal rule

The left-endpoint Riemann sum uses rectangles whose heights are the values at the left ends of the intervals:

$$\int_a^b f(x)\,dx \approx (y_0 + y_1 + \cdots + y_{n-1})\,\Delta x.$$

The trapezoidal rule replaces the graph on each interval by the chord joining its endpoints. A trapezoid with parallel sides y_{i-1} and y_i and width Δx has area $\frac{y_{i-1}+y_i}{2}\Delta x$, and summing,

$$\int_a^b f(x)\,dx \approx \Delta x\Big(\frac{y_0+y_1}{2} + \frac{y_1+y_2}{2} + \cdots + \frac{y_{n-1}+y_n}{2}\Big) = \Delta x\Big(\frac{y_0}{2} + y_1 + \cdots + y_{n-1} + \frac{y_n}{2}\Big).$$

This is the average of the left and right Riemann sums, so it treats the two ends a and b symmetrically.

24.2 Simpson's rule

Simpson's rule matches parabolas instead of lines to the graph, two intervals at a time, so it needs n even. It is often much more accurate than the trapezoidal rule.

Lemma 24.1 [Area under a parabola]. *If p is a polynomial of degree at most 2 and x_1 is the midpoint of $[x_0, x_2]$ with $x_2 - x_1 = x_1 - x_0 = \Delta x$, then*

$$\int_{x_0}^{x_2} p(x)\,dx = (2\Delta x)\,\frac{y_0 + 4y_1 + y_2}{6}, \qquad y_i = p(x_i).$$

That is, the area is the base times a weighted average height.

Proof. Translate so that $x_1 = 0$, write $h = \Delta x$, and $p(x) = A + Bx + Cx^2$. Then $\int_{-h}^{h} p\,dx = 2Ah + \frac{2}{3}Ch^3$, while $y_0 + 4y_1 + y_2 = (A - Bh + Ch^2) + 4A + (A + Bh + Ch^2) = 6A + 2Ch^2$, and $\frac{2h}{6}(6A + 2Ch^2) = 2Ah + \frac{2}{3}Ch^3$. □

Applying the lemma to the parabola through the three points of each pair of intervals and adding,

$$\int_a^b f(x)\,dx \approx \frac{2\Delta x}{6}\big[(y_0 + 4y_1 + y_2) + (y_2 + 4y_3 + y_4) + \cdots + (y_{n-2} + 4y_{n-1} + y_n)\big].$$

Each interior even-indexed point is shared by two consecutive pairs, so the coefficients add up as

$$\begin{array}{ccccccc} 1 & 4 & 1 & & & & \\ & & 1 & 4 & 1 & & \\ & & & & 1 & 4 & 1 \\ \hline 1 & 4 & 2 & 4 & 2 & 4 & 1 \end{array}$$

giving the rule, for n even,

$$\boxed{\int_a^b f(x)\,dx \approx \frac{\Delta x}{3}\big(y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \cdots + 2y_{n-2} + 4y_{n-1} + y_n\big).}$$

As a check, $f = 1$ should give $b - a = n\Delta x$. The coefficients are two 1s, $n/2$ fours and $n/2 - 1$ twos, so the rule gives $\frac{\Delta x}{3}\big(2 + 4 \cdot \frac{n}{2} + 2(\frac{n}{2} - 1)\big) = n\Delta x$.

24.3 An example

Example: $\int_0^1 \frac{dx}{1+x^2}$ *with* $\Delta x = \frac{1}{2}$. The values of $f(x) = 1/(1 + x^2)$ at $x = 0, \frac{1}{2}, 1$ are $y_0 = 1$, $y_1 = \frac{4}{5}$, $y_2 = \frac{1}{2}$. The trapezoidal rule gives

$$\Delta x\Big(\frac{1}{2}y_0 + y_1 + \frac{1}{2}y_2\Big) = \frac{1}{2}\Big(\frac{1}{2} + \frac{4}{5} + \frac{1}{4}\Big) = 0.775,$$

and Simpson's rule gives

$$\frac{\Delta x}{3}(y_0 + 4y_1 + y_2) = \frac{1/2}{3}\Big(1 + \frac{16}{5} + \frac{1}{2}\Big) = 0.78333\ldots$$

The exact value is

$$\int_0^1 \frac{dx}{1 + x^2} = \tan^{-1} x\Big|_0^1 = \frac{\pi}{4} \approx 0.785.$$

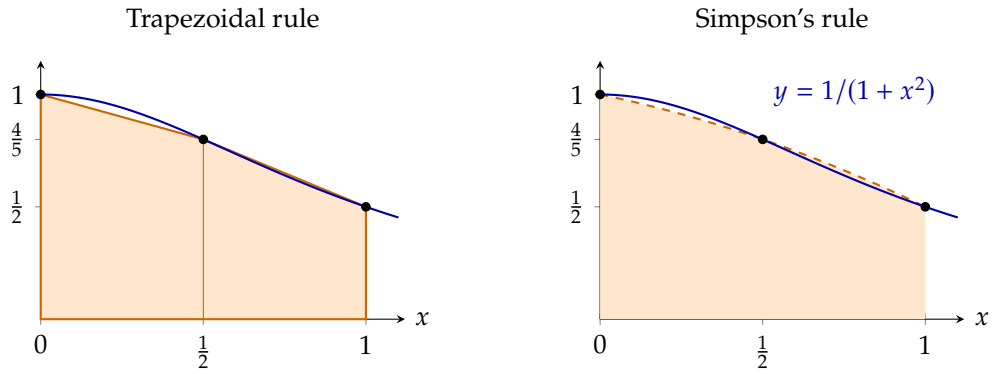


Figure 31: $f(x) = 1/(1+x^2)$ on $[0,1]$ with $\Delta x = \frac{1}{2}$. Left: the trapezoidal rule uses the two chords. Right: Simpson's rule uses the parabola $1 - 0.3x - 0.2x^2$ (dashed) through the three sample points, whose area is $0.78333\ldots$ (both curves drawn exactly).

Roughly speaking, the error |Simpson − exact| has order of magnitude $(\Delta x)^4$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 24, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 25: Exam 3 review: work and applications

25.1 Setting up volume problems

Volume problems are of two kinds: solids of revolution, and other solids, which are computed by slices, $V = \int A(y)\,dy$ with $A(y)$ the area of a cross-section. In either case there is a plane picture that shows what is going on. For a solid of revolution, the main decision is how to chop it up, into shells or disks; equivalently, whether to integrate with respect to x or y. After that choice the rest is determined. For a region rotated about the y-axis:

- *Shells*: height $y_2 - y_1$, circumference $2\pi x$, thickness dx, so $dV = (y_2 - y_1)\,2\pi x\,dx$.
- *Disks or washers*: area πx^2, or $\pi x_2^2 - \pi x_1^2$ for a washer with a hole, and thickness dy, so one integrates with respect to y.

25.2 Work

For a constant force acting along the direction of motion, work is force times distance. When the force varies, the work is the integral of force with respect to distance, $W = \int F\,ds$.

Example: *Work done by gravity on a pendulum.* A pendulum of length L carries a mass m; gravity exerts the vertical force mg. Find the work gravity does as the pendulum swings from angle θ_0 (measured from the vertical) down to the bottom of its arc.

At angle θ, the component of gravity along the circular path is $F = mg\sin\theta$ (Figure 32), and an arc step has length $ds = L\,d\theta$. The force varies, so

$$W = \int_0^{\theta_0} (mg\sin\theta)(L\,d\theta) = -Lmg\cos\theta\Big|_0^{\theta_0} = -Lmg(\cos\theta_0 - 1) = mg\big[L(1-\cos\theta_0)\big].$$

The quantity $L(1-\cos\theta_0)$ is the vertical drop of the mass, so the work equals what gravity does on a straight fall through the same height. The work depends only on how far down the mass goes, not on the path; in particular a pendulum cannot gain energy from gravity over a swing, so there is no perpetual motion machine.

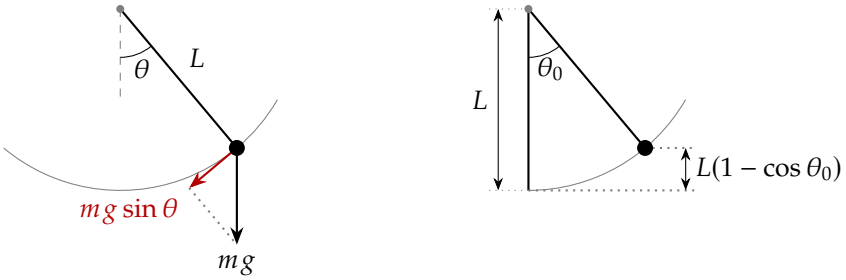


Figure 32: Left: the force mg on the pendulum bob and its component $mg \sin\theta$ along the path. Right: the vertical drop $L(1 - \cos\theta_0)$ from angle θ_0 to the bottom of the arc.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 25, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 26: Trigonometric integrals and substitution

Integrals of products of powers of trigonometric functions reduce, by identities and a substitution, to integrals of polynomials. These integrals in turn evaluate integrals involving square roots by trigonometric substitution.

26.1 Powers of sine and cosine

The goal is $\int \sin^n x \cos^m x\, dx$ for $n, m = 0, 1, 2, \ldots$, starting from $\int \sin x\, dx = -\cos x + c$ and $\int \cos x\, dx = \sin x + c$.

Method A: n or m odd. Split off one factor of the odd power and use $\sin^2 x + \cos^2 x = 1$ to write the rest in terms of the other function; then substitute.

Example: $\int \sin^3 x \cos^2 x\, dx$. Write the integrand as $f(\cos x) \sin x$:

$$\sin^3 x \cos^2 x = (1 - \cos^2 x) \cos^2 x \sin x.$$

With $u = \cos x$, $du = -\sin x\, dx$,

$$\int \sin^3 x \cos^2 x\, dx = \int (1 - u^2) u^2 (-du) = \int (-u^2 + u^4)\, du = -\frac{1}{3}\cos^3 x + \frac{1}{5}\cos^5 x + c.$$

Example: $\int \cos^3 x\, dx$. Here $\cos^3 x = (1 - \sin^2 x)\cos x$, and $u = \sin x$, $du = \cos x\, dx$ give

$$\int \cos^3 x\, dx = \int (1 - u^2)\, du = u - \frac{u^3}{3} + c = \sin x - \frac{\sin^3 x}{3} + c.$$

Method B: n and m both even. Use the double-angle formulas. From $\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$,

$$\cos^2 x = \frac{1 + \cos 2x}{2}, \qquad \sin^2 x = \frac{1 - \cos 2x}{2},$$

so

$$\int \cos^2 x\, dx = \frac{x}{2} + \frac{\sin 2x}{4} + c, \qquad \int \sin^2 x\, dx = \frac{x}{2} - \frac{\sin 2x}{4} + c.$$

In general, apply Method B repeatedly, halving the powers, until a power is odd and Method A applies.

Example: $\int \sin^2 x \cos^2 x\, dx$. Applying Method B twice,

$$\int \frac{1-\cos 2x}{2}\cdot\frac{1+\cos 2x}{2}\,dx = \int\Big(\frac{1}{4}-\frac{1}{4}\cos^2 2x\Big)dx = \int\Big(\frac{1}{4}-\frac{1}{8}(1+\cos 4x)\Big)dx = \frac{x}{8}-\frac{1}{32}\sin 4x + c.$$

A shortcut: $\sin 2x = 2\sin x\cos x$, so $\sin^2 x\cos^2 x = \frac{1}{4}\sin^2 2x = \frac{1}{4}\cdot\frac{1-\cos 4x}{2}$, which integrates to the same answer.

26.2 Powers of secant and tangent

The next family is $\int \sec^n x \tan^m x\, dx$. The basic facts are

$$\sec^2 x = 1+\tan^2 x, \qquad \int \sec^2 x\,dx = \tan x + c, \qquad \int \sec x\tan x\,dx = \sec x + c;$$

the identity follows by dividing $\cos^2 x + \sin^2 x = 1$ by $\cos^2 x$. With $u = \cos x$, $du = -\sin x\,dx$,

$$\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = -\int \frac{du}{u} = -\ln|\cos x| + c.$$

Example: *An even power of secant:* $\int \sec^4 x\,dx$. Write $\sec^4 x = (1+\tan^2 x)\sec^2 x$, a function of $\tan x$ times $\sec^2 x$. With $u = \tan x$, $du = \sec^2 x\,dx$,

$$\int \sec^4 x\,dx = \int (1+u^2)\,du = \tan x + \frac{\tan^3 x}{3} + c.$$

Example: *An odd power of tangent:* $\int \tan^3 x \sec x\,dx$. Write $\tan^3 x\sec x = (\sec^2 x - 1)\sec x\tan x$, a function of $\sec x$ times $d(\sec x)$. With $u = \sec x$, $du = \sec x\tan x\,dx$,

$$\int \tan^3 x\sec x\,dx = \int (u^2-1)\,du = \frac{\sec^3 x}{3} - \sec x + c.$$

Example: *The case* $n = 1$, $m = 0$. Knowing the answer ahead of time, multiply and divide by $\sec x + \tan x$:

$$\int \sec x\,dx = \int \frac{\sec^2 x + \sec x\tan x}{\sec x + \tan x}\,dx.$$

With $u = \sec x + \tan x$, $du = (\sec x\tan x + \sec^2 x)\,dx$, this is $\int du/u$, so

$$\int \sec x\,dx = \ln|\sec x + \tan x| + c.$$

The cases with n odd and m even, such as $\int \sec^3 x\,dx$, are more complicated and are left for later.

26.3 Trigonometric substitution

These trigonometric integrals evaluate integrals involving square roots. The area of the upper half of the disk $x^2 + y^2 \le a^2$ is $\int_{-a}^{a} \sqrt{a^2 - x^2}\, dx = \pi a^2/2$. Consider instead the area under the circle between 0 and x (Figure 33),

$$\int_0^x \sqrt{a^2 - t^2}\, dt, \qquad 0 \le x \le a.$$

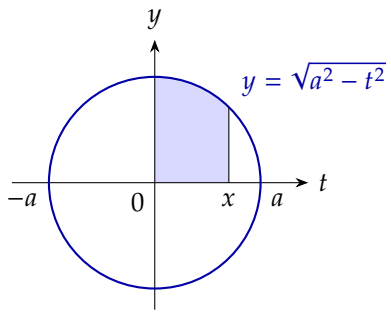


Figure 33: The area $\int_0^x \sqrt{a^2 - t^2}\, dt$ under the circle $t^2 + y^2 = a^2$ between 0 and x.

Substitute $t = a \sin u$, $dt = a \cos u\, du$, with $-\pi/2 \le u \le \pi/2$ so that $\cos u \ge 0$. Then

$$a^2 - t^2 = a^2 - a^2 \sin^2 u = a^2 \cos^2 u, \qquad \sqrt{a^2 - t^2} = a \cos u.$$

Change the limits as well: $t = 0$ gives $u = 0$, and $t = x$ gives $a \sin u = x$, i.e. $u = \sin^{-1}(x/a)$. By Method B,

$$\int_0^x \sqrt{a^2 - t^2}\, dt = a^2 \int_0^{\sin^{-1}(x/a)} \cos^2 u\, du = a^2 \left(\frac{u}{2} + \frac{\sin 2u}{4} \right) \Big|_0^{\sin^{-1}(x/a)}.$$

Using $\sin 2u = 2 \sin u \cos u$ with $\sin u = x/a$ and $\cos u = \sqrt{1 - x^2/a^2} = \sqrt{a^2 - x^2}/a$,

$$\int_0^x \sqrt{a^2 - t^2}\, dt = \frac{a^2}{2} \sin^{-1} \frac{x}{a} + \frac{x}{2} \sqrt{a^2 - x^2}.$$

At $x = a$ this is $\frac{a^2}{2} \cdot \frac{\pi}{2} = \frac{\pi a^2}{4}$, a quarter of the disk, as it should be.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 26, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 28: Integration by inverse substitution; completing the square

In an ordinary substitution the new variable is a function of the old one, $u = g(x)$. In an inverse substitution the old variable is written as a function of the new one, such as $t = a\sin u$. Trigonometric inverse substitutions remove square roots of quadratics, at the price of converting back to the original variable at the end.

28.1 Trigonometric substitution: area under a semicircle

For $-a \le x \le a$, consider the area

$$\int_0^x \sqrt{a^2 - t^2}\, dt$$

under the semicircle $y = \sqrt{a^2 - t^2}$, between $t = 0$ and $t = x$. Substitute

$$t = a\sin u, \qquad dt = a\cos u\, du, \qquad -\tfrac{\pi}{2} \le u \le \tfrac{\pi}{2}.$$

As u runs over this range, t runs from $-a$ to a exactly once, and $\cos u \ge 0$. Hence the square root disappears:

$$a^2 - t^2 = a^2 - a^2\sin^2 u = a^2\cos^2 u \quad \Longrightarrow \quad \sqrt{a^2 - t^2} = a\cos u.$$

Using $\cos^2 u = \frac{1+\cos 2u}{2}$ and $\frac{\sin 2u}{4} = \frac{1}{2}\sin u\cos u$,

$$\int \sqrt{a^2 - t^2}\, dt = a^2 \int \cos^2 u\, du = a^2\Big(\frac{u}{2} + \frac{\sin 2u}{4}\Big) + c = a^2\Big(\frac{u}{2} + \frac{1}{2}\sin u\cos u\Big) + c.$$

The limits change with the variable: $t = 0$ corresponds to $u = 0$, and $t = x$ to $u = \sin^{-1}(x/a)$. Then $\sin u = x/a$, and $\cos u$ is read off the right triangle with hypotenuse a and opposite side x (left of Figure 34): $\cos u = \sqrt{a^2 - x^2}/a$. Therefore

$$\boxed{\int_0^x \sqrt{a^2 - t^2}\, dt = \frac{a^2}{2}\sin^{-1}\Big(\frac{x}{a}\Big) + \frac{1}{2}x\sqrt{a^2 - x^2}}.$$

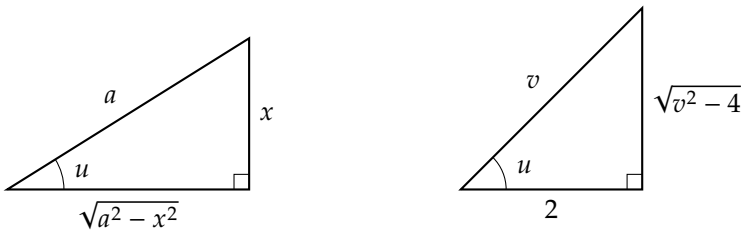


Figure 34: Reference triangles for converting back to the original variable. Left: $\sin u = x/a$, so $\cos u = \sqrt{a^2 - x^2}/a$. Right: $\sec u = v/2$, so $\tan u = \sqrt{v^2 - 4}/2$.

The answer has a geometric check. With $u = \sin^{-1}(x/a)$, the region splits into a circular sector of angle u, measured from the vertical axis, and a right triangle with base x and height $\sqrt{a^2 - x^2}$ (Figure 35). The sector has area $\frac{1}{2}a^2u$ and the triangle has area $\frac{1}{2}x\sqrt{a^2 - x^2}$, the two terms of the formula.

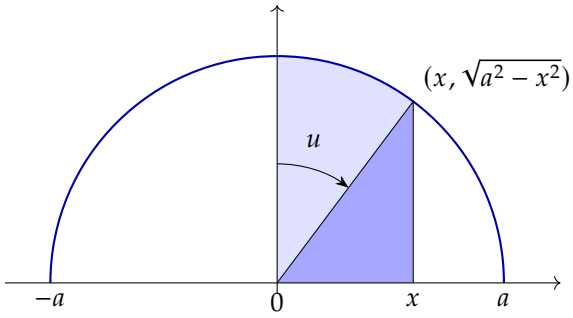


Figure 35: The area under $y = \sqrt{a^2 - t^2}$ from 0 to x is a sector of angle $u = \sin^{-1}(x/a)$ (light) plus a right triangle (dark).

28.2 Standard substitutions

Each square root of a quadratic below is removed by one trigonometric substitution and one identity.

Integral	Substitution	Identity
$\int \frac{dx}{\sqrt{x^2 + 1}}$	$x = \tan u$	$\tan^2 u + 1 = \sec^2 u$
$\int \frac{dx}{\sqrt{x^2 - 1}}$	$x = \sec u$	$\sec^2 u - 1 = \tan^2 u$
$\int \frac{dx}{\sqrt{1 - x^2}}$	$x = \sin u$	$1 - \sin^2 u = \cos^2 u$

28.3 Completing the square

When the quadratic under the root has both a linear and a squared term, first COMPLETE THE SQUARE, writing it as (something)^2 $\pm$ constant.

Example: *The integral* $\int dx/\sqrt{x^2+4x}$. Since $(x+2)^2 = x^2+4x+4$, we have $x^2+4x = (x+2)^2-4$. Put $v = x+2$, $dv = dx$, and then $v = 2\sec u$, $dv = 2\sec u \tan u\, du$. Take $v > 2$ (that is, $x > 0$) and $0 < u < \pi/2$, so $\sqrt{v^2-4} = 2\tan u$. Then

$$\int \frac{dx}{\sqrt{(x+2)^2-4}} = \int \frac{dv}{\sqrt{v^2-4}} = \int \frac{2\sec u \tan u}{2\tan u}\, du = \int \sec u\, du = \ln(\sec u + \tan u) + c.$$

From $\cos u = 2/v$, the right triangle in Figure 34 gives $\sec u = v/2$ and $\tan u = \sqrt{v^2-4}/2$. So

$$\int \frac{dv}{\sqrt{v^2-4}} = \ln\left(\frac{v}{2} + \frac{\sqrt{v^2-4}}{2}\right) + c = \ln\left(v + \sqrt{v^2-4}\right) - \ln 2 + c.$$

Absorbing $-\ln 2$ into a new constant $\tilde{c}$ and returning to x,

$$\int \frac{dx}{\sqrt{x^2+4x}} = \ln\left(x + 2 + \sqrt{x^2+4x}\right) + \tilde{c}.$$

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 28, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 29: Partial fractions

A rational function is a ratio $P(x)/Q(x)$ of polynomials. Some are easy to integrate, for instance

$$\int \Big(\frac{1}{x-1} + \frac{3}{x+2}\Big)dx = \ln|x-1| + 3\ln|x+2| + c.$$

The same integrand can be disguised by combining the fractions:

$$\frac{1}{x-1} + \frac{3}{x+2} = \frac{(x+2)+3(x-1)}{(x-1)(x+2)} = \frac{4x-1}{x^2+x-2}.$$

The goal is a systematic way to reverse this, splitting P/Q into simpler pieces, called partial fractions, that can be integrated one at a time.

29.1 Distinct linear factors and the cover-up method

First factor the denominator, then write

$$\frac{4x-1}{x^2+x-2} = \frac{4x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}.$$

The slow way clears denominators, $4x-1 = A(x+2)+B(x-1)$, and matches coefficients: $A+B=4$ and $2A-B=-1$. With three or more unknowns this system becomes tedious.

The cover-up method is faster. Multiplying both sides by $x-1$ gives

$$\frac{4x-1}{x+2} = A + \frac{B}{x+2}(x-1),$$

and setting $x=1$ kills the B term: $A = \frac{4-1}{1+2} = 1$. In practice, cover up the factor $x-1$ on the left and evaluate the rest at its root. Likewise, covering $x+2$ and setting $x=-2$ gives

$$B = \frac{4(-2)-1}{-2-1} = 3,$$

recovering $\frac{1}{x-1} + \frac{3}{x+2}$.

So far this works when $\deg P < \deg Q$ and Q factors completely into distinct linear factors.

29.2 Repeated linear factors

A repeated factor $(x-1)^2$ contributes one term for each power:

$$\frac{x^2+2}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}.$$

Cover-up works for the highest power. Multiplying by $(x-1)^2$ gives

$$\frac{x^2+2}{x+2} = B + \left[A(x-1) + \frac{C}{x+2}(x-1)^2\right],$$

and letting $x \to 1$ gives B. Covering $x+2$ and setting $x=-2$ gives C:

$$B = \frac{1^2+2}{1+2} = 1, \qquad C = \frac{(-2)^2+2}{(-2-1)^2} = \frac{2}{3}.$$

Cover-up cannot isolate A, because multiplying by $x-1$ leaves the $B/(x-1)$ term infinite at $x=1$. Instead, evaluate both sides at an easy point such as $x=0$:

$$\frac{2}{(-1)^2(2)} = \frac{A}{-1} + 1 + \frac{1}{3} \quad\Longrightarrow\quad 1 = \frac{4}{3} - A \quad\Longrightarrow\quad A = \frac{1}{3}.$$

Therefore

$$\frac{x^2+2}{(x-1)^2(x+2)} = \frac{1}{3(x-1)} + \frac{1}{(x-1)^2} + \frac{2}{3(x+2)}.$$

29.3 Irreducible quadratic factors

Not every polynomial factors into real linear factors. A quadratic factor such as x^2+1 receives a numerator that is linear rather than constant:

$$\frac{1}{(x^2+1)(x-1)} = \frac{A_1}{x-1} + \frac{B_1x + C_1}{x^2+1}.$$

Cover-up at $x=1$ gives $A_1 = \frac{1}{1^2+1} = \frac{1}{2}$. The remaining two unknowns come from evaluating at two other points. At $x=0$,

$$\frac{1}{1\cdot(-1)} = -\frac{1}{2} + \frac{C_1}{1} \quad\Longrightarrow\quad C_1 = -\frac{1}{2}.$$

At $x=-1$,

$$\frac{1}{2(-2)} = \frac{1/2}{-2} + \frac{-B_1 - 1/2}{2} \quad\Longrightarrow\quad 0 = \frac{-B_1 - 1/2}{2} \quad\Longrightarrow\quad B_1 = -\frac{1}{2}.$$

(Clearing denominators and matching coefficients also works.) Each piece now integrates directly; for the middle one substitute $w = x^2 + 1$, $dw = 2x\,dx$:

$$\begin{aligned}\int \frac{dx}{(x^2+1)(x-1)} &= \frac{1}{2}\int \frac{dx}{x-1} - \frac{1}{2}\int \frac{x\,dx}{x^2+1} - \frac{1}{2}\int \frac{dx}{x^2+1} \\ &= \frac{1}{2}\ln|x-1| - \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\tan^{-1} x + c.\end{aligned}$$

29.4 High powers

Powers of a linear factor integrate by the power rule:

$$\int \frac{dx}{(x-1)^{10}} = -\frac{1}{9}(x-1)^{-9} + c.$$

Powers of $x^2 + 1$ need a trigonometric substitution. With $x = \tan u$, $dx = \sec^2 u\,du$, and $\tan^2 u + 1 = \sec^2 u$,

$$\int \frac{dx}{(x^2+1)^{10}} = \int \frac{\sec^2 u\,du}{(\sec^2 u)^{10}} = \int \cos^{18} u\,du,$$

a trigonometric integral of the kind handled in Lecture 26.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 29, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 30: Integration by parts, reduction formulae

30.1 Integration by parts

Rearranging the product rule $(uv)' = u'v + uv'$ as $uv' = (uv)' - u'v$ and integrating gives the basic formula.

Proposition 30.1 [Integration by parts]. *For differentiable u, v,*

$$\int uv'\,dx = uv - \int u'v\,dx, \qquad \int_a^b uv'\,dx = uv\Big|_a^b - \int_a^b u'v\,dx.$$

With $du = u'\,dx$ and $dv = v'\,dx$ the indefinite form reads $\int u\,dv = uv - \int v\,du$.

Proof. Integrate $uv' = (uv)' - u'v$. For the definite form, the fundamental theorem of calculus gives $\int_a^b (uv)'\,dx = uv\big|_a^b$. □

The formula helps when $u'v$ is simpler than uv', often because u' is simpler than u.

Example: *The integral $\int \tan^{-1} x\,dx$.* Write the integrand as $\tan^{-1} x \cdot 1$, with $u = \tan^{-1} x$ and $v' = 1$. Then $u' = \frac{1}{1+x^2}$ and $v = x$, so

$$\int \tan^{-1} x\,dx = x\tan^{-1} x - \int \frac{x}{1+x^2}\,dx = x\tan^{-1} x - \frac{1}{2}\ln(1+x^2) + c.$$

Example: *The integral $\int \ln x\,dx$.* Take $u = \ln x$, $du = dx/x$, and $dv = dx$, $v = x$:

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + c.$$

The same answer comes from *advanced guessing*: the guess $x\ln x$ has derivative $\ln x + 1$, one too many, and subtracting x repairs it, $\frac{d}{dx}(x\ln x - x) = \ln x$.

30.2 Reduction formulae

Integration by parts often lowers an exponent by one, producing a reduction (or recurrence) formula that can be repeated until the exponent reaches zero.

Proposition 30.2 [Two reduction formulae]. *For $n \geq 1$,*

$$\int (\ln x)^n\,dx = x(\ln x)^n - n\int (\ln x)^{n-1}\,dx, \qquad \int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx.$$

Proof. For the first take $u = (\ln x)^n$, so $u' = n(\ln x)^{n-1} \cdot \frac{1}{x}$, and $v' = 1$, $v = x$; the factors x and $\frac{1}{x}$ cancel in $u'v$. For the second take $u = x^n$, $u' = nx^{n-1}$, and $v' = v = e^x$. □

Applying each formula repeatedly, $n \to n-1 \to n-2 \to \cdots$, evaluates the integral completely.

Small changes can make an integral impossible in closed form: neither $\int (\tan^{-1} x)^2\,dx$ nor $\int \frac{e^x}{x}\,dx$ can be expressed by elementary functions. Then an expression such as

$$\int_1^2 \frac{e^x}{x}\,dx$$

is itself the answer; setting up the right integral is what matters, and its value can always be computed numerically. Evaluation by hand remains useful for families of integrals depending on a parameter, such as $F(a) = \int_1^\infty e^{-x}x^{-a}\,dx$, where one wants to know how the value depends on a.

30.3 Arc length

Let s denote arc length along a curve $y = f(x)$, and ds its infinitesimal increment. Zooming in, the curve over a short stretch is nearly the hypotenuse of a right triangle with legs dx and dy (Figure 36), so

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx,$$

by dividing $(ds)^2 = (dx)^2 + (dy)^2$ by $(dx)^2$.

Proposition 30.3 [Arc length of a graph]. *The length of the graph of $y = f(x)$ from $P_0 = (a, f(a))$ to $P_1 = (b, f(b))$ is*

$$\int_{P_0}^{P_1} ds = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx.$$

Example: *An arc of the unit circle.* On the upper half of $x^2 + y^2 = 1$, $y = \sqrt{1-x^2}$ and $\frac{dy}{dx} = \frac{-x}{\sqrt{1-x^2}}$. Then

$$1 + \left(\frac{-x}{\sqrt{1-x^2}}\right)^2 = \frac{1 - x^2 + x^2}{1-x^2} = \frac{1}{1-x^2}, \qquad ds = \frac{dx}{\sqrt{1-x^2}}.$$

For $0 \le a \le 1$, the arc from $x = 0$ to $x = a$ has length

$$s = \int_0^a \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x\Big|_0^a = \sin^{-1} a, \qquad \text{so} \qquad \sin s = a.$$

On the unit circle arc length equals the angle in radians: s is the angle between the vertical axis and the radius to $(a, \sqrt{1-a^2})$, and a is the side opposite that angle in a right triangle with hypotenuse 1 (Figure 36).

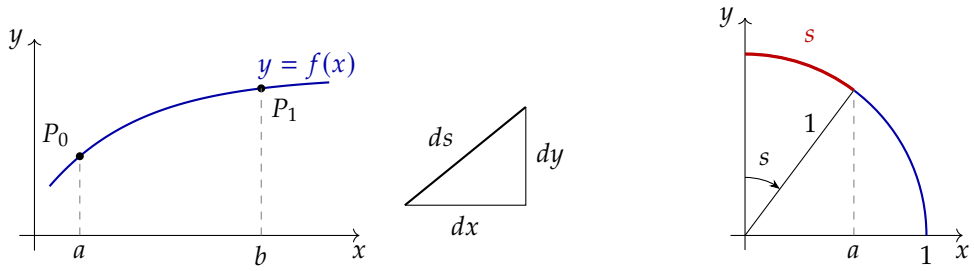


Figure 36: Left: a schematic graph $y = f(x)$ between P_0 and P_1, with the infinitesimal triangle of legs dx, dy and hypotenuse ds enlarged. Right: on the unit circle, the arc from $(0, 1)$ to $(a, \sqrt{1-a^2})$ has length s equal to its angle, and $\sin s = a$.

30.4 Parametric equations

A curve can be described PARAMETRICALLY by giving both coordinates as functions of a parameter t, here with constants such as a fixed and t varying.

Example: *A circle.* Let $x = a\cos t$, $y = a\sin t$. Eliminating t, $x^2 + y^2 = a^2\cos^2 t + a^2\sin^2 t = a^2$, so the point lies on the circle of radius a. The parametrization carries more information than the circle: at $t = 0$ the point is $(a, 0)$ and at $t = \pi/2$ it is $(0, a)$, so for $0 \le t \le \pi/2$ a quarter circle is traced counterclockwise. Its arc length element is

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{(-a\sin t\, dt)^2 + (a\cos t\, dt)^2} = a\, dt.$$

Example: *An ellipse.* Let $x = 2\sin t$, $y = \cos t$. Then $\frac{x^2}{4} + y^2 = \sin^2 t + \cos^2 t = 1$. Now $t = 0$ gives $(0, 1)$ and $t = \pi/2$ gives $(2, 0)$, so the quarter ellipse is traced clockwise (Figure 37).

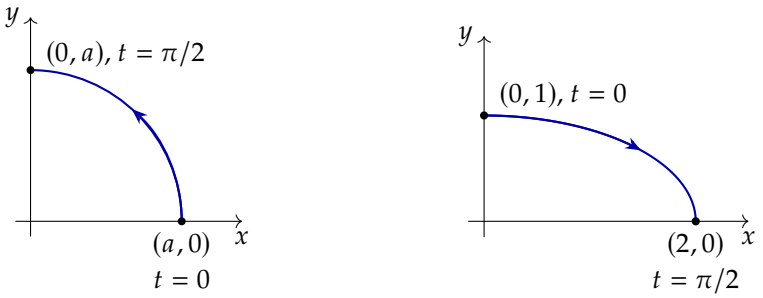


Figure 37: Left: $x = a\cos t$, $y = a\sin t$, $0 \le t \le \pi/2$, traced counterclockwise. Right: $x = 2\sin t$, $y = \cos t$, $0 \le t \le \pi/2$, traced clockwise.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 30, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 31: Parametric equations, arclength, surface area

31.1 Arc length of a parametric curve

For a curve $x = x(t)$, $y = y(t)$, the relation $(ds)^2 = (dx)^2 + (dy)^2$ still holds, now with $dx = x'(t)\,dt$ and $dy = y'(t)\,dt$:

$$ds = \sqrt{x'(t)^2 + y'(t)^2}\,dt.$$

Example: *The curve* $x = t^2$, $y = t^3$, $0 \le t \le 1$. Since $x^3 = t^6 = y^2$, this is the graph $y = x^{3/2}$, $0 \le x \le 1$. Here

$$(ds)^2 = \underbrace{(2t\,dt)^2}_{(dx)^2} + \underbrace{(3t^2\,dt)^2}_{(dy)^2} = (4t^2 + 9t^4)(dt)^2,$$

so, with $t \ge 0$,

$$\text{Length} = \int_{t=0}^{t=1} ds = \int_0^1 \sqrt{4t^2 + 9t^4}\,dt = \int_0^1 t\sqrt{4 + 9t^2}\,dt = \frac{(4 + 9t^2)^{3/2}}{27}\Big|_0^1 = \frac{1}{27}\left(13^{3/2} - 8\right).$$

When an arc length integral has no closed form, it can always be evaluated numerically.

31.2 Surfaces of revolution

Revolve a curve lying above the x-axis around the x-axis. A short piece of the curve of length ds at height y sweeps out a thin band of width ds and circumference $2\pi y$ (Figure 38).

Proposition 31.1 [Surface area of revolution]. *The area of the surface obtained by revolving a curve around the x-axis is*

$$\text{Area} = \int 2\pi y\,ds.$$

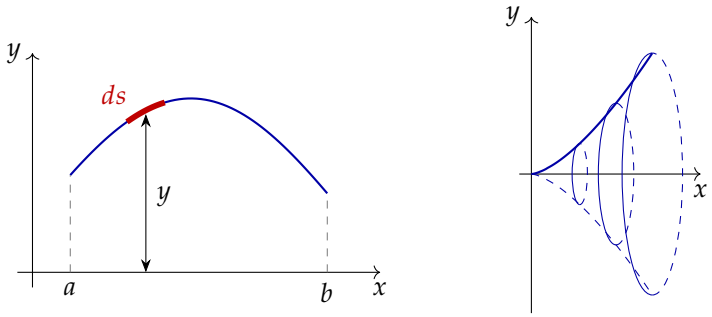


Figure 38: Left: a piece ds of a schematic curve at height y; revolved around the x-axis it sweeps a band of area $2\pi y\,ds$. Right: the trumpet obtained by revolving $y = x^{3/2}$, $0 \le x \le 1$.

Example: *A trumpet.* Revolve $x = t^2$, $y = t^3$, $0 \le t \le 1$, around the x-axis. Then

$$\text{Area} = \int_0^1 2\pi \underbrace{t^3}_{y} \underbrace{t\sqrt{4+9t^2}\,dt}_{ds} = 2\pi \int_0^1 t^4\sqrt{4+9t^2}\,dt.$$

Dropping the factor 2π, substitute $t = \frac{2}{3}\tan u$, $dt = \frac{2}{3}\sec^2 u\,du$. Since $4+9t^2 = 4(1+\tan^2 u) = 4\sec^2 u$,

$$\int t^4(4+9t^2)^{1/2}\,dt = \int \left(\frac{2}{3}\tan u\right)^4 (2\sec u)\left(\frac{2}{3}\sec^2 u\,du\right) = 2\left(\frac{2}{3}\right)^5 \int \tan^4 u \sec^3 u\,du,$$

a tan-sec integral of the kind introduced in Lecture 26. It can be evaluated, but the computation is long.

31.3 The area of a spherical zone

The upper half of the unit circle is $y = \sqrt{1-x^2}$, and by Lecture 30 its arc length element is $ds = dx/\sqrt{1-x^2}$. Revolving the arc over $-1 \le a \le x \le b \le 1$ around the x-axis gives a slice of the spherical surface (the peel only, not the solid), of area

$$A = \int_{x=a}^{x=b} 2\pi y\,ds = \int_a^b 2\pi\sqrt{1-x^2}\,\frac{dx}{\sqrt{1-x^2}} = \int_a^b 2\pi\,dx = 2\pi(b-a).$$

The factors of $\sqrt{1-x^2}$ cancel, so the area depends only on the width $b-a$ of the slice. For the whole sphere, $a = -1$ and $b = 1$ give surface area 4π; for a hemisphere, $a = 0$ and $b = 1$ give 2π.

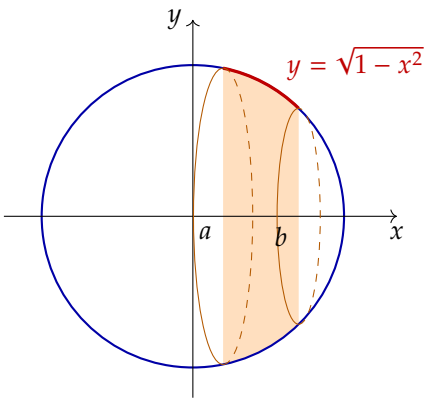


Figure 39: Revolving the arc of the unit circle over $a \le x \le b$ around the x-axis gives a spherical zone of area $2\pi(b-a)$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 31, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 32: Polar coordinates; area in polar coordinates; Exam 4 review

32.1 Polar coordinates

Polar coordinates (r, θ) locate a point by its distance r from the origin and the angle θ that the ray from the origin to the point makes with the positive x-axis. Whatever conventions are used,

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta}.$$

Usually $r \geq 0$ and $0 \leq \theta < 2\pi$, or $r \geq 0$ and $-\pi < \theta \leq \pi$, but every θ and even negative r are allowed: a negative r points backwards, at angle $\theta + \pi$. Conversely, $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1}(y/x)$, with the quadrant of θ chosen to match the signs of x and y.

Example: *The point* $(x, y) = (1, -1)$. Here $r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$ and $\theta = -\pi/4$. The same point is also $r = -\sqrt{2}$, $\theta = 3\pi/4$, since $-\sqrt{2}\cos\frac{3\pi}{4} = 1$ and $-\sqrt{2}\sin\frac{3\pi}{4} = -1$.

32.2 Curves in polar coordinates

To find a polar equation from a Cartesian one, substitute $x = r\cos\theta$, $y = r\sin\theta$ and solve for r. The main issue is then the range of θ that traces the curve exactly once.

Example: *A circle through the origin.* The circle of radius a centered at $(a, 0)$ is $(x - a)^2 + y^2 = a^2$. Substituting,

$$(r\cos\theta - a)^2 + (r\sin\theta)^2 = a^2 \implies r^2 - 2ar\cos\theta = 0 \implies \boxed{r = 2a\cos\theta}.$$

(Dividing by r loses nothing: the origin is still on the curve, at $\theta = \pm\pi/2$.) At $\theta = 0$, $r = 2a$, the point $(2a, 0)$; at $\theta = \pi/4$, $r = a\sqrt{2}$. The range $0 \leq \theta \leq \pi/2$ traces the top half and $-\pi/2 \leq \theta \leq 0$ the bottom half, so $-\pi/2 < \theta < \pi/2$ traces the circle once (Figure 40). The range $\pi/2 < \theta < 3\pi/2$ should be avoided: there $r < 0$ (for example $r = -2a$ at $\theta = \pi$), and the same circle is traced a second time with the radius pointing backwards.

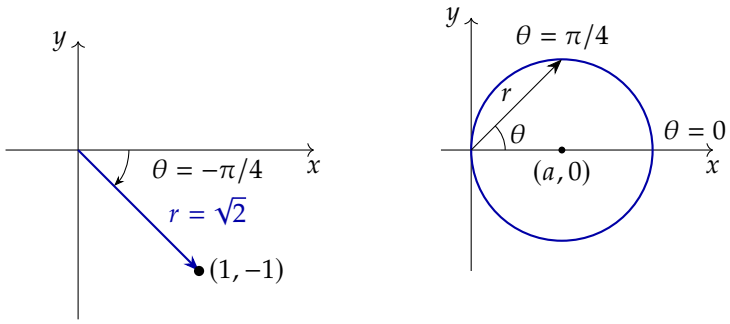


Figure 40: Left: polar coordinates of the point $(1, -1)$. Right: the circle $r = 2a\cos\theta$, $-\pi/2 < \theta < \pi/2$, with the radius at $\theta = \pi/4$.

Simple sets have simple polar descriptions. The circle $x^2 + y^2 = a^2$ is $r = a$, traced once by $0 \le \theta < 2\pi$. The ray at angle b is $\theta = b$, $0 \le r < \infty$, that is, $x = r\cos b$, $y = r\sin b$. The horizontal line $y = 1$ becomes

$$r\sin\theta = 1 \quad \Longrightarrow \quad r = \frac{1}{\sin\theta}, \qquad 0 < \theta < \pi.$$

Example: *From polar back to Cartesian: an ellipse.* Start with

$$r = \frac{1}{1 + \frac{1}{2}\sin\theta}, \qquad \text{so} \qquad r + \frac{r}{2}\sin\theta = 1.$$

With $r = \sqrt{x^2 + y^2}$ and $r\sin\theta = y$,

$$\sqrt{x^2 + y^2} = 1 - \frac{y}{2} \implies x^2 + y^2 = \left(1 - \frac{y}{2}\right)^2 = 1 - y + \frac{y^2}{4} \implies x^2 + \frac{3y^2}{4} + y = 1.$$

This is an ellipse, with the origin at one focus.

Example: *A rose.* The curve $r = \cos 2\theta$ has four petals (Figure 41). The first petal, along the positive x-axis, is traced for $-\pi/4 < \theta < \pi/4$, where $r > 0$. For $\pi/4 < \theta < 3\pi/4$, $r < 0$, so the petal traced then lies in the opposite direction, along the negative y-axis. In all, the petals along the x-axis have $r > 0$ and those along the y-axis have $r < 0$.

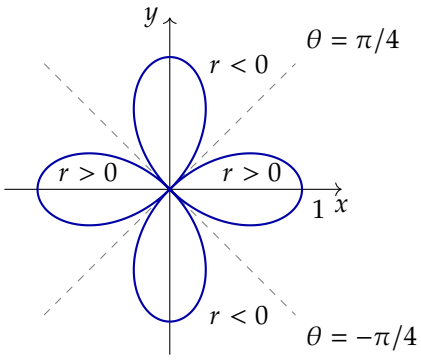


Figure 41: The rose $r = \cos 2\theta$. The petals along the x-axis are traced with $r > 0$, those along the y-axis with $r < 0$; the first petal corresponds to $-\pi/4 < \theta < \pi/4$.

32.3 Area in polar coordinates

For a region bounded by the rays $\theta = \theta_1$, $\theta = \theta_2$ and the curve $r = f(\theta)$, integrate in θ. A thin slice of angle $d\theta$ is nearly a right triangle with base r and height $r\,d\theta$ (Figure 42), so its area is about $\frac{1}{2}r(r\,d\theta)$.

Proposition 32.1 [Area in polar coordinates]. *The region swept by the radius as θ runs from θ_1 to θ_2 along $r = f(\theta)$ has area*

$$\boxed{\int_{\theta_1}^{\theta_2} \frac{1}{2} r^2 \, d\theta}.$$

The range of θ must cover the region once, neither double-counting parts of it nor missing them.

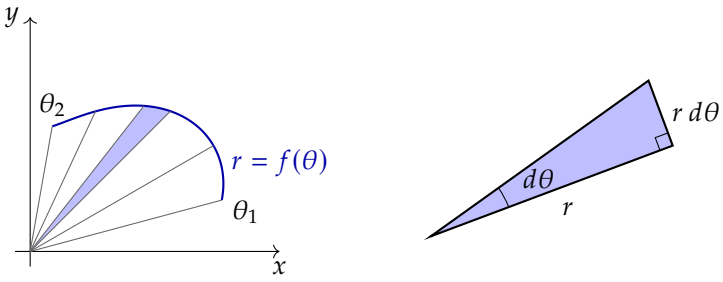


Figure 42: Left: a schematic region under $r = f(\theta)$, cut into thin slices by rays. Right: one slice of angle $d\theta$ is nearly a right triangle with legs r and $r\,d\theta$.

Example: *The circle $r = 2a\cos\theta$, $-\pi/2 < \theta < \pi/2$.* Using $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$,

$$A = \int_{-\pi/2}^{\pi/2} \frac{1}{2}(2a\cos\theta)^2\,d\theta = a^2\int_{-\pi/2}^{\pi/2}(1+\cos 2\theta)\,d\theta = \pi a^2 + \frac{a^2}{2}\sin 2\theta\Big|_{-\pi/2}^{\pi/2} = \pi a^2,$$

since $\sin\pi = \sin(-\pi) = 0$.

Example: *The circle $r = a$.* Here $A = \int_0^{2\pi} \frac{1}{2}a^2\,d\theta = \frac{1}{2}a^2 \cdot 2\pi = \pi a^2$.

32.4 Arc length and surface area in other variables

The arc length element $ds = \sqrt{(dx)^2 + (dy)^2}$ can be written with respect to whichever variable describes the curve:

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\,dy.$$

For example, $x = t^4$, $y = 1 + t$ has $ds = \sqrt{(4t^3)^2 + 1}\,dt$, to be integrated in t. Revolving around the x-axis gives surface area $\int 2\pi y\,ds$, with ds in terms of dx when $y = y(x)$; revolving around the y-axis gives $\int 2\pi x\,ds$, with $ds = \sqrt{(dx/dy)^2 + 1}\,dy$ when $x = x(y)$.

32.5 Integrating rational functions in general

Partial fractions apply to P/Q only when $\deg P < \deg Q$. Otherwise, first use LONG DIVISION: if $\deg P = n \geq m = \deg Q$, then

$$\frac{P(x)}{Q(x)} = P_1(x) + \frac{R(x)}{Q(x)},$$

where the quotient P_1 is a polynomial of degree $n - m$, easy to integrate, and the remainder R has degree at most $m - 1$, so R/Q has a partial fraction decomposition. For example, dividing without factoring the denominator,

$$\frac{x^3}{x^2 - 2x + 1} = x + 2 + \frac{3x - 2}{x^2 - 2x + 1}, \qquad \frac{3x - 2}{(x-1)^2} = \frac{(3x-3)+1}{(x-1)^2} = \frac{3}{x-1} + \frac{1}{(x-1)^2}.$$

Then factor Q completely into real linear and quadratic factors. Each factor $(x - a)^m$ contributes

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \cdots + \frac{A_m}{(x-a)^m},$$

and each irreducible factor $(Ax^2 + Bx + C)^p$ contributes

$$\frac{a_1 x + b_1}{Ax^2 + Bx + C} + \frac{a_2 x + b_2}{(Ax^2 + Bx + C)^2} + \cdots + \frac{a_p x + b_p}{(Ax^2 + Bx + C)^p}.$$

For instance,

$$\frac{3x^2 + 1}{(x-1)(x+2)^2(x^2+9)} = \frac{A}{x-1} + \frac{B_1}{x+2} + \frac{B_2}{(x+2)^2} + \frac{Cx + D}{x^2 + 9}.$$

In practice the hard part is factoring the denominator.

The pieces integrate as follows. For linear factors,

$$\int \frac{dx}{(x-a)^n} = \begin{cases} \ln|x - a| + c, & n = 1, \\ \dfrac{-1}{n-1}(x-a)^{1-n} + c, & n \neq 1. \end{cases}$$

For quadratic factors, complete the square,

$$Ax^2 + Bx + C = A\left(x + \frac{B}{2A}\right)^2 + C - \frac{B^2}{4A},$$

for example $x^2 + 4x + 13 = (x+2)^2 + 9$. For $A > 0$ and an irreducible quadratic, $u = \sqrt{A}\,(x + \frac{B}{2A})$ reduces the pieces to combinations of

$$\int \frac{u\,du}{(u^2 + k^2)^n} \qquad \text{and} \qquad \int \frac{du}{(u^2 + k^2)^n}.$$

The first yields to $w = u^2 + k^2$, $dw = 2u\,du$. The second yields to $u = k\tan\theta$, $du = k\sec^2\theta\,d\theta$,

which leads to sec-tan integrals that are long for large n; a term such as $D/(x^2+9)$ needs only $x = 3\tan u$. Hence every rational function can be integrated in closed form.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 32, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 34: Indeterminate forms: L'Hospital's rule

A limit of a quotient whose numerator and denominator both tend to 0, or both tend to $\pm\infty$, is an INDETERMINATE FORM $0/0$ or ∞/∞: the limits of the pieces do not determine the limit of the quotient. L'Hospital's rule (also spelled L'Hôpital) replaces the quotient by the quotient of derivatives.

34.1 The easy version

Algebra sometimes removes the indeterminacy:

$$\lim_{x\to 1}\frac{x^3-1}{x^2-1}=\lim_{x\to 1}\frac{(x-1)(x^2+x+1)}{(x-1)(x+1)}=\lim_{x\to 1}\frac{x^2+x+1}{x+1}=\frac{3}{2}.$$

A general method comes from difference quotients. Suppose $f(a)=g(a)=0$ and f, g are differentiable at a. Dividing numerator and denominator by $x-a$,

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{\big(f(x)-f(a)\big)/(x-a)}{\big(g(x)-g(a)\big)/(x-a)}=\frac{f'(a)}{g'(a)},$$

provided $g'(a)\neq 0$, since the two difference quotients tend to $f'(a)$ and $g'(a)$.

Theorem 34.1 [L'Hospital's rule, easy version**].** *If $f(a)=g(a)=0$, f and g are differentiable at a, and $g'(a)\neq 0$, then* $\lim_{x\to a}\dfrac{f(x)}{g(x)}=\dfrac{f'(a)}{g'(a)}$.

For the example above, $f(x)=x^3-1$ and $g(x)=x^2-1$ give $f'(1)/g'(1)=3/2$. Every derivative $\lim_{\Delta x\to 0}\Delta y/\Delta x$ is itself a $0/0$ limit.

34.2 The full rule

Theorem 34.2 [L'Hospital's rule**].** *Let a be a real number or $\pm\infty$, and suppose that as $x\to a$ either $f(x)\to 0$ and $g(x)\to 0$, or $f(x)\to\pm\infty$ and $g(x)\to\pm\infty$. If f and g are differentiable near a, with $g'\neq 0$ there, and $\lim_{x\to a} f'(x)/g'(x)$ exists (finite or infinite), then*

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

The rule may be applied repeatedly, as long as each new quotient is still of the form $0/0$ or ∞/∞.

Example: *Comparison with linear approximation.*

$$\lim_{x\to 1}\frac{x^{15}-1}{x^3-1}=\lim_{x\to 1}\frac{15x^{14}}{3x^2}=\frac{15}{3}=5.$$

Linear approximation near $x=1$ gives the same answer: $x^{15}-1\approx 15(x-1)$ and $x^3-1\approx 3(x-1)$, so the quotient is approximately $15(x-1)/\big(3(x-1)\big)=5$.

Example: *Trigonometric limits.*

$$\lim_{x\to 0}\frac{\sin 3x}{x}=\lim_{x\to 0}\frac{3\cos 3x}{1}=3,\qquad \lim_{x\to \pi/4}\frac{\sin x-\cos x}{x-\pi/4}=\lim_{x\to \pi/4}\frac{\cos x+\sin x}{1}=\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}=\sqrt{2}.$$

Both are derivatives in disguise: the first is $\frac{d}{dx}\sin 3x$ at $x=0$, and the second is $f'(\pi/4)$ for $f(x)=\sin x-\cos x$. Similarly,

$$\lim_{x\to 0}\frac{\cos x-1}{x}=\lim_{x\to 0}\frac{-\sin x}{1}=0,\qquad \lim_{x\to 0}\frac{\cos x-1}{x^2}=\lim_{x\to 0}\frac{-\sin x}{2x}=\lim_{x\to 0}\frac{-\cos x}{2}=-\frac{1}{2},$$

where the second limit applies the rule twice. The quadratic approximation $\cos x\approx 1-\frac{1}{2}x^2$ confirms it: $(\cos x-1)/x^2\approx(-\frac{1}{2}x^2)/x^2=-\frac{1}{2}$.

Example: *Look before you L'Hôp.* One application of the rule gives

$$\lim_{x\to 0}\frac{\sin x}{x^2}=\lim_{x\to 0}\frac{\cos x}{2x}.$$

Applying the rule again would give $\lim_{x\to 0}(-\sin x)/2=0$, which is wrong: linear approximation gives $\sin x/x^2\approx x/x^2=1/x\to\infty$ as $x\to 0^+$. The error is the second step. The quotient $\cos x/(2x)$ has the form $1/0$, not $0/0$, so the rule does not apply to it; the limit is infinite ($+\infty$ from the right, $-\infty$ from the left).

34.3 Rates of growth

The rule also applies to ∞/∞ forms and to limits as $x\to\pm\infty$, which compares how fast functions grow. For $a>0$,

$$\lim_{x\to\infty}\frac{e^{ax}}{x}=\lim_{x\to\infty}\frac{ae^{ax}}{1}=+\infty.$$

Ten applications give

$$\lim_{x\to\infty}\frac{e^{ax}}{x^{10}}=\lim_{x\to\infty}\frac{ae^{ax}}{10x^9}=\lim_{x\to\infty}\frac{a^2e^{ax}}{10\cdot 9\,x^8}=\cdots=\lim_{x\to\infty}\frac{a^{10}e^{ax}}{10!}=\infty.$$

It is quicker to write $e^{ax}/x^{10}=\big(e^{ax/10}/x\big)^{10}$; the inner quotient tends to ∞ by the first limit with $a/10$ in place of a, so the tenth power does too. At the slow end,

$$\lim_{x\to\infty}\frac{\ln x}{x^{1/3}}=\lim_{x\to\infty}\frac{1/x}{\frac{1}{3}x^{-2/3}}=\lim_{x\to\infty}3x^{-1/3}=0.$$

Together these give the ordering, as $x \to \infty$ with $a > 0$,

$$\ln x \ll x^{1/3} \ll x \ll x^{10} \ll e^{ax},$$

where $f \ll g$ means $f(x)/g(x) \to 0$.

34.4 Other indeterminate forms

The forms $0 \cdot \infty$, 1^∞, and 0^0 are also indeterminate. Algebra, exponentials, and logarithms convert them to $0/0$ or ∞/∞.

Example: *The limit of x^x as $x \to 0^+$.* Since the exponent varies, write $x^x = e^{x \ln x}$ and first find the limit of the exponent, a $0 \cdot \infty$ form. Writing it as $x/(1/\ln x)$ gives a $0/0$ form whose derivative quotient is no easier. Writing it as an ∞/∞ form works:

$$\lim_{x \to 0^+} x \ln x = \lim_{x \to 0^+} \frac{\ln x}{1/x} = \lim_{x \to 0^+} \frac{1/x}{-1/x^2} = \lim_{x \to 0^+} (-x) = 0.$$

By continuity of the exponential, $\lim_{x \to 0^+} x^x = e^{\lim x \ln x} = e^0 = 1$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 34, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 35: Improper integrals

35.1 Integrals over infinite intervals

Definition 35.1 [Improper integral]. The improper integral of f over $[a, \infty)$ is

$$\int_a^\infty f(x)\,dx = \lim_{M\to\infty} \int_a^M f(x)\,dx.$$

It converges if the limit exists and is finite, and diverges otherwise.

Example: *Exponential decay.* For $k > 0$,

$$\int_0^M e^{-kx}\,dx = -\frac{1}{k}e^{-kx}\Big|_0^M = \frac{1}{k}\left(1 - e^{-kM}\right) \longrightarrow \frac{1}{k} \qquad (M \to \infty),$$

since $e^{-kM} \to 0$. Informally, $\int_0^\infty e^{-kx}\,dx = -\frac{1}{k}e^{-kx}\Big|_0^\infty = \frac{1}{k}(1 - e^{-k\infty}) = \frac{1}{k}$. The infinite integral has a simpler formula than the finite one, and for large M the term e^{-kM} is negligible, so $1/k$ also approximates the finite integral well. Infinite integrals are often easier than finite ones, just as derivatives are easier than difference quotients.

Example: *Radioactive decay.* Let t be time in seconds and let R be the number of atoms decaying per second at time 0. At time t the decay rate is Re^{-kt}. Every atom eventually decays, so the total number of atoms is

$$N = \int_0^\infty Re^{-kt}\,dt = \frac{R}{k}.$$

The half-life H is the time at which the decay rate has halved: $e^{-kH} = \frac{1}{2}$, so $k = (\ln 2)/H$ and $R = Nk = N(\ln 2)/H$. For polonium 210, $H = 138$ days $= 138 \cdot 24 \cdot 60^2$ seconds. One gram contains about $6 \times 10^{23}/210$ atoms, so it emits about 1.66×10^{14} decays per second (about 4500 curies). At 5.3 MeV per decay this is about 140 watts per gram.

Example: *Arctangent.*

$$\int_0^M \frac{dx}{1+x^2} = \tan^{-1} x\Big|_0^M = \tan^{-1} M \longrightarrow \frac{\pi}{2},$$

because $\theta = \tan^{-1} M$ is the angle in $(-\pi/2, \pi/2)$ with $\tan\theta = M$, and $\theta \to \pi/2$ as $M \to \infty$ (Figure 43). So $\int_0^\infty dx/(1+x^2) = \pi/2$.

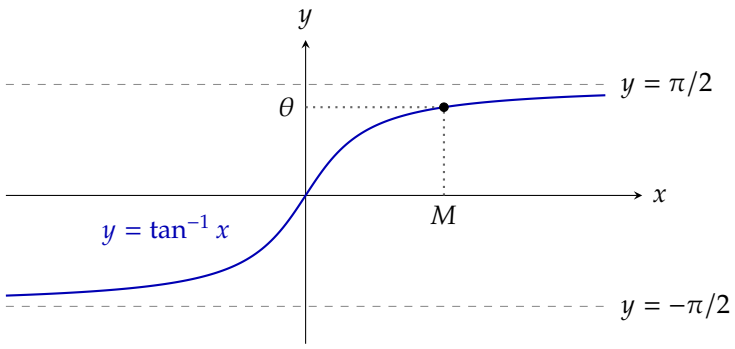


Figure 43: The graph of $\theta = \tan^{-1} M$ approaches the horizontal asymptote $\theta = \pi/2$ as $M \to \infty$.

Example: *The Gaussian.* $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$, found earlier in the course by computing a volume in two ways (slices and shells). The finite integral $\int_0^M e^{-x^2}\,dx$ has no elementary formula and can only be evaluated numerically, so here too the infinite integral is simpler.

35.2 Powers

For $p \neq 1$,

$$\int_1^M \frac{dx}{x^p} = \frac{x^{1-p}}{1-p}\Big|_1^M = \frac{M^{1-p}-1}{1-p}, \qquad \int_1^M \frac{dx}{x} = \ln M.$$

As $M \to \infty$, $\ln M \to \infty$; $M^{1-p} \to 0$ when $p > 1$ and $M^{1-p} \to \infty$ when $p < 1$. Hence

$$\int_1^\infty \frac{dx}{x^p} = \frac{1}{p-1} \quad \text{for } p > 1, \qquad \int_1^\infty \frac{dx}{x^p} \text{ diverges for } 0 < p \leq 1.$$

35.3 Comparison

Convergence can often be decided without evaluating the integral, by comparing with a simpler integral that can be evaluated.

Theorem 35.2 [Comparison]. *Suppose $0 \leq f(x) \leq g(x)$ for $x \geq a$. If $\int_a^\infty g\,dx$ converges, so does $\int_a^\infty f\,dx$. Equivalently, if $\int_a^\infty f\,dx$ diverges, so does $\int_a^\infty g\,dx$.*

Proof. Since $f \geq 0$, $F(M) = \int_a^M f\,dx$ increases with M, so it has a limit (possibly $+\infty$). The area under f is at most the area under g: $F(M) \leq \int_a^M g\,dx \leq \int_a^\infty g\,dx$. A bounded increasing function has a finite limit. □

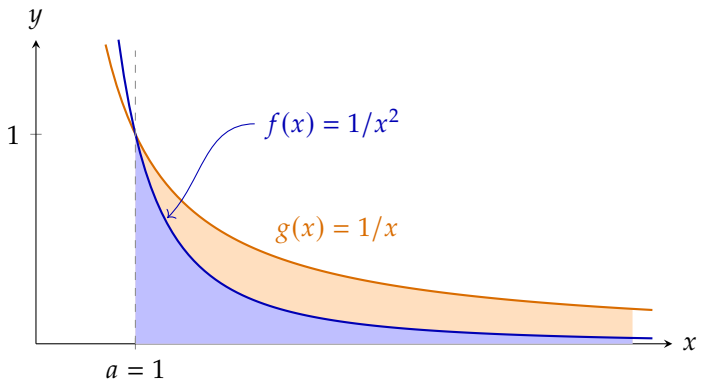


Figure 44: Comparison with $f(x) = 1/x^2 \le g(x) = 1/x$ for $x \ge 1$: the area under f (blue) lies inside the area under g (blue and orange). Here the area under f is finite and the area under g is infinite.

To show convergence, trap the integrand from above by a function with a finite integral; to show divergence, trap it from below by one with an infinite integral.

Example: *Excluding a bad interval.* For $\int_0^\infty dx/\sqrt{x^3+1}$, the natural bound $1/\sqrt{x^3+1} \le x^{-3/2}$ fails over $(0,\infty)$, because $\int_0^\infty x^{-3/2}\,dx$ is infinite owing to its behavior near $x = 0$. Split instead:

$$\int_0^\infty \frac{dx}{\sqrt{x^3+1}} = \int_0^1 \frac{dx}{\sqrt{x^3+1}} + \int_1^\infty \frac{dx}{\sqrt{x^3+1}}.$$

The first integral is an ordinary integral of a continuous function, and the second is at most $\int_1^\infty x^{-3/2}\,dx < \infty$ since $3/2 > 1$. So the integral converges.

Example: *A faster decay.* For $x \ge 1$, $x^3 \ge x$, so

$$\int_1^\infty e^{-x^3}\,dx \le \int_1^\infty e^{-x}\,dx = e^{-1} < \infty.$$

The interval $[0,1]$ has finite length and e^{-x^3} is bounded there, so $\int_0^\infty e^{-x^3}\,dx$ converges.

Proposition 35.3 [Limit comparison]. *Let $f, g > 0$ and suppose $f(x)/g(x) \to L$ as $x \to \infty$, with $0 < L < \infty$. Then $\int_a^\infty f\,dx$ and $\int_a^\infty g\,dx$ either both converge or both diverge.*

Proof. For x beyond some b, $\frac{1}{2}L\,g(x) \le f(x) \le 2L\,g(x)$. The integrals over $[a,b]$ are finite, and Theorem 35.2 applies on $[b,\infty)$ in each direction. □

We write $f \sim g$ in this situation; only the behavior for large x matters.

Example: *A divergent integral.* As $x \to \infty$,

$$\frac{x+10}{x^2+1} \sim \frac{x}{x^2} = \frac{1}{x},$$

and $\int_1^\infty dx/x = \infty$, so $\int_0^\infty (x+10)\,dx/(x^2+1)$ diverges.

Example: *Polynomial times exponential.* $\int_0^\infty x^n e^{-x}\,dx$ converges. The comparison $x^n \ll e^x$ is not enough, but $x^n/e^{x/2} \to 0$ by L'Hospital's rule, so $x^n \le e^{x/2}$ for large x and

$$x^n e^{-x} \le e^{x/2} e^{-x} = e^{-x/2}.$$

Since $\int_0^\infty e^{-x/2}\,dx = 2$ converges, so does the integral.

35.4 Improper integrals of the second type

An integrand that becomes infinite at an endpoint also gives an improper integral, defined by a limit at that endpoint. For example $1/\sqrt{x} \to \infty$ as $x \to 0^+$, and

$$\int_0^1 \frac{dx}{\sqrt{x}} = \lim_{a\to 0^+} \int_a^1 x^{-1/2}\,dx = \lim_{a\to 0^+} \left(2 - 2a^{1/2}\right) = 2.$$

In general, for $p < 1$,

$$\int_0^1 x^{-p}\,dx = \lim_{a\to 0^+} \frac{1-a^{1-p}}{1-p} = \frac{1}{1-p},$$

which gives 2 for $p = \frac{1}{2}$. For $p \ge 1$ the integral diverges: for $p = 1$ it is $\lim_{a\to 0^+}(-\ln a) = \infty$, and for $p > 1$ the term a^{1-p} tends to ∞. The behavior is the reverse of $\int_1^\infty x^{-p}\,dx$; in particular $\int_0^\infty x^{-p}\,dx$ diverges for every p.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 35, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 36: Infinite series and convergence tests

36.1 Geometric series and convergence

To sum the geometric series $S = 1 + a + a^2 + a^3 + \cdots$, multiply by a to get $aS = a + a^2 + a^3 + \cdots$ and subtract:

$$S - aS = 1 \quad \Longrightarrow \quad S = \frac{1}{1-a}.$$

This is valid only for $|a| < 1$. For $a = 1$ the series is $1 + 1 + 1 + \cdots = \infty$, and the subtraction reads $\infty - \infty = 1$, which is meaningless. For $a = -1$ the formula would give $1 - 1 + 1 - 1 + \cdots = \frac{1}{2}$, but the series has no sum.

Definition 36.1 [Convergence of a series]. The nth partial sum of $\sum_{k=0}^{\infty} a_k$ is $S_n = \sum_{k=0}^{n} a_k = a_0 + \cdots + a_n$. The statement $\sum_{k=0}^{\infty} a_k = s$ means $\lim_{n\to\infty} S_n = s$; the series converges to s if this limit exists and is finite, and diverges otherwise.

For the geometric series, $S_n = (1 - a^{n+1})/(1 - a)$ for $a \neq 1$, which tends to $1/(1-a)$ exactly when $|a| < 1$; for $a = -1$ the partial sums alternate between 1 and 0.

36.2 Series and integrals

Whether $\sum_{n\geq 1} 1/n^p$ converges can be decided by comparing it with $\int_1^{\infty} dx/x^p$, written

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \sim \int_1^{\infty} \frac{dx}{x^p}.$$

This decides convergence only, not the value of the sum, which is a much harder problem. For example, $\sum_{n\geq 1} 1/n^2 = \pi^2/6$ takes a lot of work, and $\sum_{n\geq 1} 1/n^3$ was only fairly recently shown to be irrational.

Example: *The harmonic series.* Let $s_N = 1 + \frac{1}{2} + \cdots + \frac{1}{N}$. On each interval $[n, n+1]$, the function $1/x$ lies between $1/(n+1)$ and $1/n$ (Figure 45). The upper Riemann sum gives

$$\ln N = \int_1^N \frac{dx}{x} \leq 1 + \frac{1}{2} + \cdots + \frac{1}{N-1} = s_{N-1} < s_N,$$

so $s_N \to \infty$: the harmonic series $\sum 1/n$ diverges. The lower Riemann sum gives

$$s_N = 1 + \sum_{n=2}^{N} \frac{1}{n} \leq 1 + \int_1^N \frac{dx}{x} = 1 + \ln N.$$

Hence $\ln N < s_N \le 1 + \ln N$, so s_N tends to ∞ very slowly.

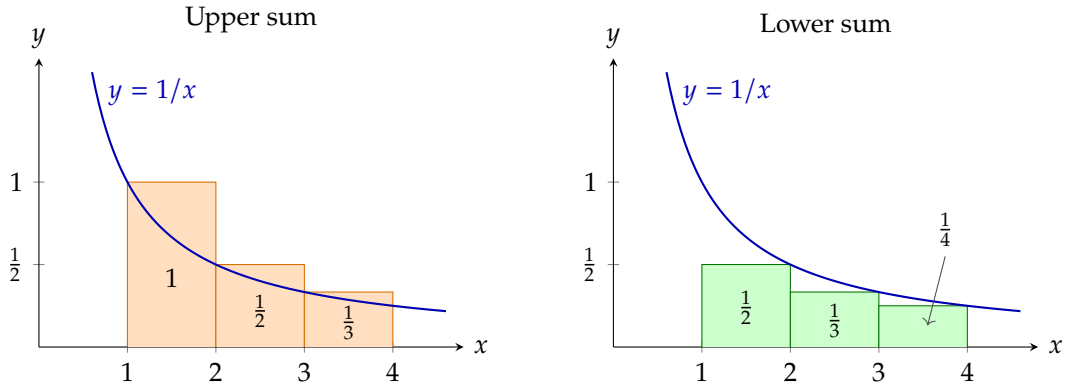


Figure 45: Upper and lower Riemann sums for $y = 1/x$ on unit intervals. The upper rectangles have heights $1, \frac{1}{2}, \frac{1}{3}$ and the lower ones $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$.

Theorem 36.2 [Integral comparison]. *Let f be positive and decreasing on $[1, \infty)$. Then*

$$0 \le \sum_{n=1}^{N} f(n) - \int_1^N f(x)\,dx \le f(1) \quad \text{for every } N,$$

so $\sum_{n=1}^{\infty} f(n)$ and $\int_1^{\infty} f(x)\,dx$ either both converge or both diverge.

Proof. On $[n, n+1]$, $f(n+1) \le f(x) \le f(n)$. Summing the integrals over $n = 1, \dots, N-1$ gives

$$\sum_{n=2}^{N} f(n) \le \int_1^N f(x)\,dx \le \sum_{n=1}^{N-1} f(n),$$

the lower and upper Riemann sums of the example above. The left inequality gives the upper bound $f(1)$ on the difference, and the right inequality together with $f(N) > 0$ gives the lower bound 0. Both the partial sums and $\int_1^N f$ increase with N, and they differ by a bounded amount, so they are both bounded or both unbounded. □

Corollary 36.3 [p-series]. *$\sum_{n \ge 1} 1/n^p$ diverges for $p \le 1$ and converges for $p > 1$.*

This follows from Theorem 36.2 and the corresponding facts for $\int_1^{\infty} dx/x^p$ (Lecture 35) when $p > 0$; for $p \le 0$ the terms do not even tend to 0.

Proposition 36.4 [Limit comparison]. *Let $f, g > 0$, and write $f(x) \sim g(x)$ when $\lim_{x \to \infty} f(x)/g(x) = c$ for some c with $0 < c < \infty$. If $f \sim g$, then $\sum f(n)$ and $\sum g(n)$ either both converge or both diverge.*

Proof. For n beyond some n_0, $\frac{c}{2}g(n) \le f(n) \le 2c\,g(n)$. Finitely many terms do not affect convergence, and a series of positive terms converges exactly when its partial sums are bounded, so each series is controlled by the other. □

This leaves a lot of room in comparisons. For example,

$$\frac{1}{\sqrt{n^2+10}} \sim \frac{1}{(n^2)^{1/2}} = \frac{1}{n}, \qquad \frac{n}{\sqrt{n^5-10}} \sim \frac{n}{n^{5/2}} = \frac{1}{n^{3/2}},$$

so $\sum_{n\ge1} 1/\sqrt{n^2+10}$ diverges, and $\sum_{n\ge2} n/\sqrt{n^5-10}$ converges since $3/2 > 1$.

36.3 Stacking blocks

Can identical blocks be stacked so that the top block lies farther out than the bottom block, and how far out can it go? A stack is stable when, for each block, the combined center of mass of all blocks above it lies over that block. Build from the top down, measuring horizontally in units of half a block length, so each block has length 2. Let $C_0 = 0$ be the left end of the top block and C_n the center of mass of the top n blocks. The strategy is to put the left end of block $n+1$ directly under C_n, as far right as stability allows (Figure 46).

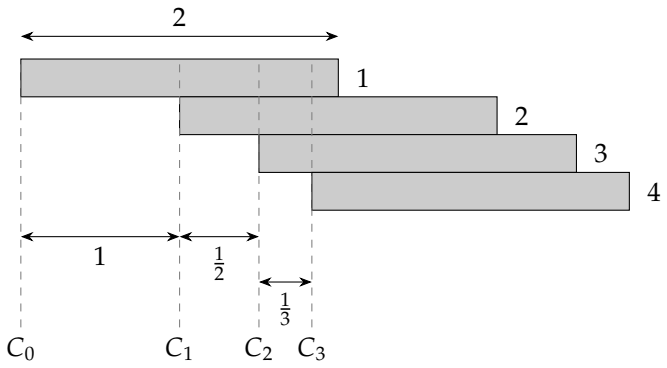


Figure 46: Four blocks of length 2, numbered from the top. The left end of block $n+1$ sits at C_n, the center of mass of blocks $1, \dots, n$.

Block $n+1$ then has its center at $C_n + 1$, so

$$C_{n+1} = \frac{nC_n + 1\cdot(C_n+1)}{n+1} = C_n + \frac{1}{n+1}, \qquad C_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

Already $C_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12} > 2$, so with five blocks the top block lies entirely beyond the bottom one. Since $C_n \ge \ln n$ grows without bound, the stack can reach as far as desired, given enough blocks.

The growth is extremely slow. The blocks measure 30 cm by 3 cm, so a unit is 15 cm. A stack spanning two lab tables, 26 units long, needs an overhang $C_{N-1} \approx 26 - 2 = 24$ units, so $\ln N \approx 24$

and $N \approx e^{24}$. The stack would be $3\text{ cm} \cdot e^{24} \approx 8 \times 10^8$ m high, roughly twice the distance to the moon.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 36, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 37: Taylor's series

What *is* $\cos x$, as a computable quantity? This lecture represents functions by power series, starting from the geometric series $1 + a + a^2 + \cdots = 1/(1-a)$ for $|a| < 1$.

37.1 Power series

Definition 37.1 [Power series]. A power series is an infinite sum

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots .$$

It represents f for $|x| < R$, where R is the radius of convergence: for $|x| < R$ the terms satisfy $|a_n x^n| \to 0$ geometrically as $n \to \infty$, while for $|x| > R$ they do not tend to 0.

For the geometric series ($a_n = 1$, $R = 1$) at $x = \frac{1}{2}$, the terms are $1/2^n$, and the tail of the series is negligible. At $x = -1$ the terms have $|(-1)^n| = 1 \not\to 0$, and the partial sums of $1 - 1 + 1 - 1 + \cdots$ bounce between 1 and 0; outside $-1 < x < 1$ the series diverges.

37.2 Basic tools

Within the radius of convergence, power series obey the rules of polynomials: they can be substituted into, multiplied, differentiated term by term, and integrated term by term.

Substitution. Starting from $1/(1-x) = 1 + x + x^2 + \cdots$, the substitutions $x = -u$ and $x = -v^2$ give

$$\frac{1}{1+u} = 1 - u + u^2 - u^3 + \cdots , \qquad \frac{1}{1+v^2} = 1 - v^2 + v^4 - v^6 + \cdots ,$$

valid for $|u| < 1$ and $|v| < 1$.

Multiplication. Multiplying term by term and collecting powers,

$$\frac{1}{(1-x)^2} = (1 + x + x^2 + \cdots)(1 + x + x^2 + \cdots) = 1 + 2x + 3x^2 + \cdots .$$

Differentiation. Differentiating $1/(1-x) = 1 + x + x^2 + x^3 + \cdots$ term by term gives the same result by a different method:

$$\frac{d}{dx}\frac{1}{1-x} = \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots .$$

Integration. If $f(x) = a_0 + a_1x + a_2x^2 + \cdots$, then

$$\int f(x)\,dx = c + a_0x + \frac{a_1}{2}x^2 + \frac{a_2}{3}x^3 + \cdots.$$

Example: *Logarithm and arctangent.* Integrating the two substituted series from 0 to x (so that the constant is 0) gives, for $|x| < 1$,

$$\ln(1+x) = \int_0^x \frac{du}{1+u} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \qquad \tan^{-1}x = \int_0^x \frac{dv}{1+v^2} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots.$$

37.3 Taylor's formula

If $f(x) = a_0 + a_1x + a_2x^2 + \cdots$, differentiating repeatedly gives

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + \cdots, \quad f''(x) = 2\cdot1\,a_2 + 3\cdot2\,a_3x + \cdots, \quad f'''(x) = 3\cdot2\cdot1\,a_3 + 4\cdot3\cdot2\,a_4x + \cdots.$$

At $x = 0$ all terms but the first vanish: $f(0) = a_0$, $f'(0) = a_1$, $f''(0) = 2a_2$, $f'''(0) = 3!\,a_3$, and in general $f^{(n)}(0) = n!\,a_n$.

Theorem 37.2 [Taylor's formula]. *If $f(x) = \sum_{n\ge0} a_nx^n$ within its radius of convergence, then*

$$a_n = \frac{f^{(n)}(0)}{n!} \qquad (n = 0, 1, 2, \ldots;\ 0! = 1).$$

The series $\sum_n f^{(n)}(0)x^n/n!$ is the Taylor series *of f at 0.*

Example: *Exponential.* For $f(x) = e^x$, every derivative is e^x, so $f^{(n)}(0) = 1$ and $a_n = 1/n!$:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \qquad e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots,$$

which computes e to any accuracy.

Example: *Cosine and sine.* For $f(x) = \cos x$ the derivatives cycle through $-\sin x$, $-\cos x$, $\sin x$, $\cos x$, so the values at 0 cycle through $1, 0, -1, 0$. Only even coefficients are nonzero and their signs alternate:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}.$$

Like $\cos x$, the series is even: it has only even powers. Differentiating term by term,

$$-\sin x = \frac{d}{dx}\cos x = -x + \frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!} - \cdots, \qquad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!},$$

where the odd power $n = 2k + 1$ has coefficient $(-1)^k/n!$. Taylor's formula gives the same series directly. The first terms recover the earlier approximations $\cos x \approx 1 - \frac{1}{2}x^2$ and $\sin x \approx x$.

Example: *Binomial expansion.* For $f(x) = (1 + x)^a$, $f^{(n)}(0) = a(a - 1)\cdots(a - n + 1)$, so

$$(1 + x)^a = 1 + \frac{a}{1}x + \frac{a(a-1)}{2!}x^2 + \frac{a(a-1)(a-2)}{3!}x^3 + \cdots .$$

37.4 Taylor series with another base point

The Taylor series of f with base point b is

$$f(x) = f(b) + f'(b)(x - b) + \frac{f''(b)}{2!}(x - b)^2 + \frac{f'''(b)}{3!}(x - b)^3 + \cdots .$$

For $\sqrt{x}$, the base point $b = 0$ is a bad choice because $\sqrt{x}$ is not differentiable at 0. With $b = 1$, the derivatives at 1 are those of $(1 + u)^{1/2}$ at $u = 0$ with $u = x - 1$, so

$$x^{1/2} = 1 + \frac{1}{2}(x - 1) + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}(x - 1)^2 + \cdots = 1 + \frac{1}{2}(x - 1) - \frac{1}{8}(x - 1)^2 + \cdots .$$

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 37, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 38: Final review: power series and applications

38.1 Differentiating and integrating series

Within the radius of convergence, if $f(x) = \sum_{n=0}^{\infty} a_n x^n$, then

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \qquad \int f(x)\,dx = C + \sum_{n=0}^{\infty} \frac{a_n x^{n+1}}{n+1}.$$

Example: *The normal distribution.* The integral $\int_0^x e^{-t^2}\,dt$, which gives probabilities for the normal (Gaussian) distribution, is not an elementary function, but substituting $-t^2$ into the series for e^u and integrating term by term computes it:

$$\int_0^x e^{-t^2}\,dt = \int_0^x \left(1 - t^2 + \frac{t^4}{2!} - \frac{t^6}{3!} + \frac{t^8}{4!} - \cdots\right)dt = x - \frac{x^3}{3} + \frac{1}{2!}\frac{x^5}{5} - \frac{1}{3!}\frac{x^7}{7} + \cdots.$$

Elementary functions are still a little better. The series $\sin\frac{\pi}{2} = \frac{\pi}{2} - \frac{(\pi/2)^3}{3!} + \frac{(\pi/2)^5}{5!} - \cdots$ hides the simple value $\sin\frac{\pi}{2} = 1$; nothing in the series shows that $\sin x$ has to do with angles.

38.2 A series for π

Reading the last fact backwards gives a series for $\pi/2$. Since $\sin^{-1} 1 = \pi/2$,

$$\int_0^1 \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x\Big|_0^1 = \frac{\pi}{2}.$$

To expand the integrand, first take the binomial series with exponent $-\frac{1}{2}$:

$$(1+u)^{-1/2} = 1 + \left(-\frac{1}{2}\right)u + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{1\cdot 2}u^2 + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{1\cdot 2\cdot 3}u^3 + \cdots = 1 - \frac{1}{2}u + \frac{1\cdot 3}{2\cdot 4}u^2 - \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}u^3 + \cdots.$$

Odd numbers appear on top, even numbers on the bottom, and the signs alternate. Setting $u = -x^2$ makes every sign positive, and integrating from 0 gives, for $|x| < 1$,

$$(1-x^2)^{-1/2} = 1 + \frac{1}{2}x^2 + \frac{1\cdot 3}{2\cdot 4}x^4 + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}x^6 + \cdots,$$

$$\sin^{-1} x = x + \frac{1}{2}\frac{x^3}{3} + \frac{1\cdot 3}{2\cdot 4}\frac{x^5}{5} + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\frac{x^7}{7} + \cdots.$$

At $x = 1$ this becomes

$$\frac{\pi}{2} = 1 + \frac{1}{2}\left(\frac{1}{3}\right) + \frac{1\cdot 3}{2\cdot 4}\left(\frac{1}{5}\right) + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\left(\frac{1}{7}\right) + \cdots . \tag{38.1}$$

Note. The point $x = 1$ is the endpoint of the interval of convergence, so (38.1) needs a check. Write the nth term as $c_n = p_n/(2n+1)$ with $p_n = \frac{1\cdot 3\cdots(2n-1)}{2\cdot 4\cdots 2n}$. Since $\frac{2k-1}{2k} \le \frac{2k}{2k+1}$,

$$p_n^2 \le \prod_{k=1}^{n} \frac{2k-1}{2k}\cdot\frac{2k}{2k+1} = \frac{1}{2n+1}, \qquad c_n \le \frac{1}{(2n+1)^{3/2}},$$

so the series converges by comparison with a p-series with $p = 3/2$ (Lecture 36). All terms are positive. For $0 \le x < 1$ and every N, $\sum_{n\le N} c_n x^{2n+1} \le \sin^{-1} x \le \sum_n c_n$; letting $x \to 1^-$ gives $\sum_{n\le N} c_n \le \frac{\pi}{2} \le \sum_n c_n$, and letting $N \to \infty$ gives equality.

38.3 Weighted averages

Definition 38.1 [Weighted average]. The weighted average of f on $[a, b]$ with weighting function $w \ge 0$ is

$$\text{Average}(f) = \frac{\int_a^b w(x) f(x)\,dx}{\int_a^b w(x)\,dx},$$

where $\int_a^b w(x)\,dx$ is the total weight. The interval may be infinite.

Example: *Betting on a decay.* You are paid t dollars if a certain particle decays at time t seconds. The likelihood that it has not yet decayed at time t, the weighting function, is $w(t) = e^{-kt}$, and $\int_0^\infty e^{-kt}\,dt = 1/k$. The fair price is the average payoff with $f(t) = t$:

$$\frac{\int_0^\infty te^{-kt}\,dt}{\int_0^\infty e^{-kt}\,dt} = k\int_0^\infty te^{-kt}\,dt = \int_0^\infty (kt)e^{-kt}\,dt = \int_0^\infty ue^{-u}\,\frac{du}{k} = \frac{1}{k},$$

using $u = kt$ and $\int_0^\infty ue^{-u}\,du = 1$ (integration by parts). With half-life H, $k = (\ln 2)/H$, so the expected payoff is $1/k = H/\ln 2$.

The same calculation prices an annuity, which pays a fixed amount per unit time until the holder dies. With t in years and $w(t) = e^{-kt}$ the likelihood of being alive at time t, the payout accumulated by time t is proportional to $f(t) = t$, and the expected number of years of payments is again $1/k$.

Source. David Jerison, MIT 18.01, *Single Variable Calculus*, Lecture 38, Fall 2006. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.